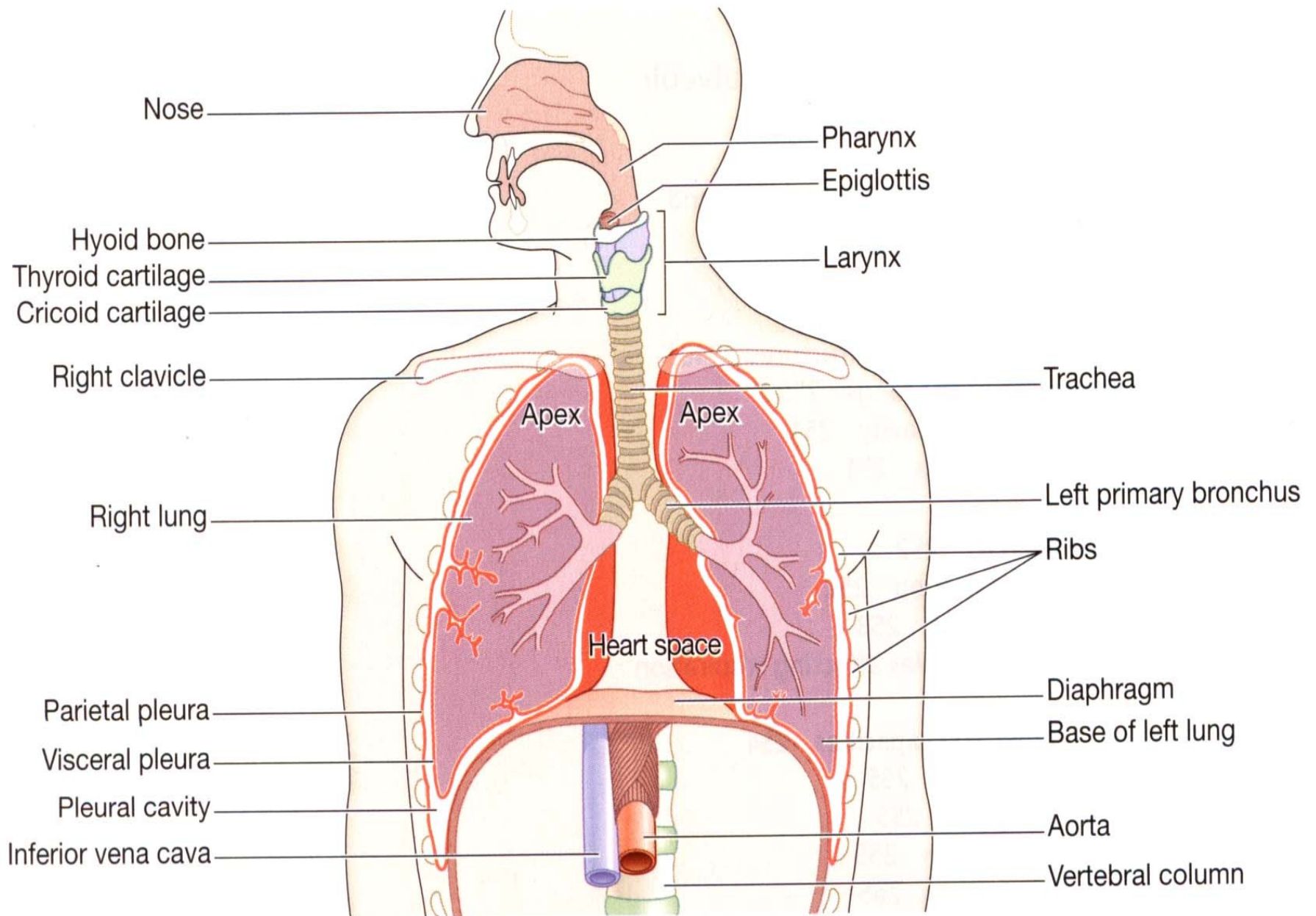


Histology of

RESPIRATORY SYSTEM / TRACT

The Respiratory System

- Consists of the two **lungs** and a series of air passage ways that lead to and from the air spaces (alveoli) in the lungs.
- is formed of **two functional components**:
 - **Conducting portion**: includes airways that deliver air to and from the lungs.
 - Extends from the nose to terminal bronchioles
 - **Respiratory portion**: includes structures within the lungs (respiratory bronchioles to alveoli inclusive) where oxygen in inhaled air is exchanged for carbon dioxide in the pulmonary blood.
- The conducting tubes possess lining epithelia, supporting CTs (loose areolar tissue rich in collagen and elastic fibers and cartilages), sero-mucous glands, smooth muscle tissue and other features that are characteristics of each part.
- Each type of airway has its own characteristic structural features but there is a gradual, rather than abrupt, transition from one airway to the next along the whole length of the tract.



The Conducting parts include:

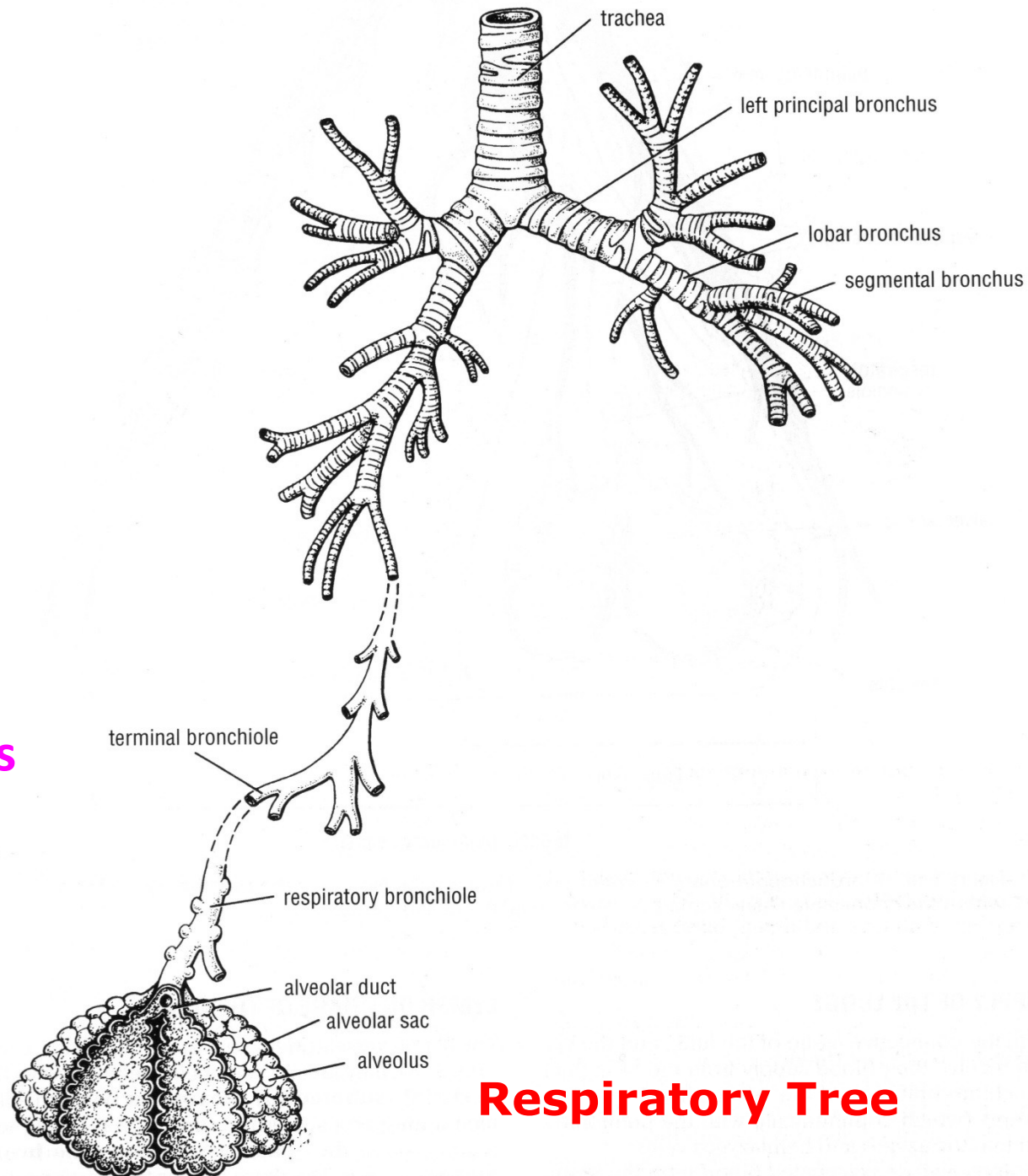
- Nasal cavity & surrounding paranasal sinuses
- Pharynx
- Larynx
- Trachea
- Bronchi (1^o, 2^o, 3^o and so on)
- Bronchioles, and
- Terminal bronchioles

The Respiratory zone includes:

- Respiratory bronchioles
- Alveolar ducts
- Alveolar sacs and
- Alveoli

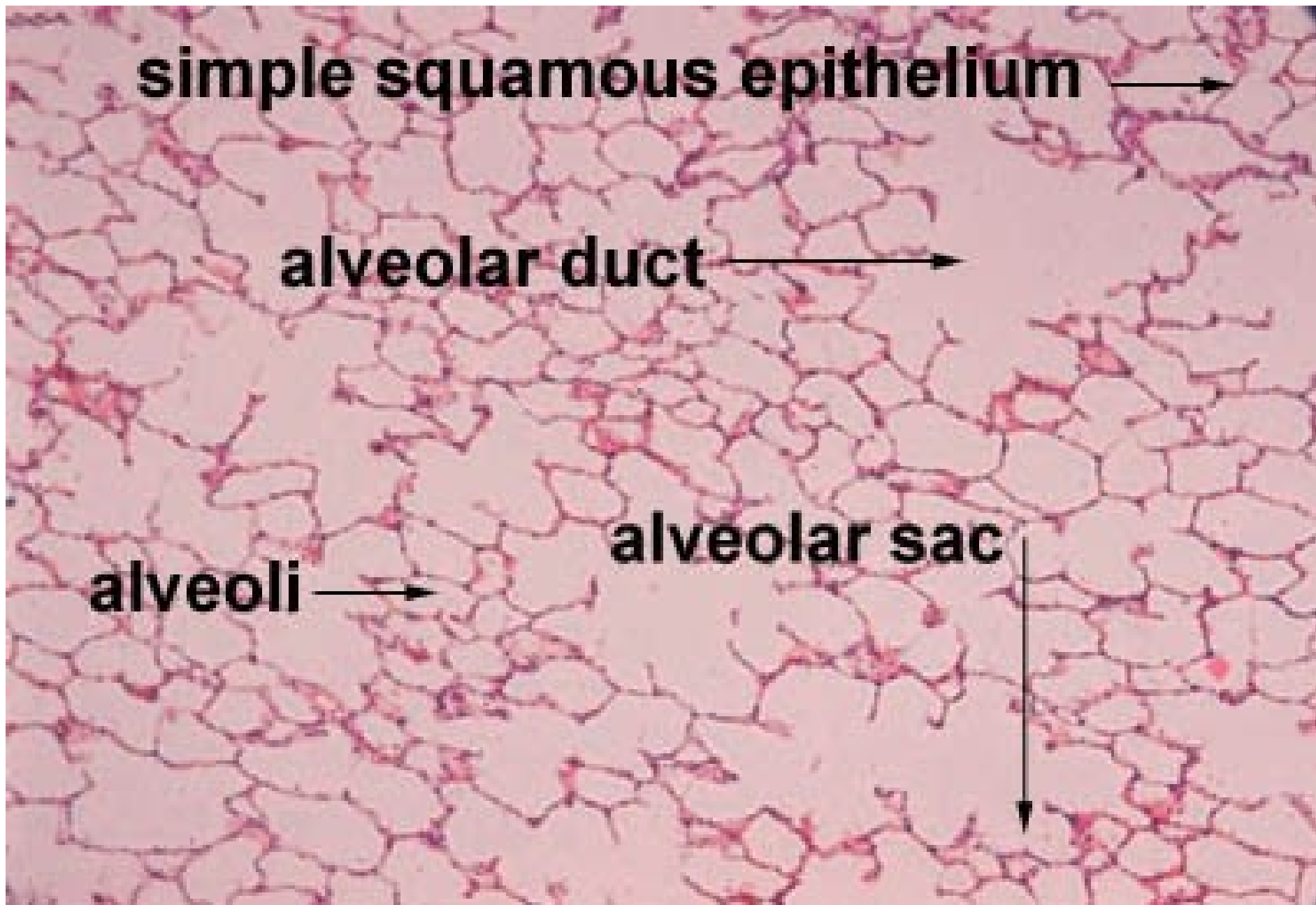
Trachea
↓
Main bronchi
↓
Lobar bronchi
↓
Segmental bronchi
↓

Bronchioles
↓
Terminal bronchioles
↓
Respiratory bronchioles
↓
Alveolar ducts
↓
Alveolar sacs
↓
Alveoli



Respiratory Tree

Histological section of LUNG, Medium power, H&E

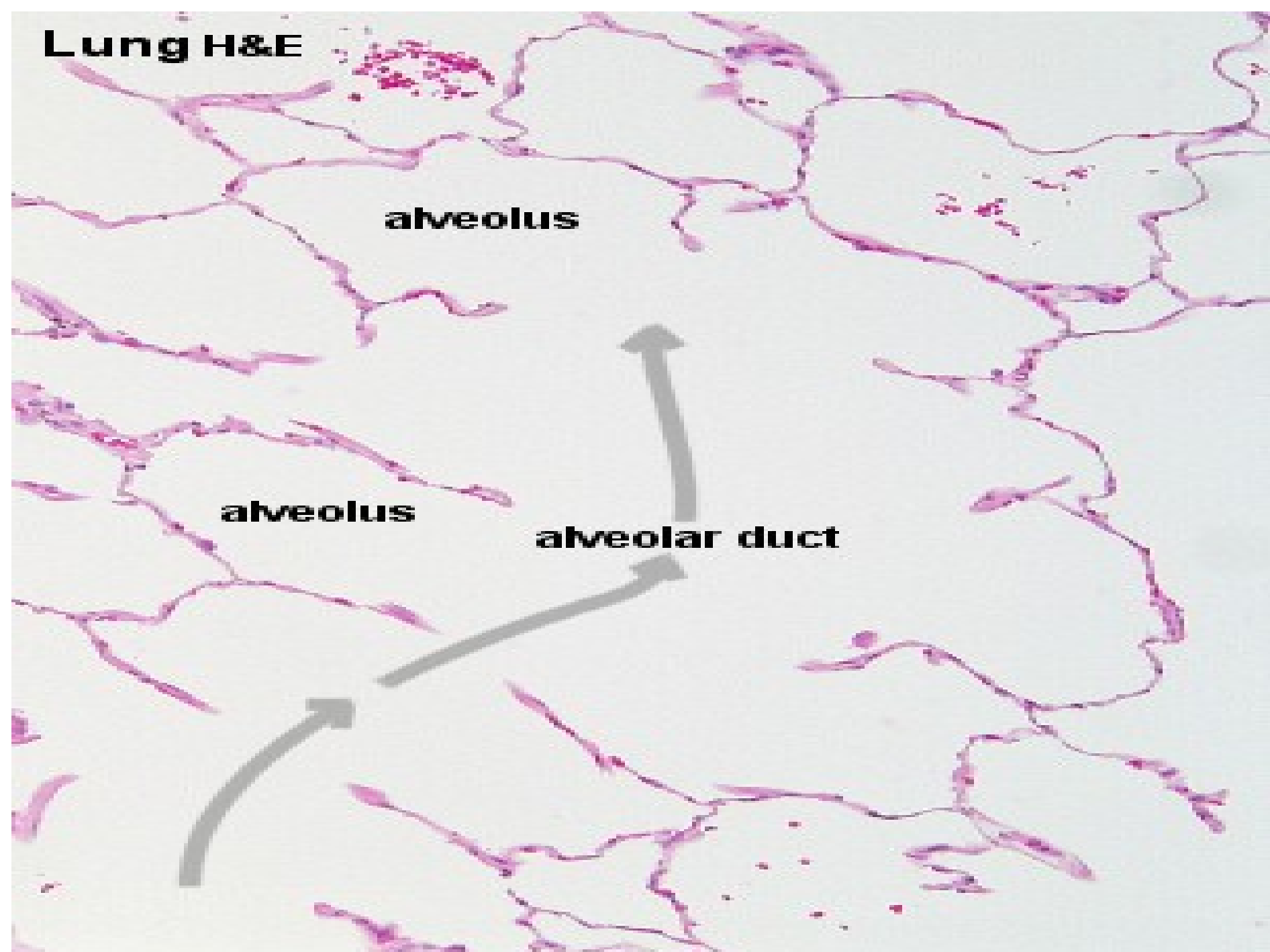


Lung H&E

alveolus

alveolus

alveolar duct



GENERAL STRUCTURE OF THE CONDUCTING PORTION OF RESPIRATORY TRACT

- Conducting respiratory airways are pliable tubes lined by respiratory mucosa and containing variable amounts of muscle and/or cartilage in their walls.
- The principal structural features include:

1. The Respiratory Epithelium

- undergoes progressive transition from a tall, pseudo-stratified columnar ciliated epithelium with goblet cells in the nasal cavity, nasopharynx, larynx, trachea, bronchi and bronchioles to a simple columnar in terminal bronchioles.
- The goblet cells are numerous in the trachea but decrease in number down the tree and are absent from the terminal bronchioles and distally.

2. The lamina propria

- **is subepithelial** loose fibroelastic CT rich in lymphoid aggregations (BALT) of variable size as part of MALT.

- These diffuse and nodular masses of immune cells and lymphocytes (BALT), produce secretory immunoglobulin A, which kills the bacteria and viruses and prevents them from penetrating the epithelium.

3. Muscle Layer

- A layer of smooth muscle lying deep to the mucosa (except in the trachea).

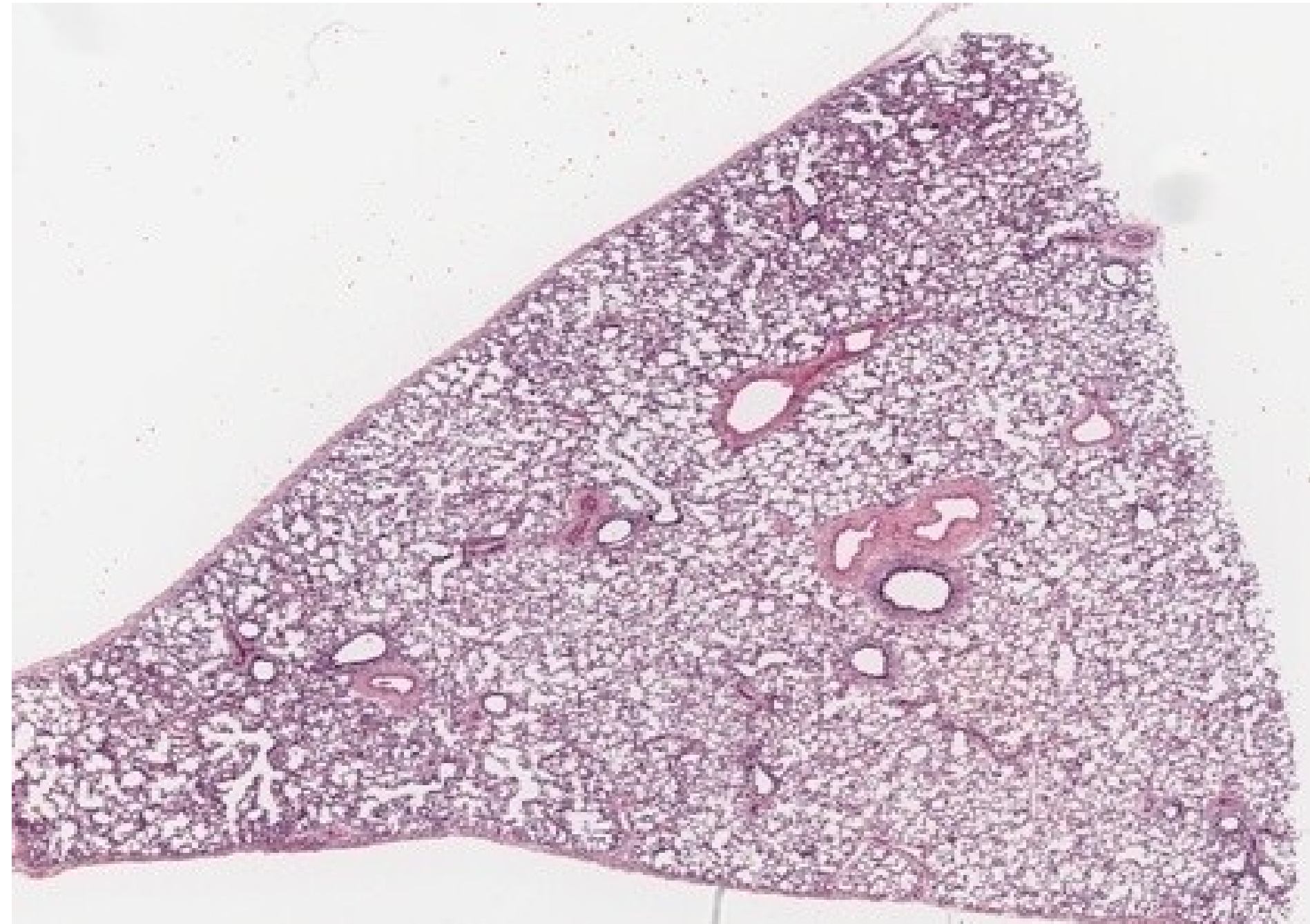
4. Submucosa (present in extrapulmonary airways only)

- contains sero-mucous glands whose secretion has moistening, filtering, and detoxifying action on the soluble gases.

5. Adventitia: outermost layer

- loose fibroelastic CT continuous with the surrounding tissue of lung parenchyma.
- In the smaller airways may contain glands and/or cartilage plates.

A Bronchopulmonary segment & its Intrapulmonary Bronchi

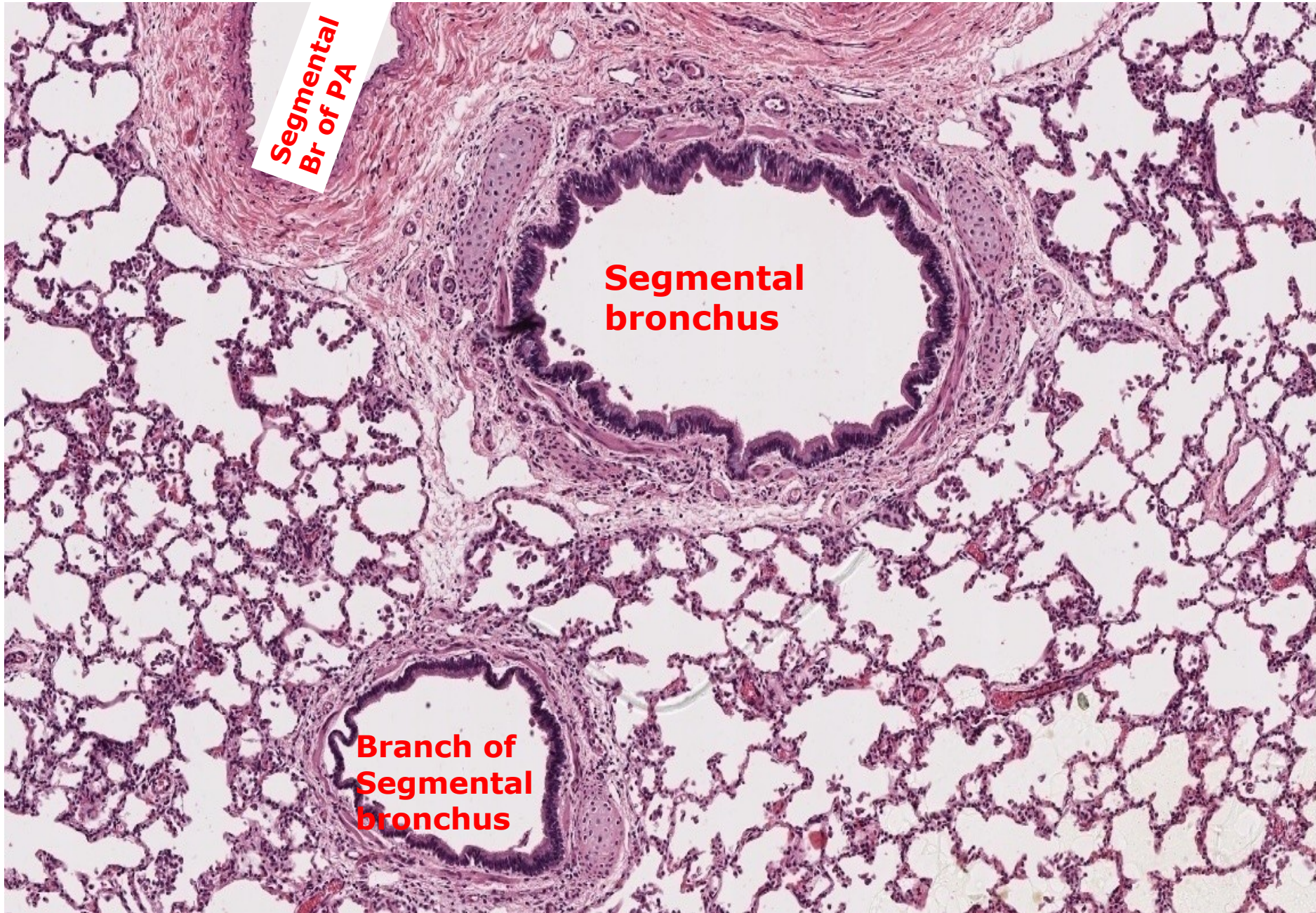


Lung section, medium power, H&E

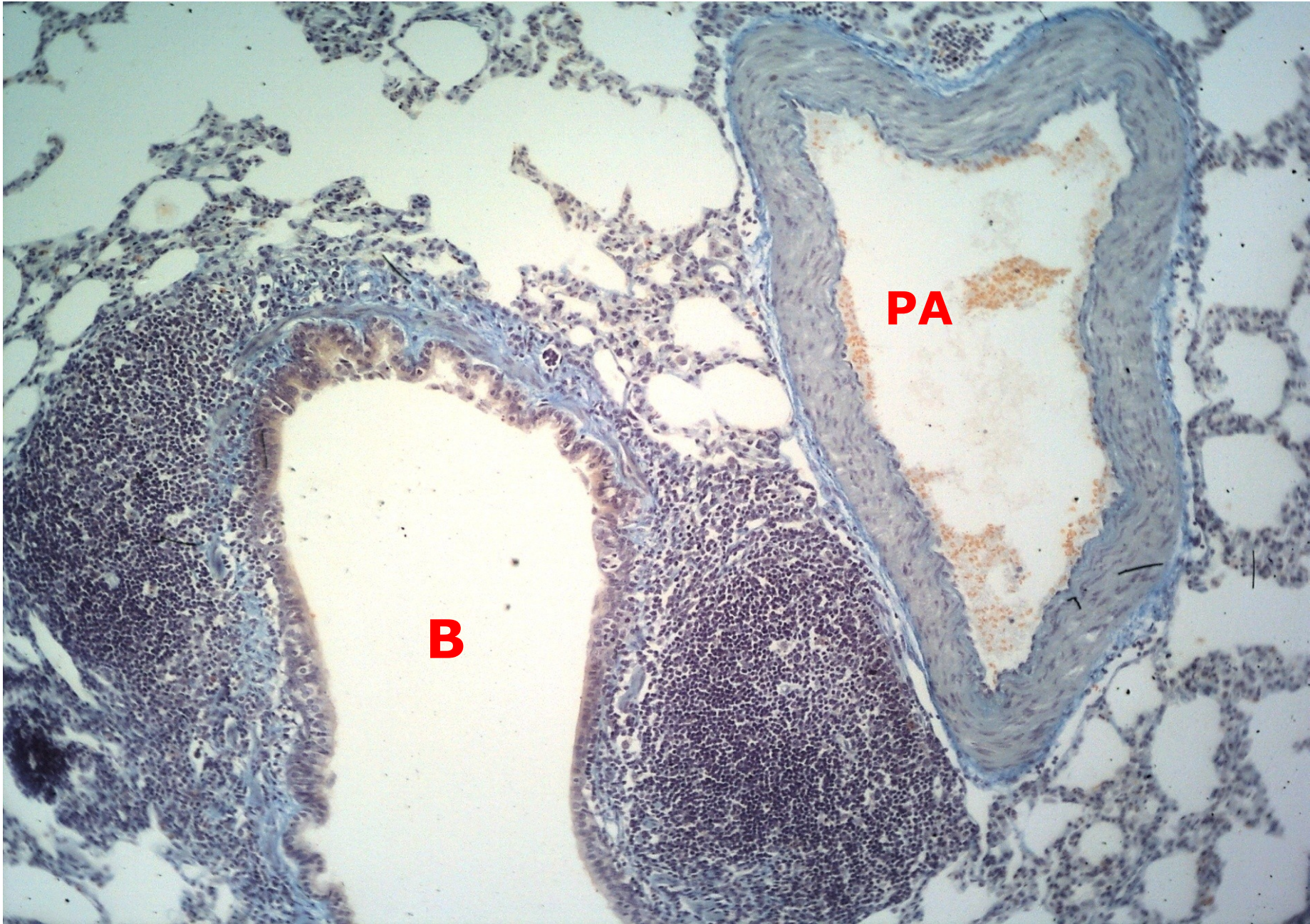
**Segmental
Br of PA**

**Segmental
bronchus**

**Branch of
Segmental
bronchus**



Bronchus with BALT and its PA

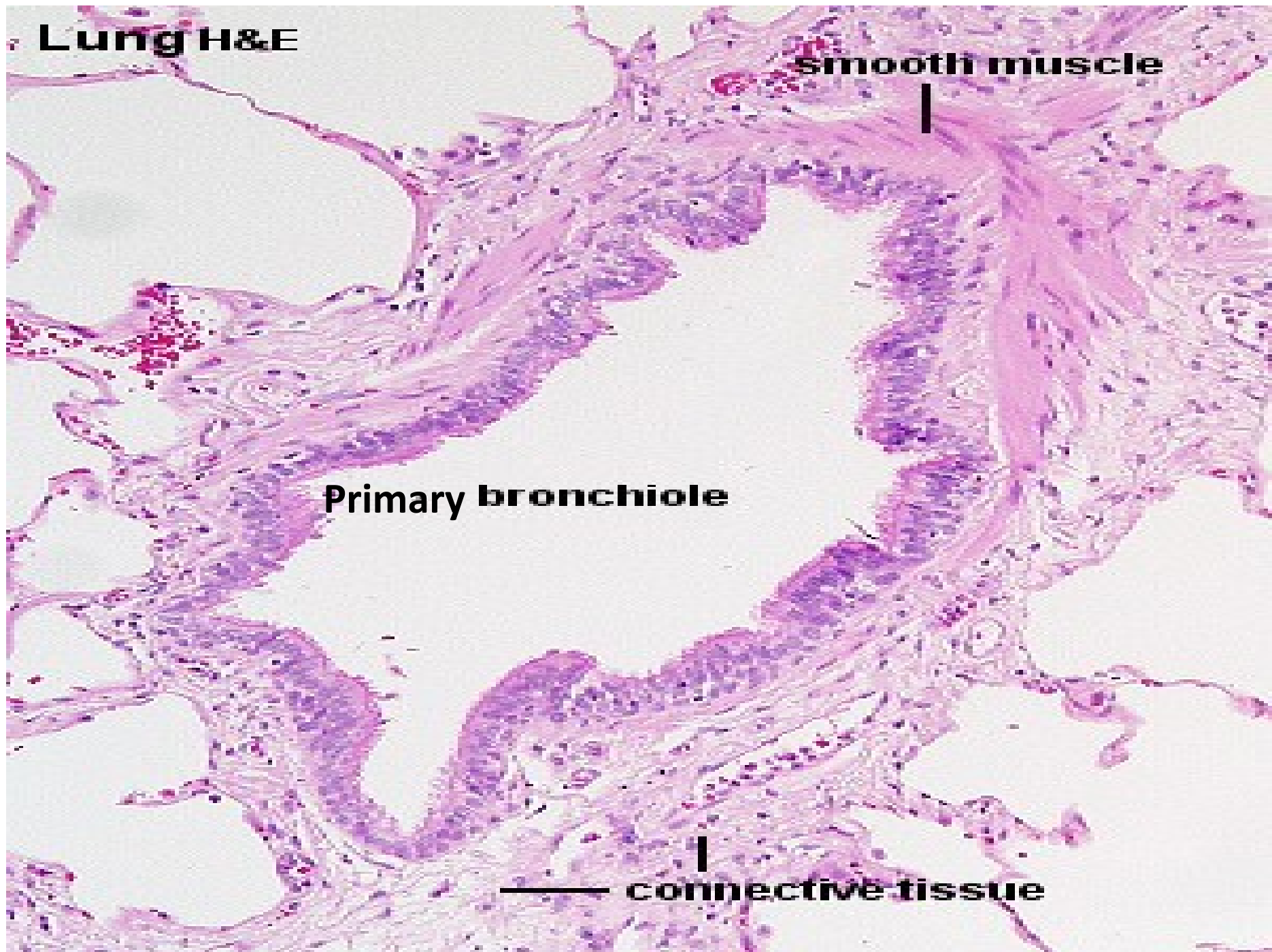


Lung H&E

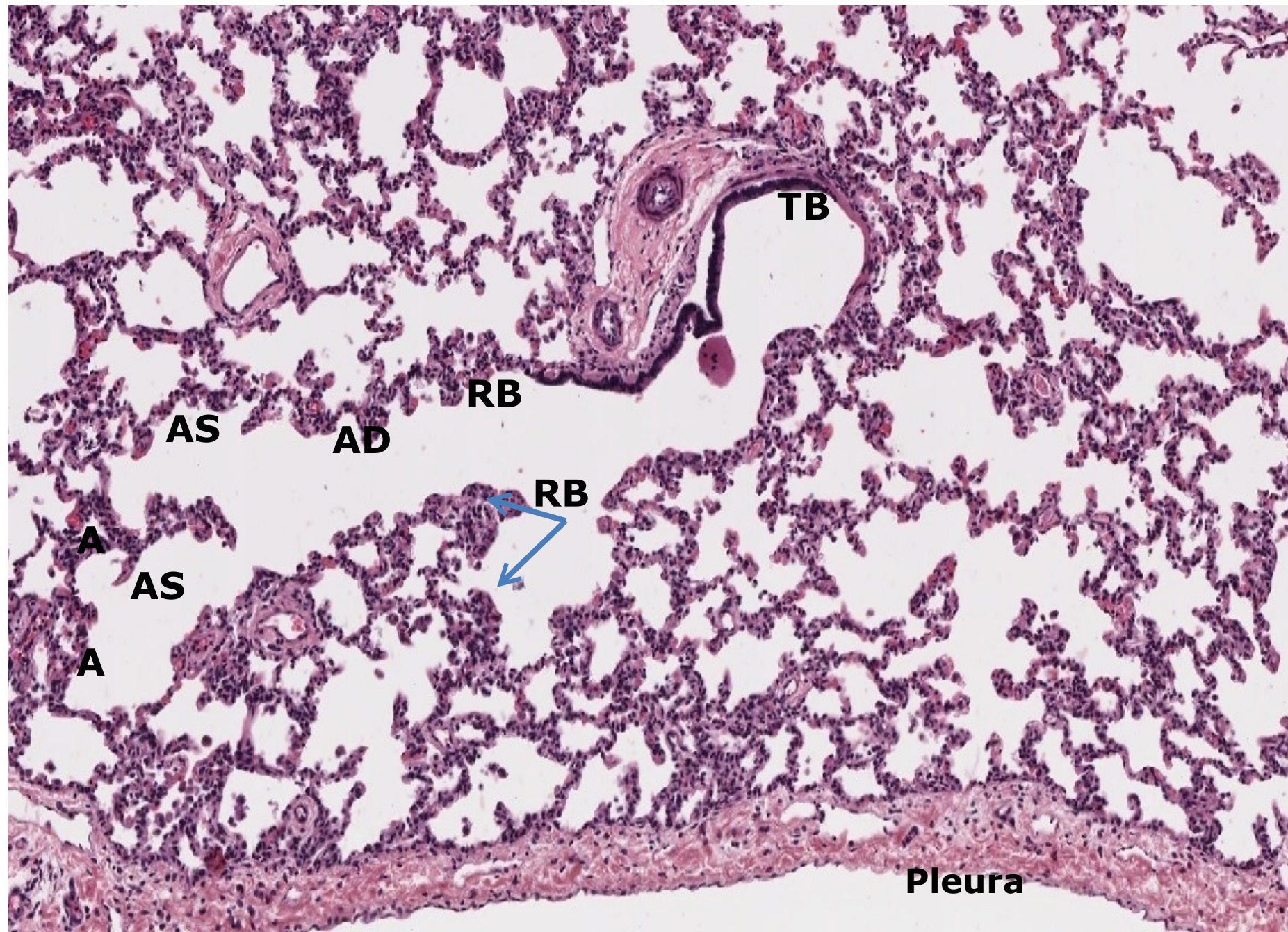
smooth muscle

Primary bronchiole

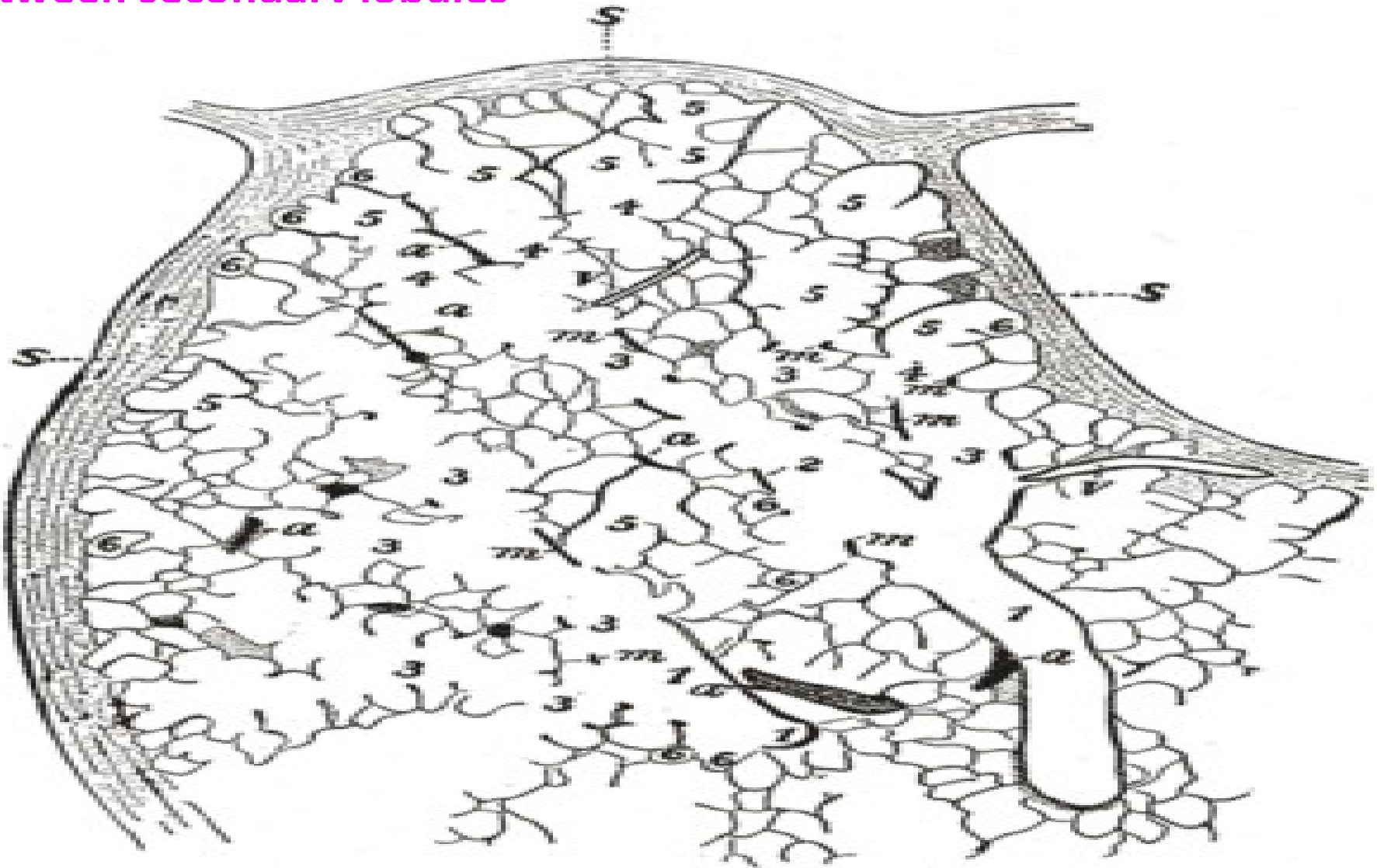
connective tissue



A Lung Acinus



1- Terminal bronchiole 2- Respiratory bronchiole 3- Alveolar duct
4- Atria 5-Alveolar sac 6- Alveolus or air cell m: smooth muscle
a: pulmonary artery branch v: pulmonary vein branch S: septum
between secondary lobules



Respiratory mucosa

- Lines most parts of respiratory tract
- Refers to the lining epithelium and underlying lamina propria.

Typical Respiratory Epithelium: pseudostratified columnar ciliated epithelium with goblet cells.

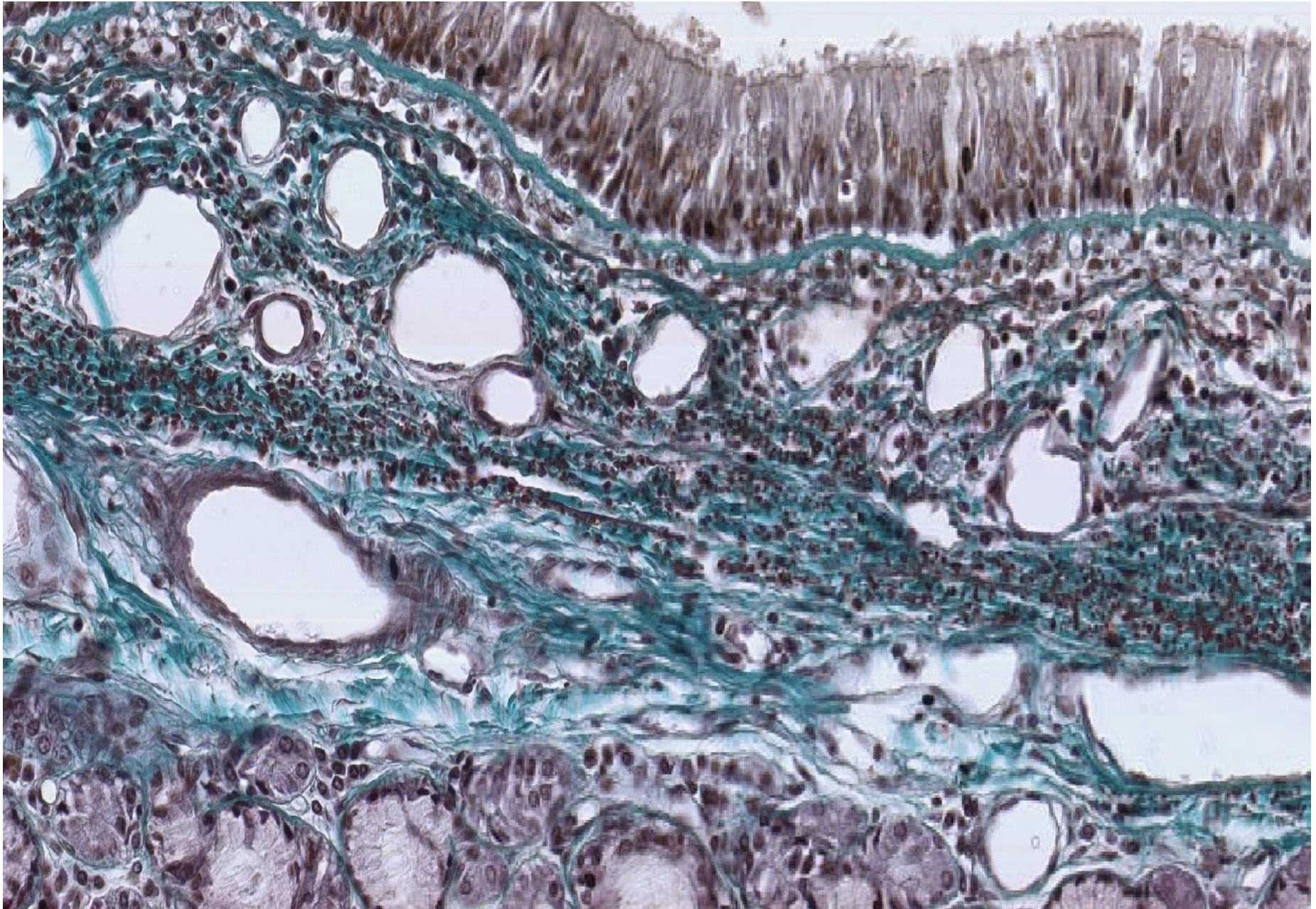
Lamina propria: loose areolar CT rich in:

- superficial blood vessels with thin wall and wide lumen for warming of incoming air.
- Lymphocytes, plasma cells and mast cells for immune response.

Note: In some regions of the nasal mucosa, there are veins resembling **erectile tissue** termed ***swell or cavernous bodies***.

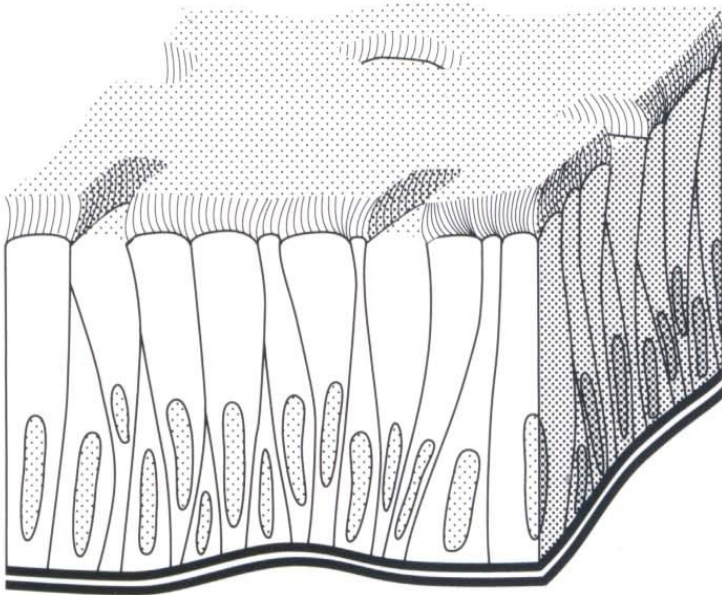
- These are specialized veins which engorge periodically and alternatively close one side of the nasal cavity, thus giving it time to recover from drying.
- This cyclic process is controlled by autonomic nerves.

Respiratory mucosa

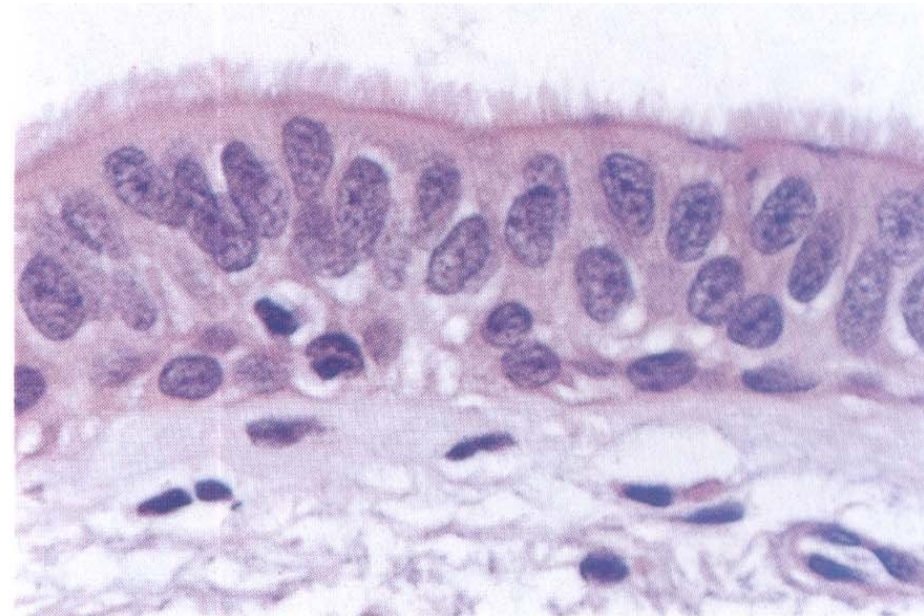


Respiratory Epithelium

- The mucosa of the conducting respiratory tract is lined by pseudostratified columnar ciliated epithelium with goblet cells.
- All the cells rest on the basement membrane but not all reach the top (the luminal surface).
- By this way two rows of nuclei are visible which gives this epithelium the name "pseudostratified". The columnar cells bear cilia on the top, which beat to push the mucus and trapped particles towards the pharynx. The goblet cells secrete mucous & appear hollow (white) in H & E.



(a)



(b)

Types of Epithelial Cells in Typical Respiratory Epithelium:

- **Three types of cells (more than 90% of cell population) are seen by light microscope:**

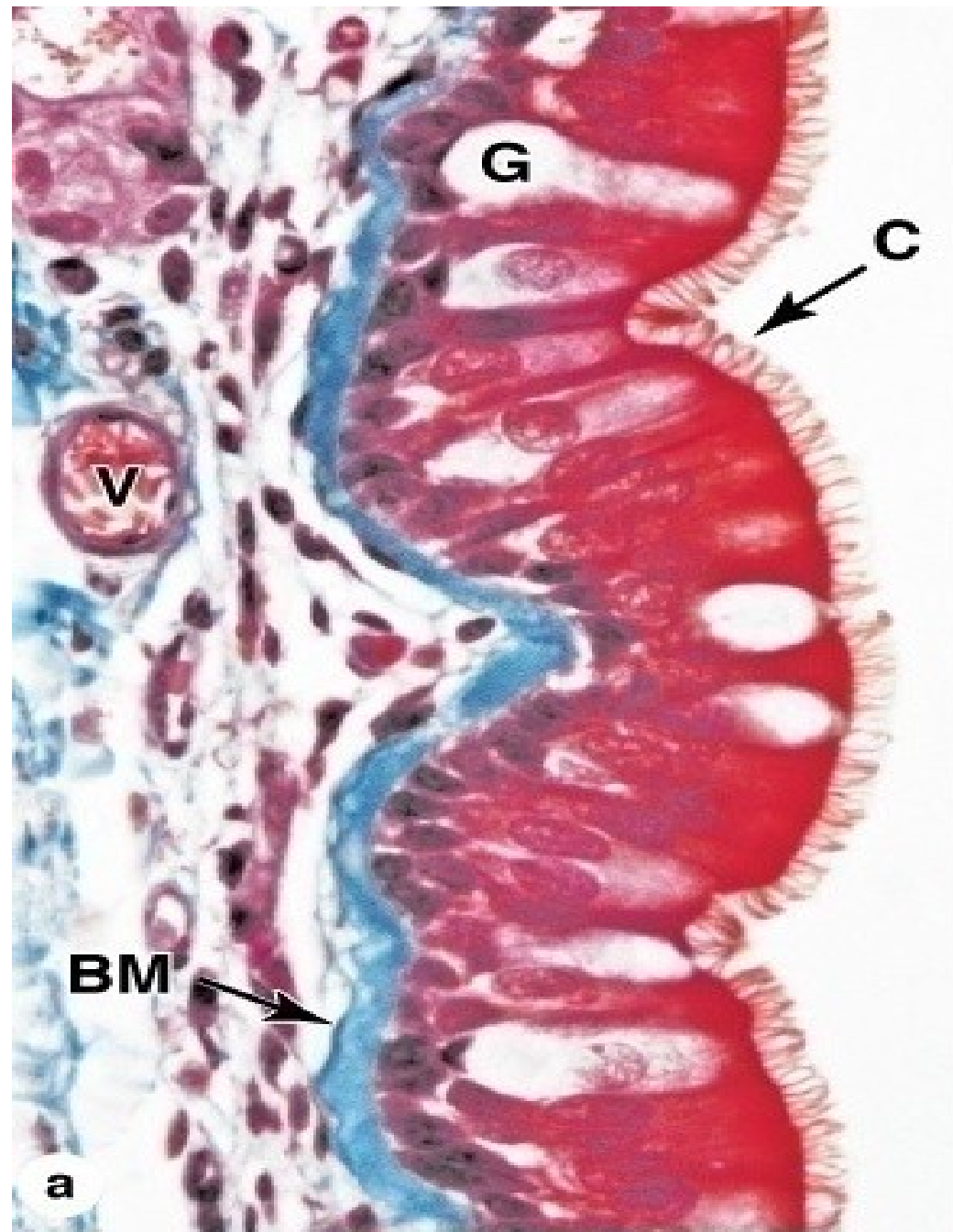
1. **Ciliated columnar cells.**
2. **Goblet cells** which produce mucous secretion.
3. **Basal cells:** these are undifferentiated cells which can divide by mitosis and can give other types of cells.

- **Two types of cells (less than 10% of cell population) seen by electron microscope or by special staining technique:**

4. **Brush cells** which contain cytoplasmic granules and carry microvilli.
5. **Granule cells** which are of 2 types according to function of granules:
 - Catecholamine secreting cells (neurosecretory cells).
 - Protein hormone secreting cells (enteroendocrine like cells).
 - They secrete amines and amine precursors.

Respiratory Epithelium: a classic example of pseudostratified columnar ciliated epithelium: rests on a thick **basement membrane (BM)** and has several cell types, most are columnar, some basal and all contacting the basal lamina. Ciliated columnar cells are the most abundant, with hundreds of long robust **cilia (C)** on each of their bulging apical ends which provide a lush cover of cilia on the luminal surface. Most of the small rounded cells at the basement membrane are **stem/basal cells** and their differentiating progeny.

Intraepithelial lymphocytes and **dendritic cells** are also present in respiratory epithelium as occasional wondering cells. Mucus-secreting **goblet cells (G)** are also present. The lamina propria is well-vascularized (**V**). X400. Mallory Azan trichrome.

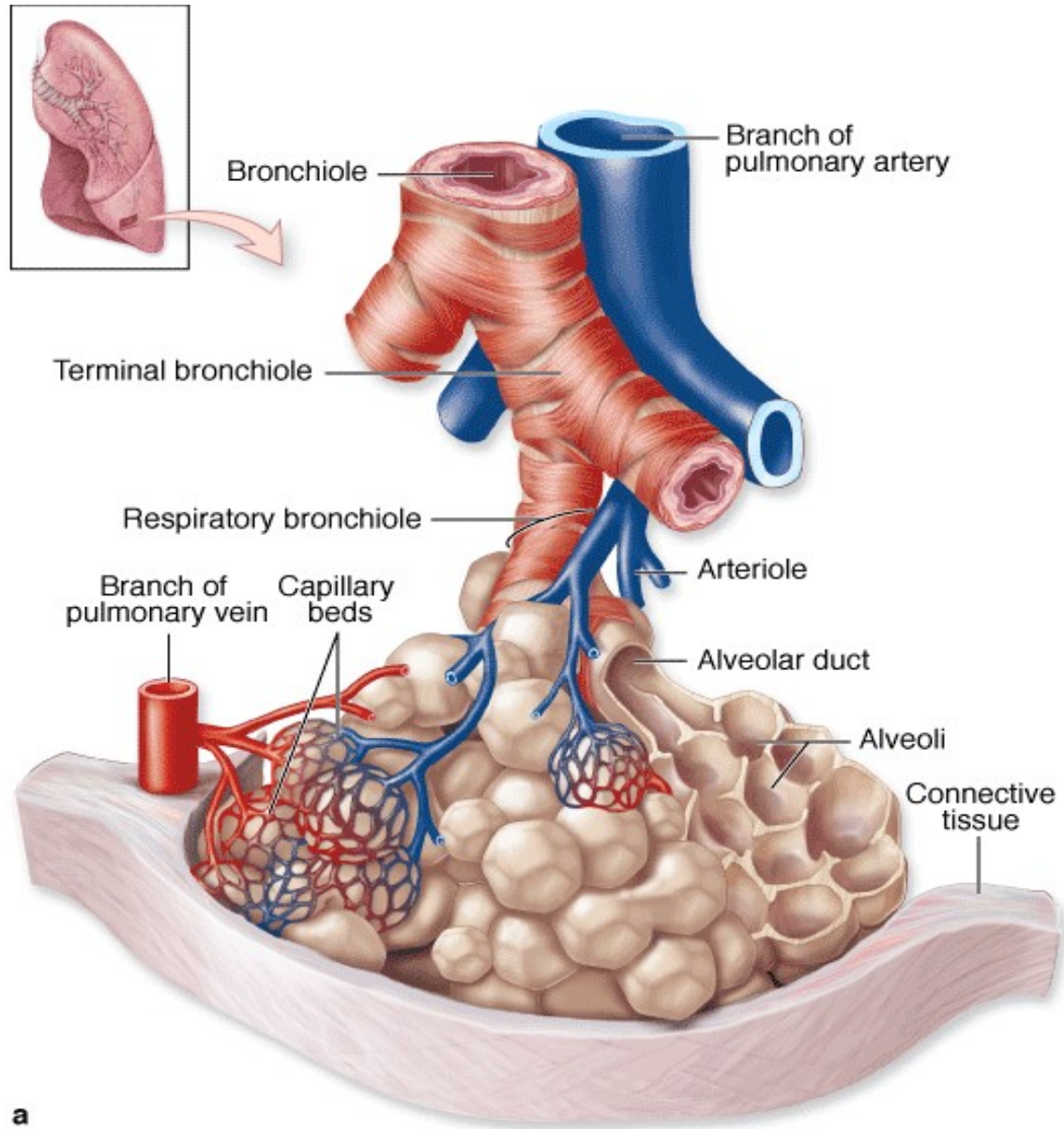


Pulmonary Vasculature & Nerves

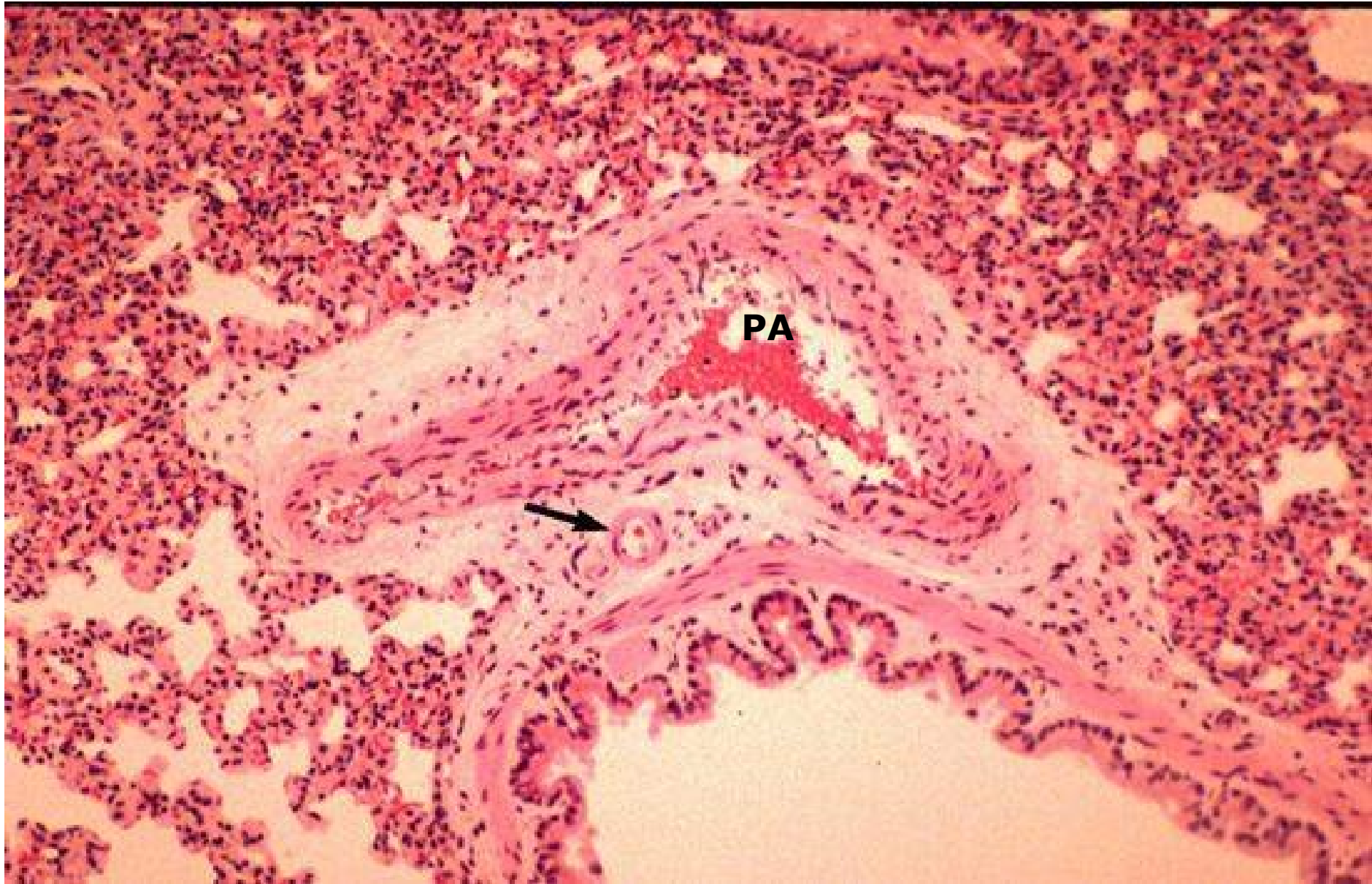
- Lungs are among the most vascularized organs of the body.
- Circulation in the lungs includes both nutrient (systemic) and functional (pulmonary) vessels.
- Pulmonary arteries and veins represent the functional circulation and the arteries are relatively thin-walled as a result of the low pressures (25 mmHg systolic, 5 mmHg diastolic) encountered in the pulmonary circuit.
- Within the lung , the pulmonary artery branches and closely accompanies the bronchial tree, with its branches surrounded by adventitia of the bronchi and bronchioles.
- At the level of the alveolar duct, the branches of this artery form a rich capillary network in the interalveolar septum and in close contact with the alveoli.
- The lung has the best-developed capillary network of any organ, with capillaries between all alveoli, including those in the respiratory bronchioles.

Pulmonary Vasculature

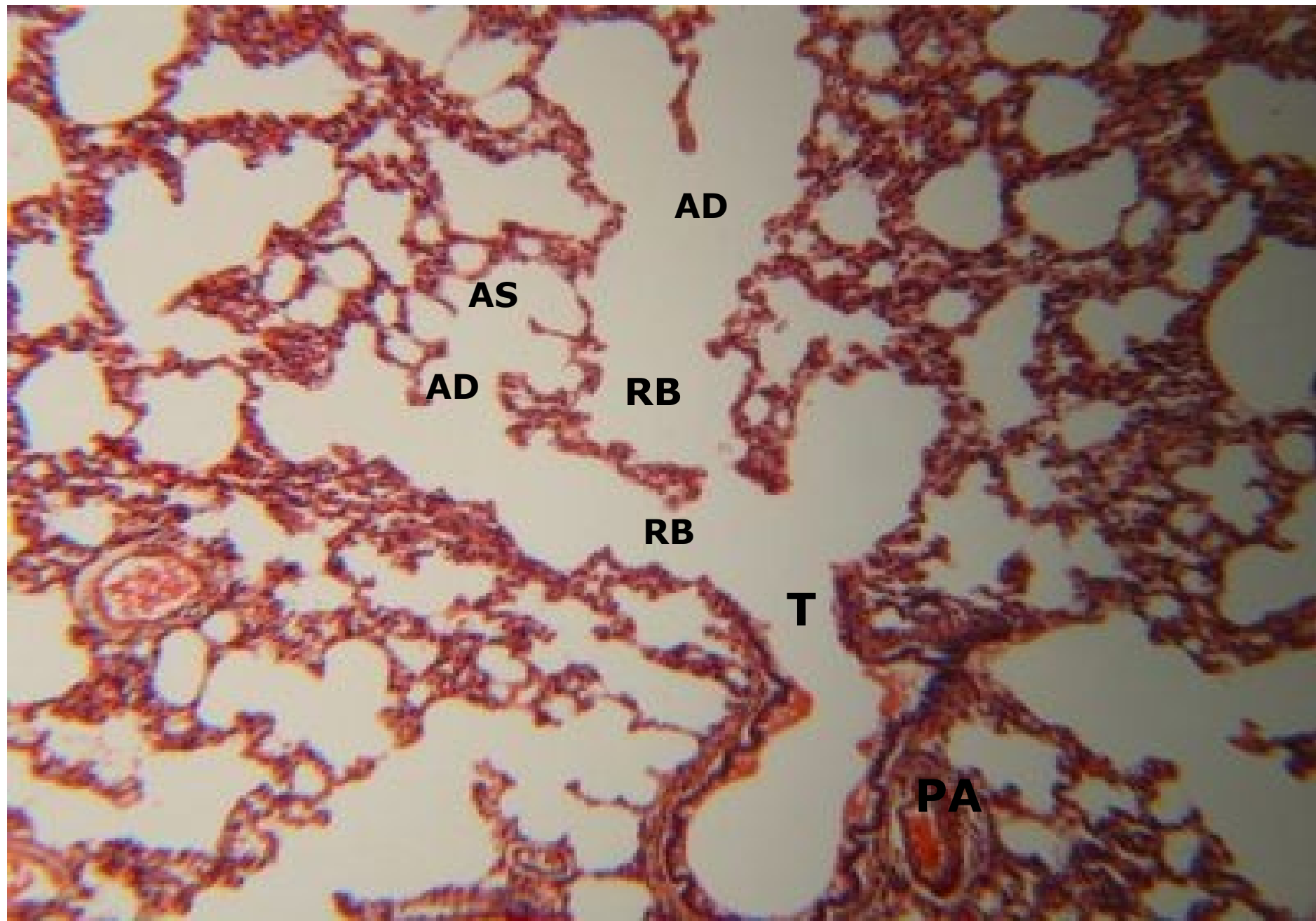
- Diagram shows the branching relationship, as well as the pulmonary blood vessels that travel with the bronchioles and the dense layer of branching capillaries that surrounds each alveolus for gas exchange between blood and air.



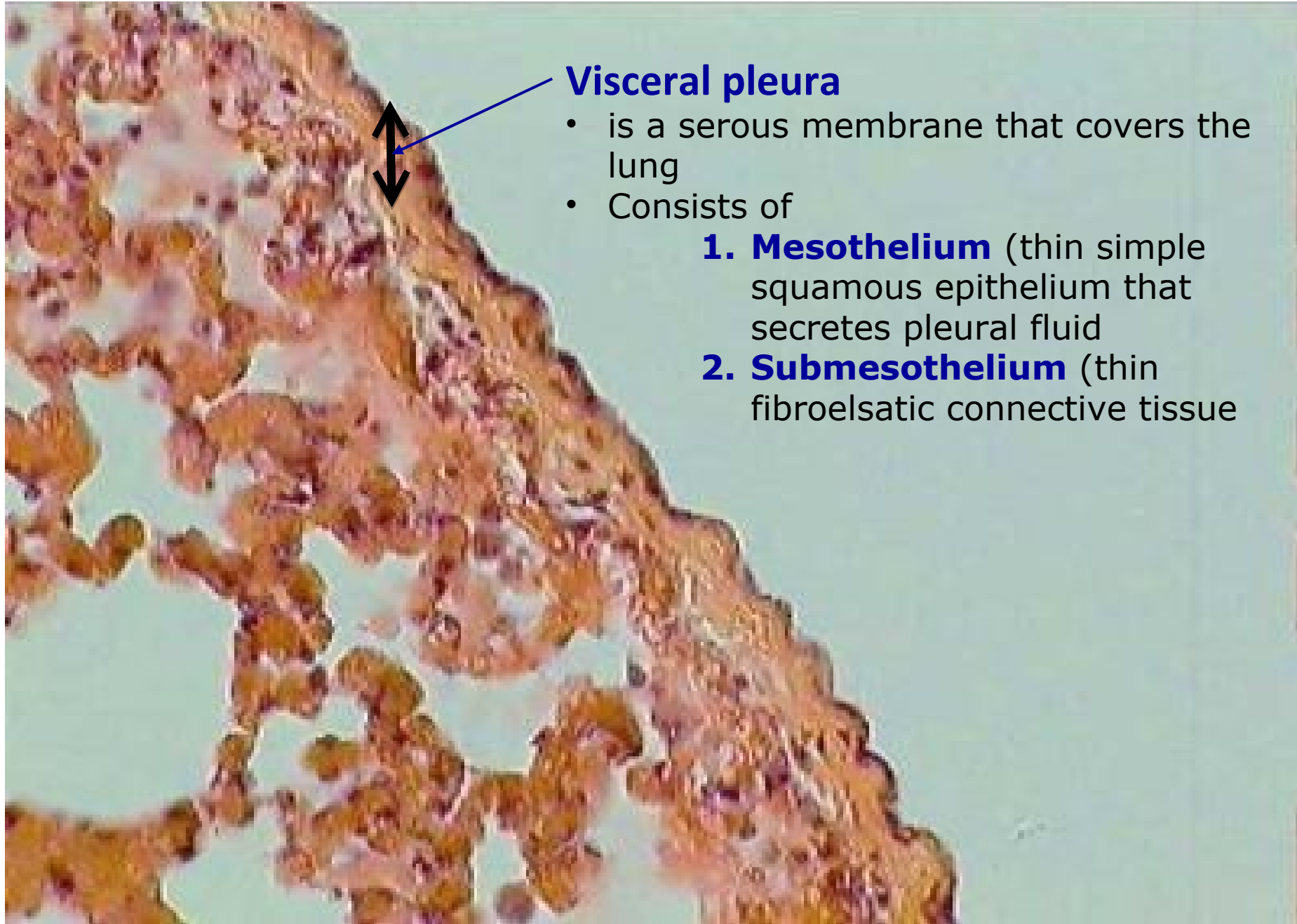
Branch of pulmonary artery (PA) and a bronchiolar arteriole in the adventitia of a bronchiole



LUNG BY HAEMATOXYLIN & EOSIN



Normal visceral pleura

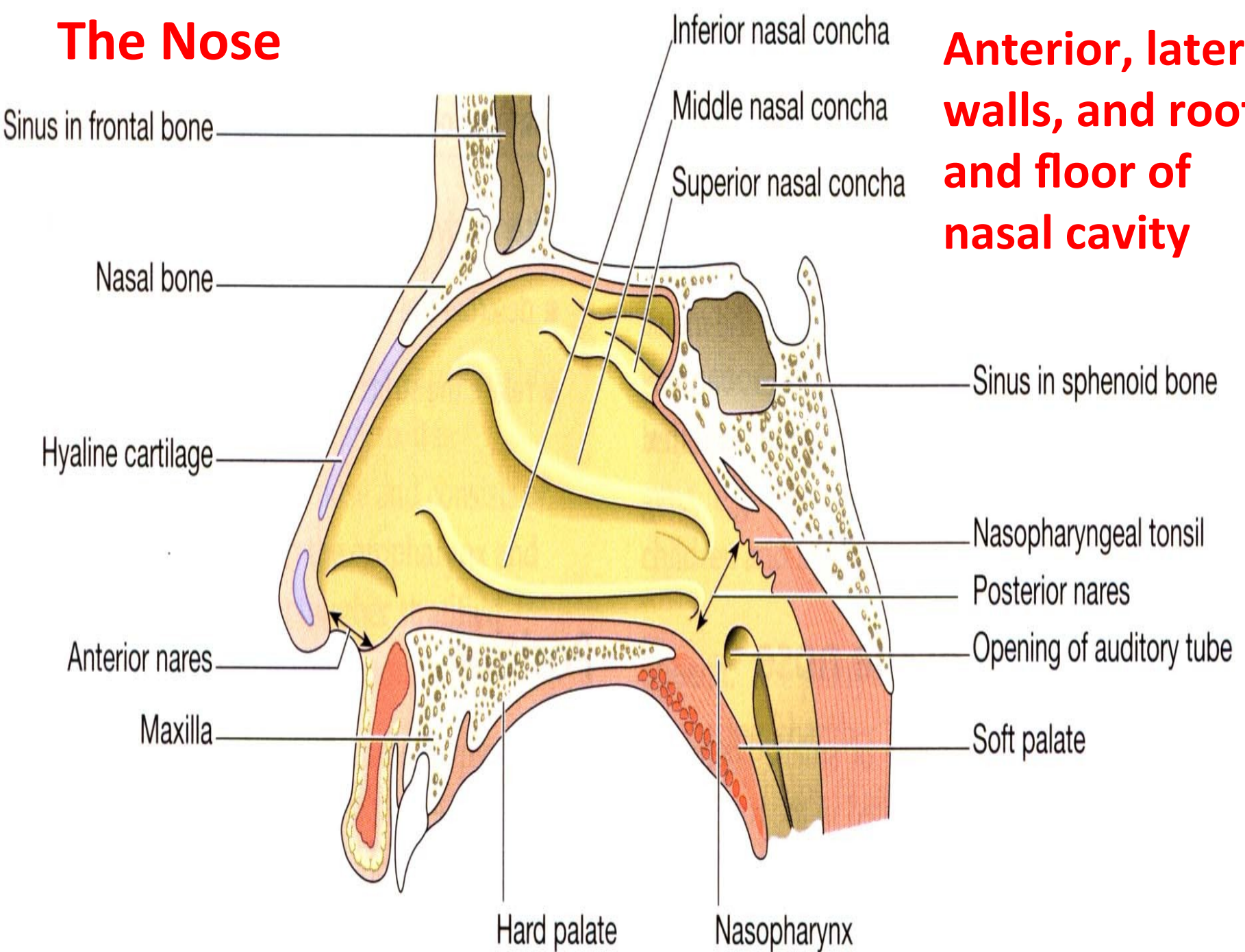


Visceral pleura

- is a serous membrane that covers the lung
- Consists of
 - 1. Mesothelium** (thin simple squamous epithelium that secretes pleural fluid)
 - 2. Submesothelium** (thin fibroelastic connective tissue)

The Nose

Anterior, lateral walls, and roof and floor of nasal cavity



The Conchae (Turbinates)

- By increasing surface area, increase the efficiency with which the inspired air is warmed.
- also increase the efficiency of filtration of inspired air through the process of turbulent precipitation.
- The air stream is broken into eddies by the turbinates.
- Particulate matter suspended in the air stream is thrown out of the stream and adheres to the mucus-covered wall of the nasal cavity.
- Particles trapped in this layer of mucus are transported to the pharynx by means of coordinated sweeping movements of cilia and are then swallowed

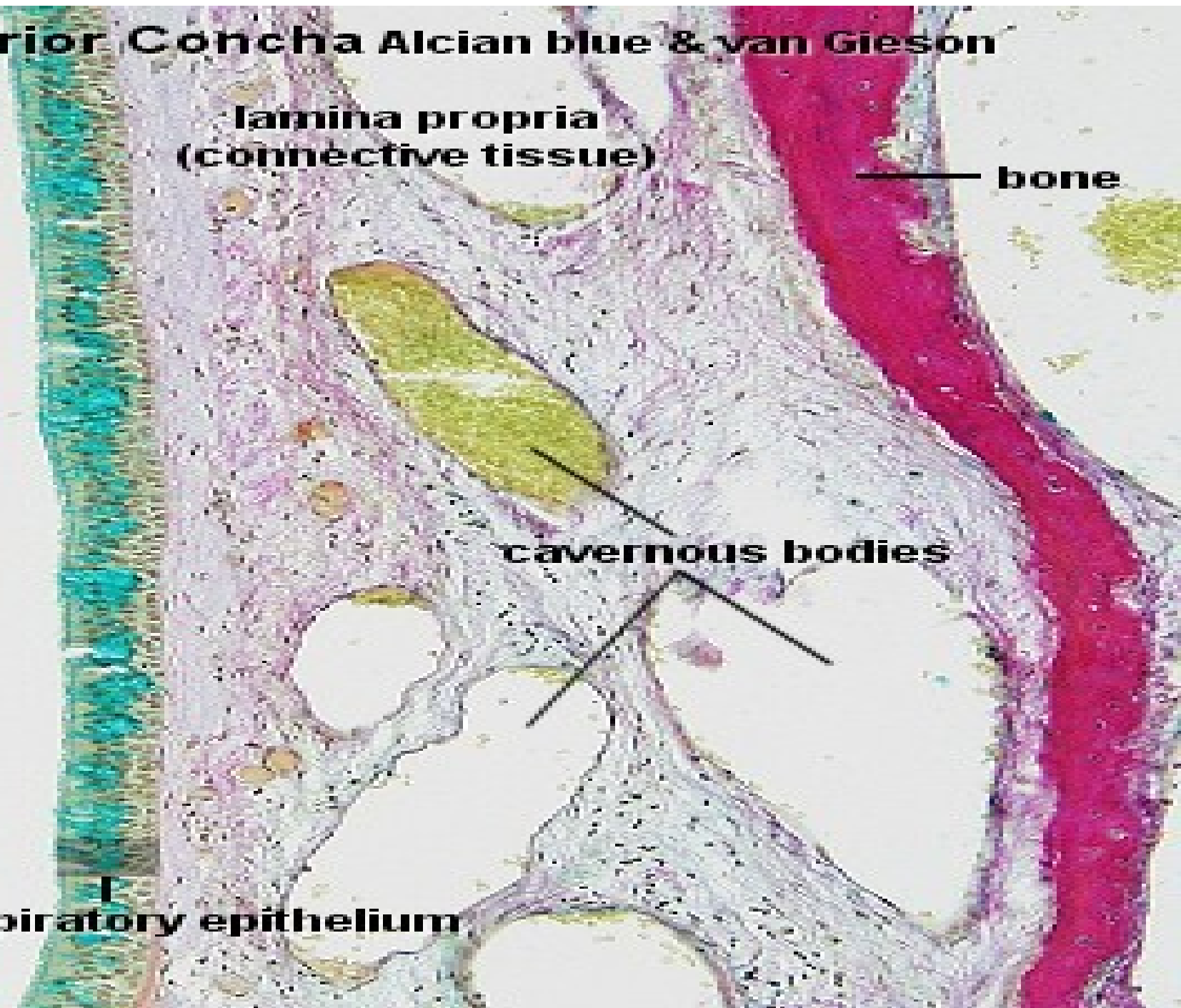
Inferior Concha Alcian blue & van Gieson

lamina propria
(connective tissue)

bone

cavernous bodies

respiratory epithelium



Inferior Concha Alcian blue & van Gieson

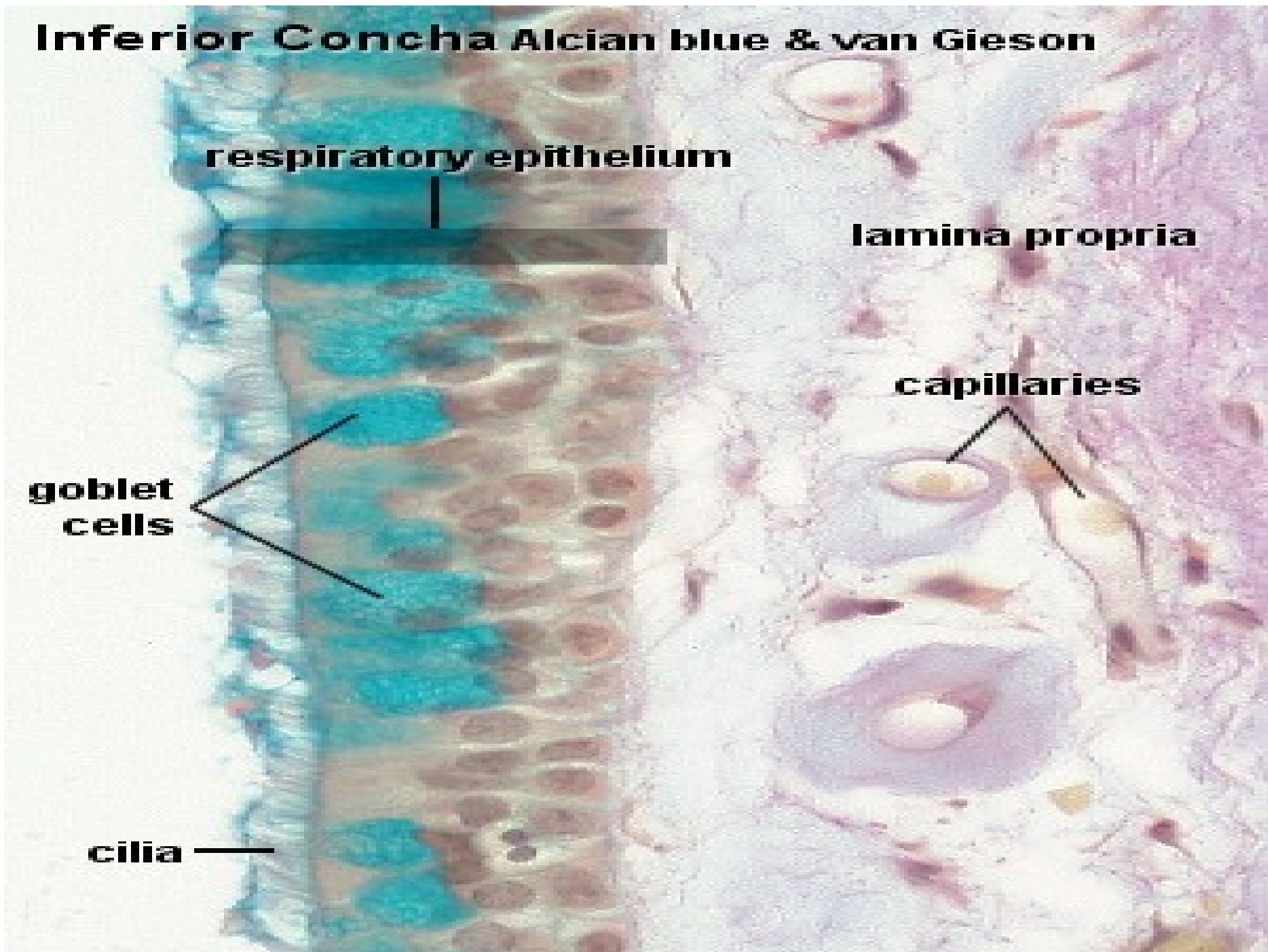
respiratory epithelium

lamina propria

capillaries

goblet
cells

cilia



3- Paranasal Sinuses:

- are bilateral air-filled cavities in the **frontal, maxillary, ethmoid, and sphenoid** bones of the skull.
- communicate with the nasal cavities through small openings, & mucus produced in the sinuses is drained into the nasal cavities by activity of ciliated lining epithelium.
- lined with a thinner respiratory mucosa with a thinner **respiratory epithelium** with **fewer goblet cells** and a thin **lamina propria**.
- This lamina propria is continuous with the underlying periosteum and contains only a few small mucous glands.
- **Sinusitis** is an inflammatory process of the sinuses that may persist for long periods of time, mainly because of obstruction of drainage orifices.
- **Chronic sinusitis and bronchitis are components of immotile cilia syndrome, characterized by defective ciliary action.**

The Olfactory Mucosa (Receptor Organ of Smell)

- It is the mucous membrane lining the olfactory areas of the nasal cavity, notably the roof of the nasal cavity, either sides of the superior part of nasal septum, & the superior nasal concha.
- In humans, it is about 10 cm² area & thicker than the respiratory mucosa but lacks the Goblet cells.
- Consists of **olfactory epithelium [neuroepithelium]** and glandular **lamina propria**.
- The lamina propria is highly vascular (with many venous plexuses) and consists of **serous glands (Bowman's glands or olfactory glands)** and **unmyelinated nerve fibers** (axons of olfactory cells).
- In living tissue, this mucosa is distinguished by its slight yellowish brown color caused by pigment in the olfactory epithelium and the associated olfactory glands.

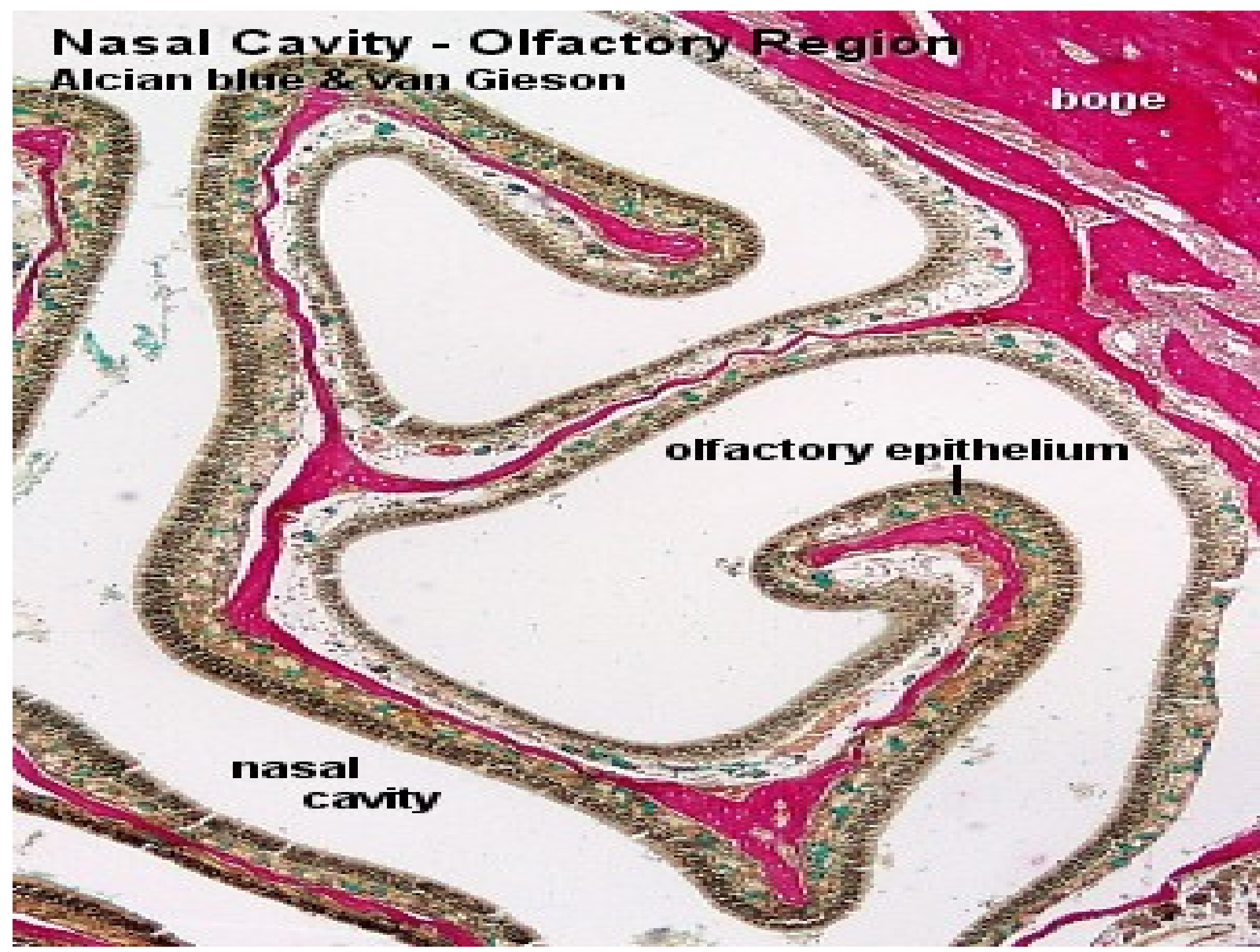
Nasal Cavity - Olfactory Region

Alcian blue & van Gieson

bone

olfactory epithelium

nasal
cavity



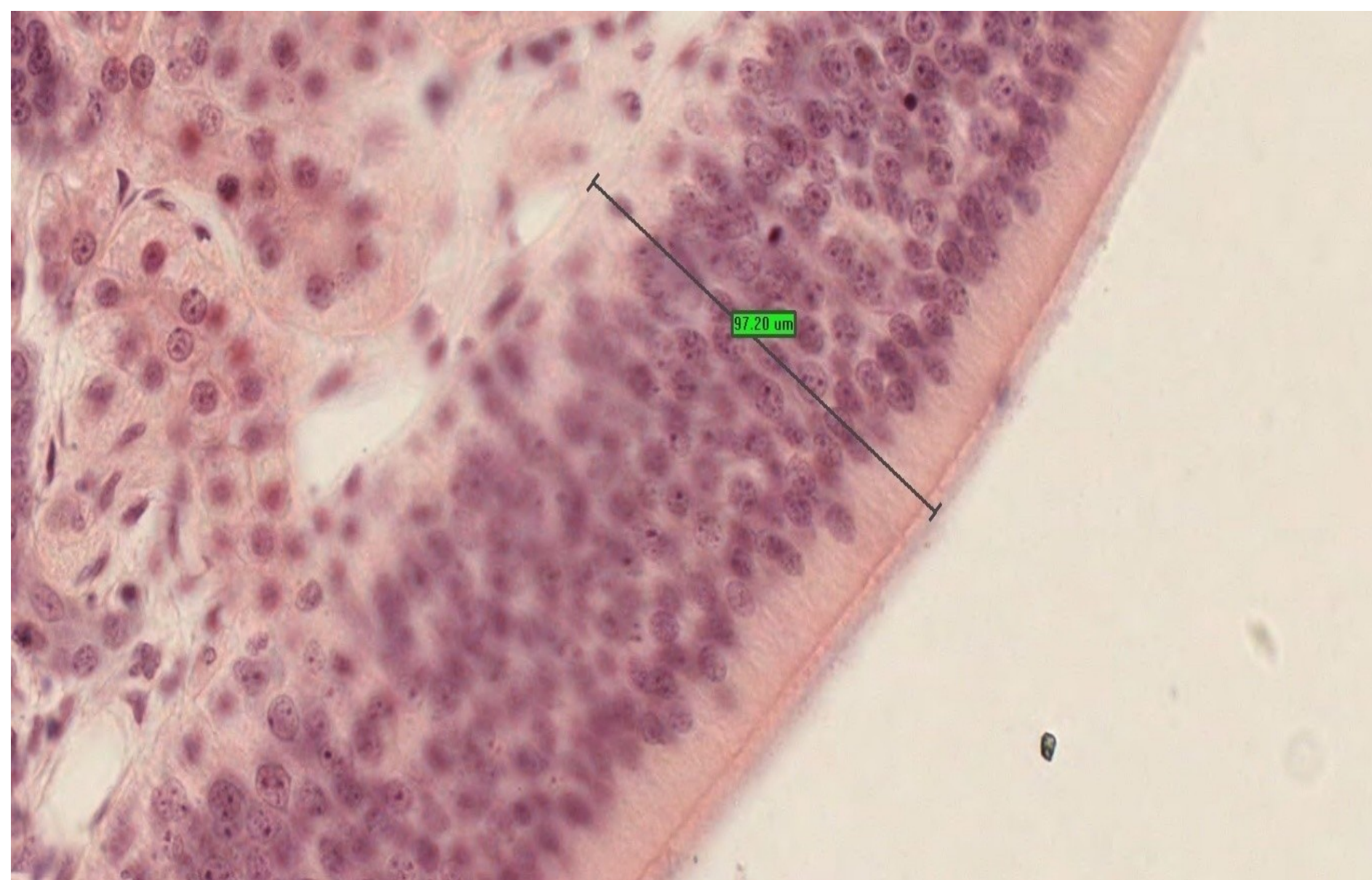
The Lamina propria of Olfactory mucosa

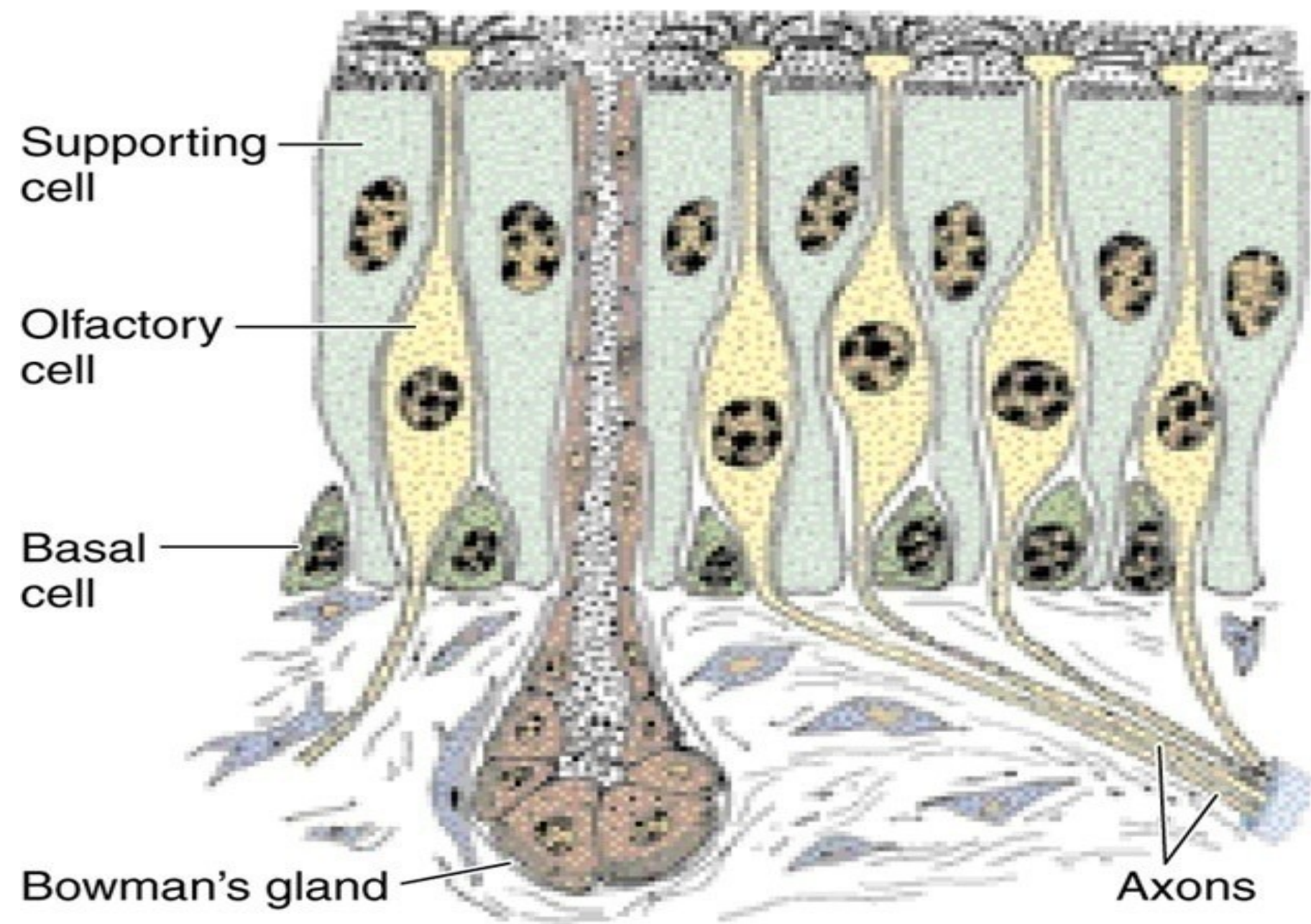


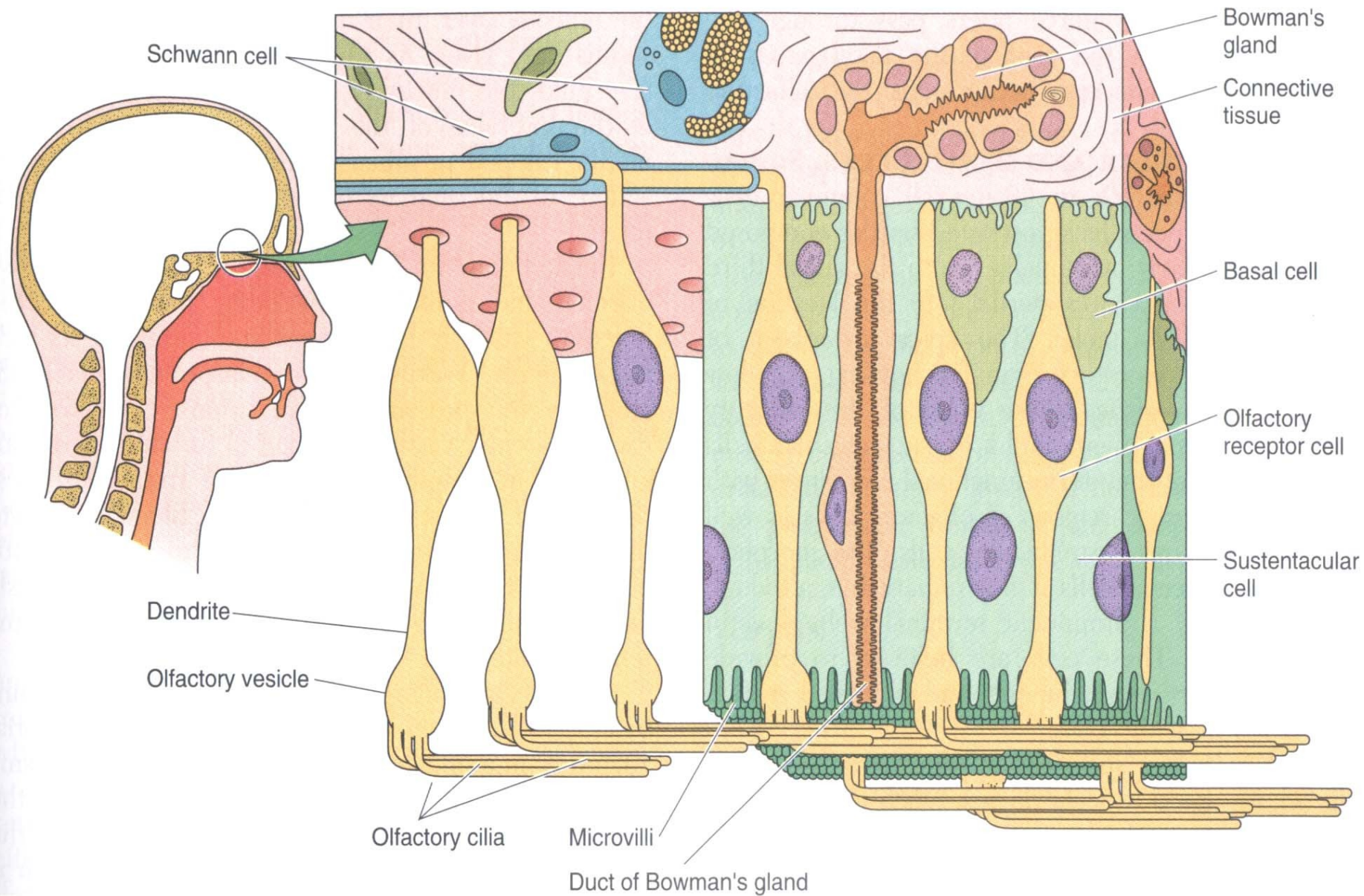
The Olfactory Epithelium (Neuroepithelium)

- is tall (up to 100 μ m) pseudostratified columnar ciliated epithelium without goblet cells, and consists of four types of cells:
 - 1. Olfactory cells (bipolar nerve cells)- sensory cells**
 - 2. Sustentacular (supporting) cells**
 - 3. Basal cells- stem cells,**
 - 4. Brush cells: recently described columnar cells specialized for transduction of general sensation.**

The Olfactory Epithelium





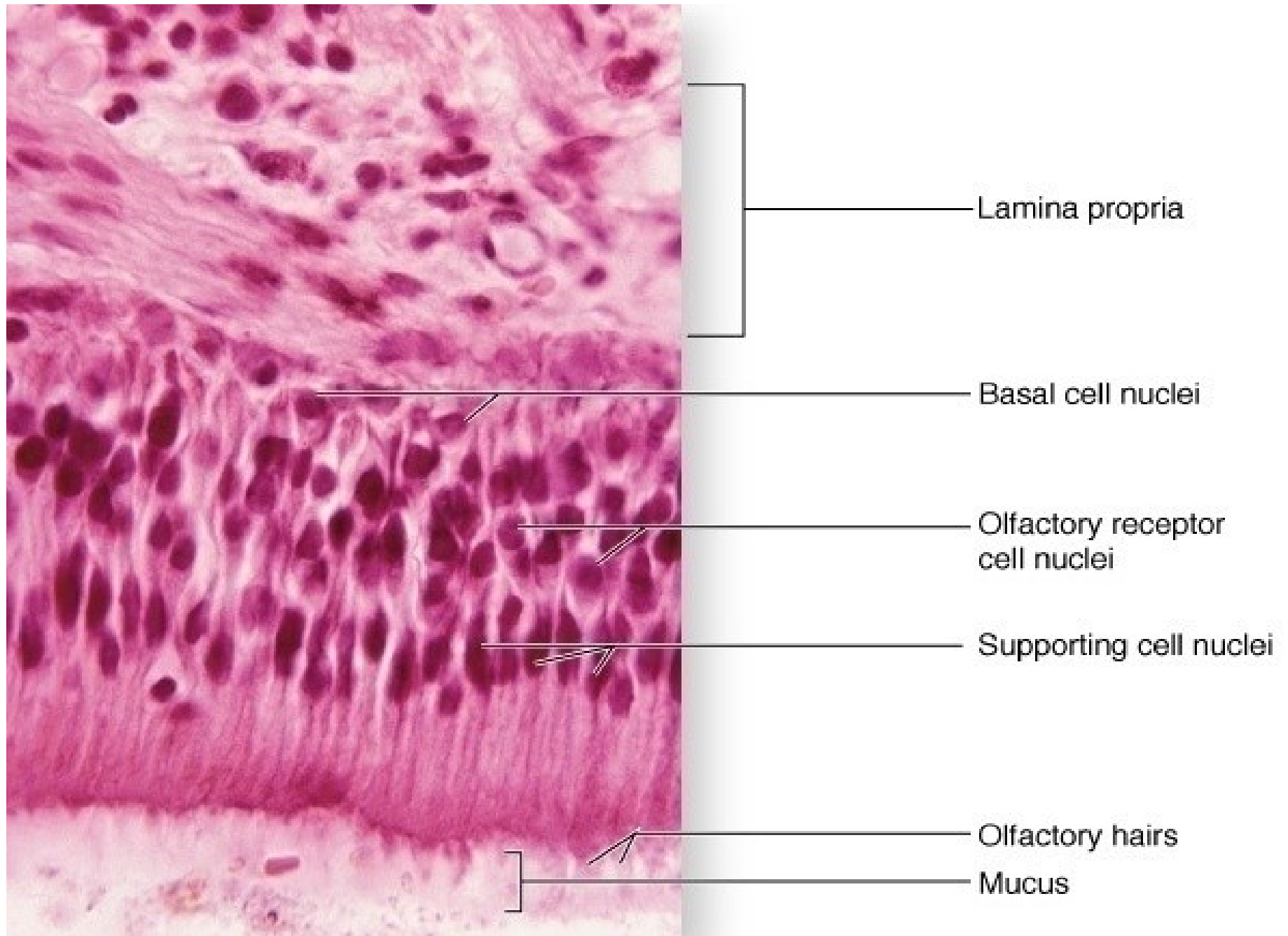


Schematic diagram of the olfactory epithelium showing basal, olfactory, and sustentacular cells.

2. Sustentacular (*Supporting*) cells:

- are columnar cells with narrower tapering bases, and broad cylindrical apices that carry microvilli, arising from a prominent terminal web.
- are the most numerous cells in the olfactory epithelium.
- possess nuclei more apically and superficially located than those of the other cell types.
- on their free apical surface have numerous microvilli submerged in a fluid layer, and abundant mitochondria.
- Numerous profiles of smooth endoplasmic reticulum (sER) and, to a more limited extent, RER) are observed in their cytoplasm.
- synthesize and secrete odorant-binding proteins
- also possess lipofuscin granules, light yellowish brown pigments that impart the olfactory mucosa yellowish coloration.
- Adhering junctions are present between these cells and the olfactory receptor cells, but gap and tight junctions are absent.
- The supporting cells function in a manner comparable to that of neuroglial cells, providing both metabolic support (secretion of OBP) and physical support to the olfactory receptor cells.

The Olfactory Mucosa



Bowman's Glands (Olfactory Glands)

- found only in the olfactory region of nasal cavity.
- are pure serous (**branched tubulo-alveolar**) glands located in lamina propria of olfactory mucosa.
- Lipofuscin granules are prevalent in the glandular cells, and in combination with the lipofuscin granules in the supporting cells of the olfactory epithelium, they give the mucosa its natural slightly yellow-brown coloration.
- produce thin watery secretion which released on to the olfactory epithelial surface via narrow ducts.
- odorous substances dissolved in this watery secretion are detected by the olfactory hairs.
- also wash the cilia and flushes the surface, preparing the receptors to receive new odorous stimuli.

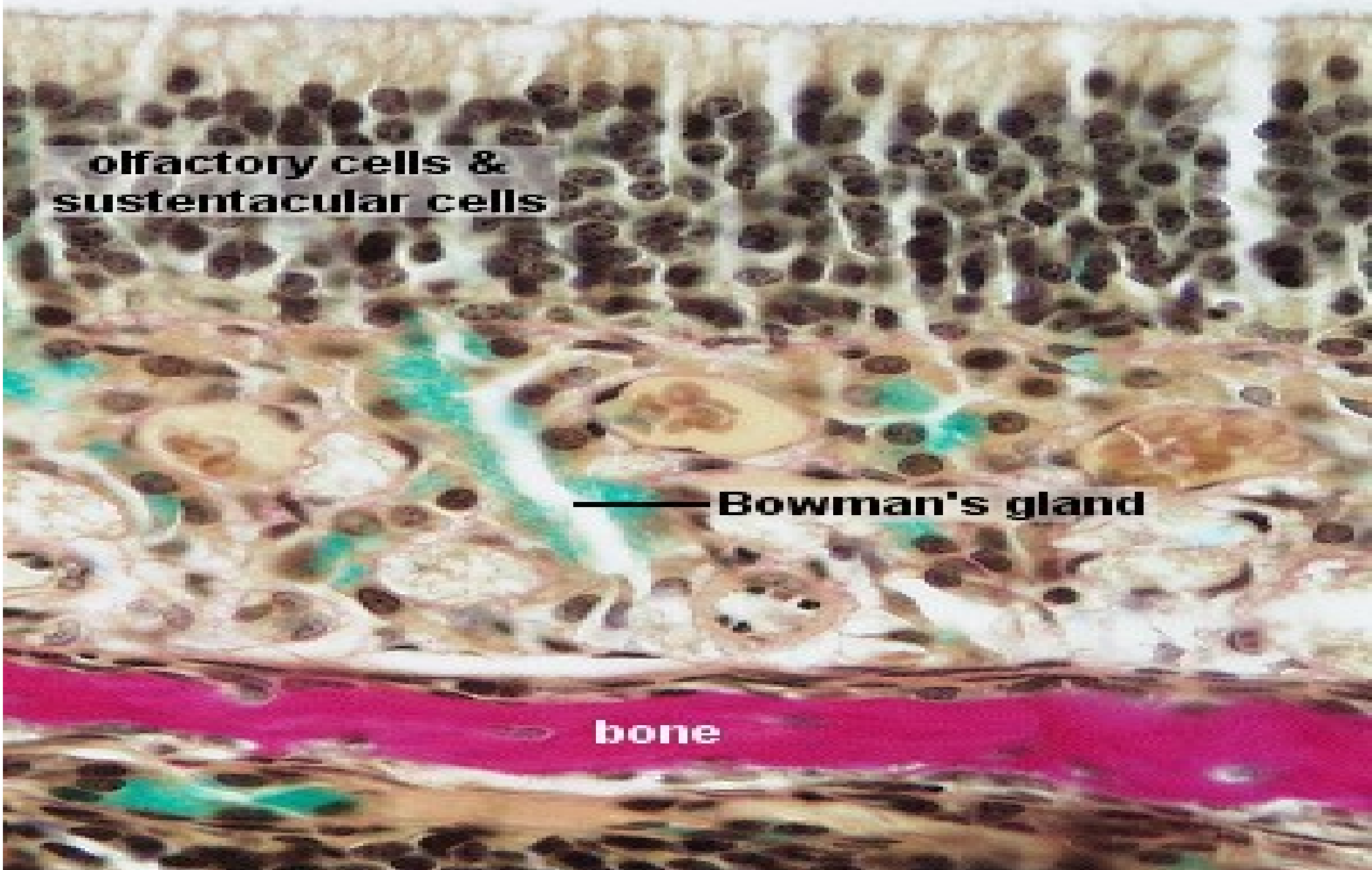
Nasal Cavity - Olfactory Region

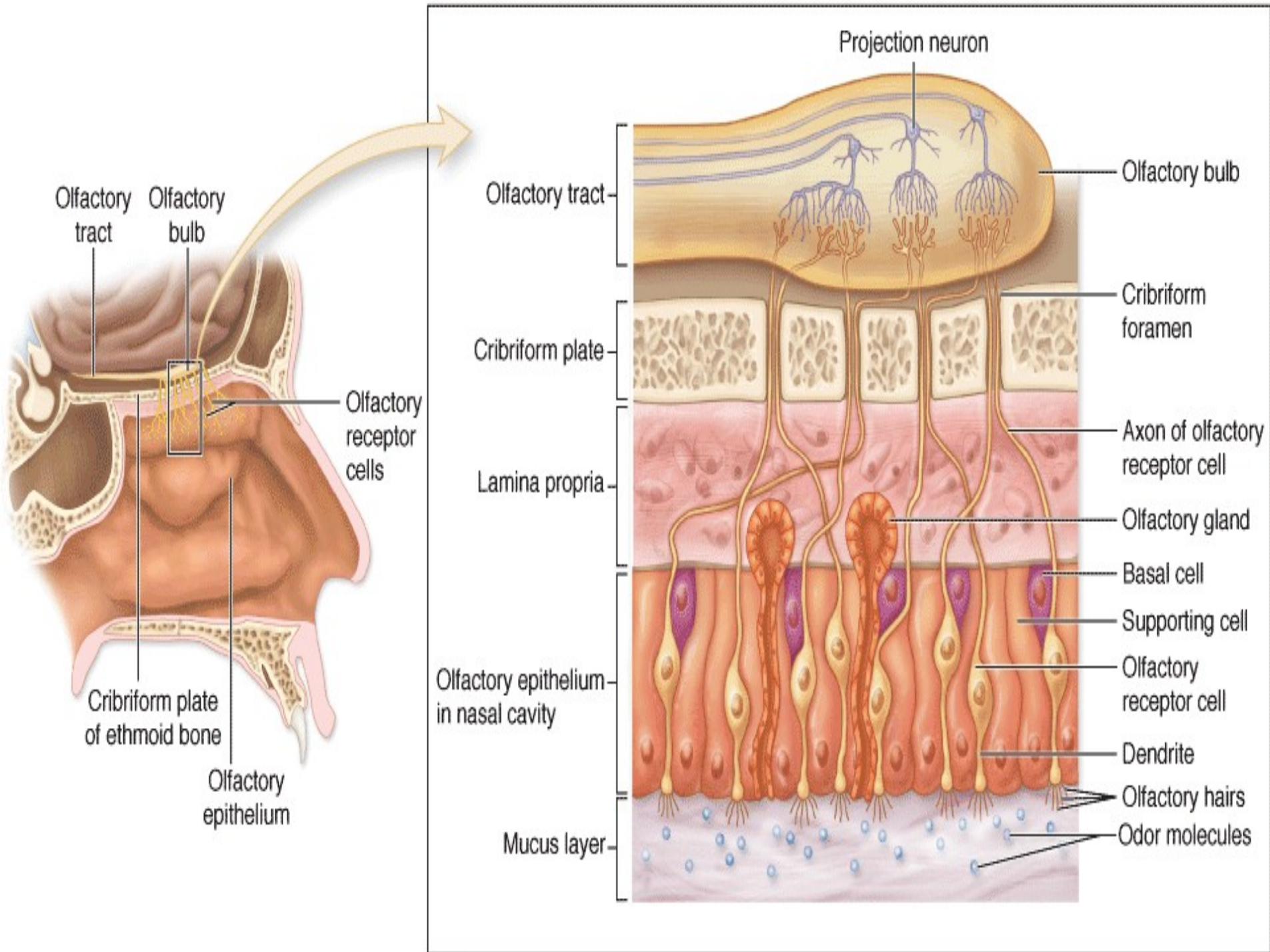
Alcian blue & van Gieson

olfactory cells &
sustentacular cells

Bowman's gland

bone





A. Nasopharynx

- lies posterior to the nasal cavity
- Contains tubal and pharyngeal tonsils
- Receives the openings the **auditory tubes (Eustachian tubes also called pharyngotympanic tubes)** which connect the nasopharynx to the middle ear cavities, an arrangement which permits equilibration of air pressure in the middle ear with that of the external environment.
- **The Eustachian tubes**
 - are lined by typical respiratory epithelium and have elastic cartilage support in their walls.
 - Near its opening called tubal elevation contains **tubal tonsils**.

– **The wall of pharynx comprises three principal layers:**

1. Mucosa: consists of

a. Lining Epithelium: ciliated pseudostratified columnar with goblet cells in nasopharynx, nonkeratinized stratified squamous epithelium in oropharynx and laryngopharynx.

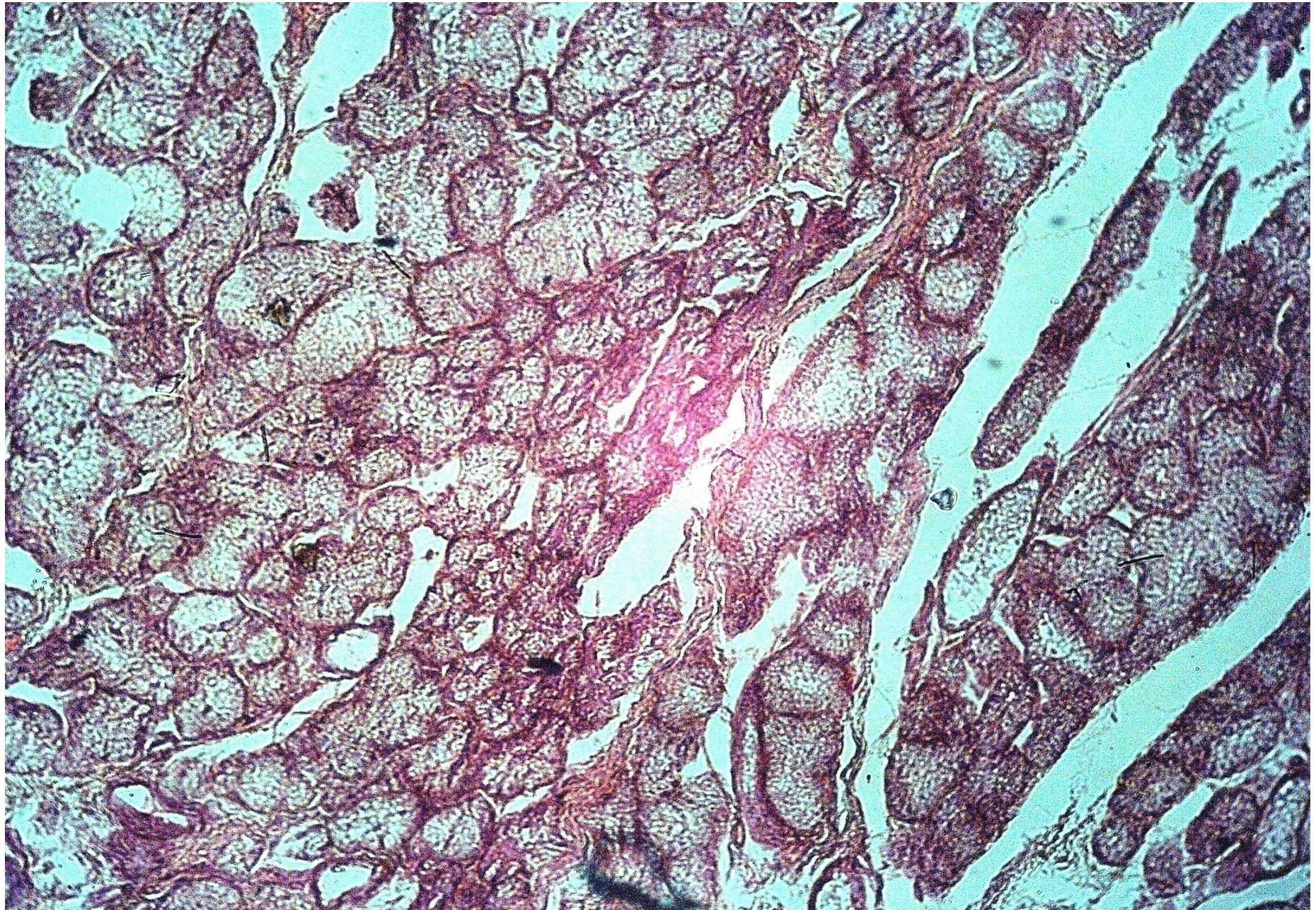
b. Extensive Lamina Propria

– loose fibroelastic CT containing lymphoid organs (Tonsils) followed by a well-developed layer of elastic fibers called **elastic lamina** that is continuous with the muscularis mucosa of the esophagus.

2. The Muscularis: consists of skeletal muscles of the pharyngeal constrictors externally and straight longitudinal muscles internally & abundant numerous minor salivary glands of mucous type interspersed between the muscles.

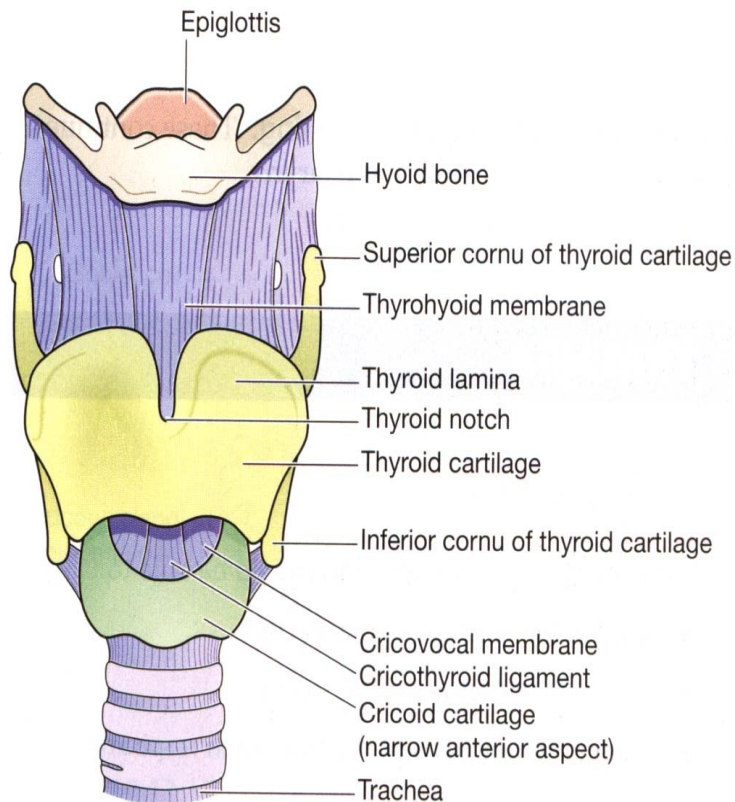
3. Adventitia: fibroelastic CT, covering the muscle layer.

Numerous mucous glands in the Muscular layer of pharynx



THE LARYNX (THE VOICE BOX)

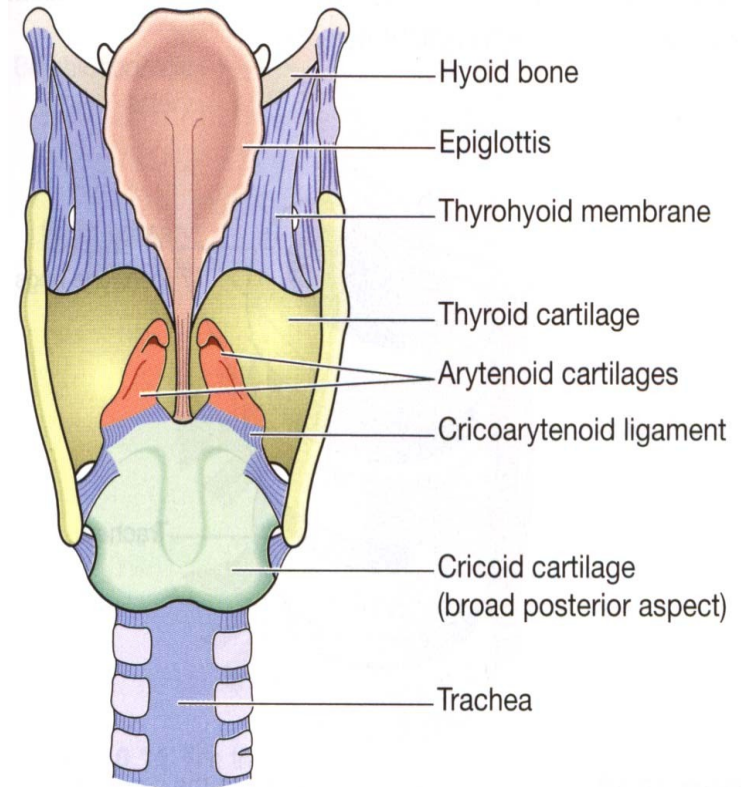
- Below the oropharynx the respiratory & digestive tracts are separate.
- The larynx is an irregular, short tube that connects the oropharynx with the trachea.
- The epiglottis** (which contains elastic cartilage core) closes the opening into the larynx called **glottis** during swallowing.
- The larynx is lined with respiratory epithelium except the vocal cords, which are covered with nonkeratinized stratified squamous



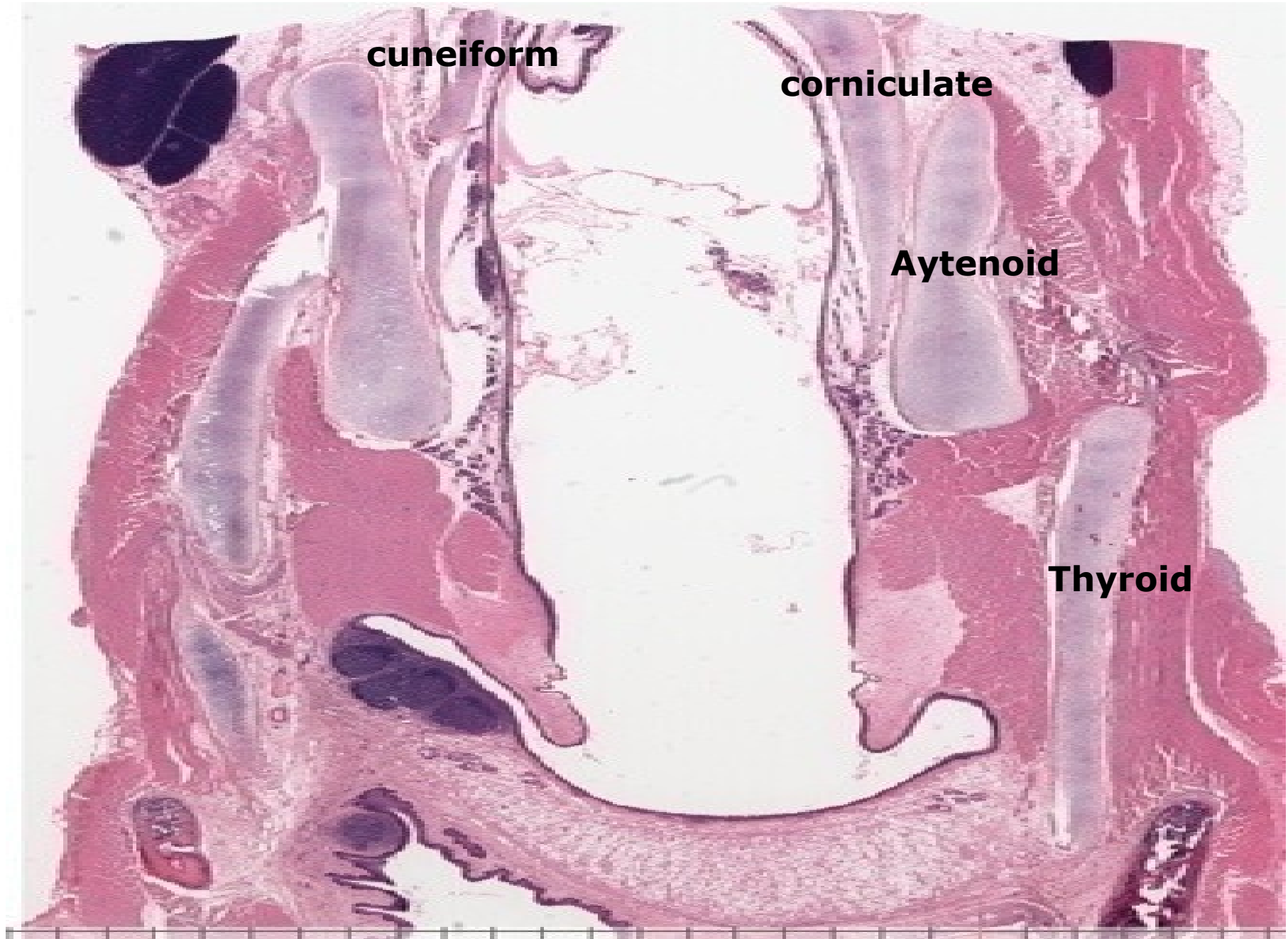
Anterior
View



Posterior
View



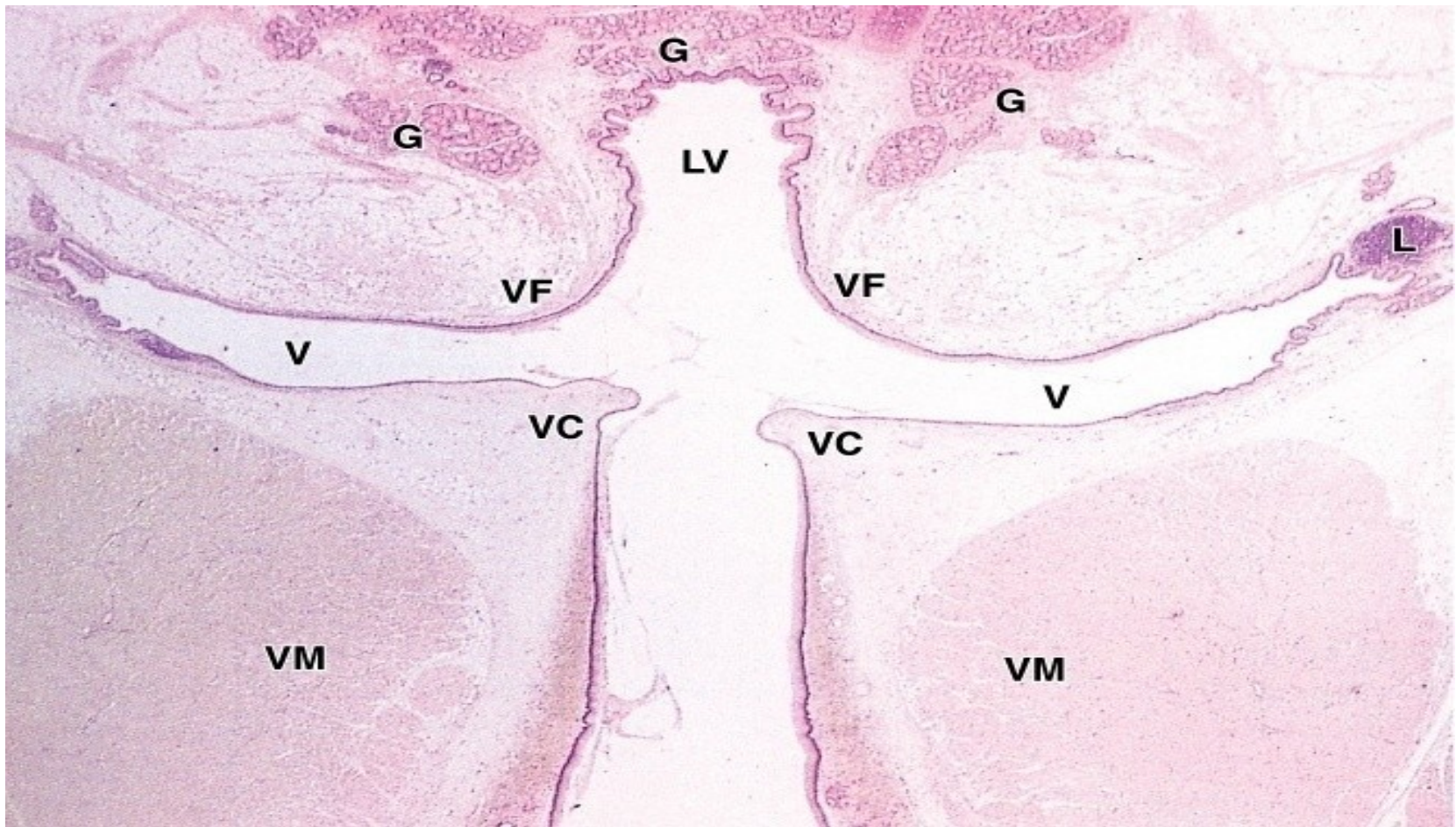
Coronal section of Larynx: see BALT, Laryngeal folds & cartilages



– Two pairs of mucosal folding are found below the epiglottis and extend through the cavity of larynx:

1- Upper paired Folds (vestibular folds or False vocal cords), covered by **respiratory epithelium**.

2- Lower paired Folds (Vocal folds or True vocal cords) covered by **stratified squamous epithelium**.



Larynx: The low-power micrograph shows the upper laryngeal vestibule (LV), which is surrounded by seromucous glands (G). The lateral walls of this region bulge as a pair of broad folds, the vestibular folds (VF). These also contain seromucous glands and areolar tissue with MALT, often with lymphoid nodules (L) and are largely covered by respiratory epithelium, with regions near the epiglottis having stratified squamous epithelium. Below each large vestibular fold is a narrow space or ventricle (V), below which is another pair of lateral folds, the vocal folds or cords (VC). These are covered by stratified squamous epithelium and project more sharply into the lumen, defining the rim of the opening into the larynx itself. Each contains a large striated vocalis muscle (VM) and nearer the surface a small ligament, which is cut transversely and therefore difficult to see here. X15. H & E

True Vocal Cords (Lower laryngeal folds)

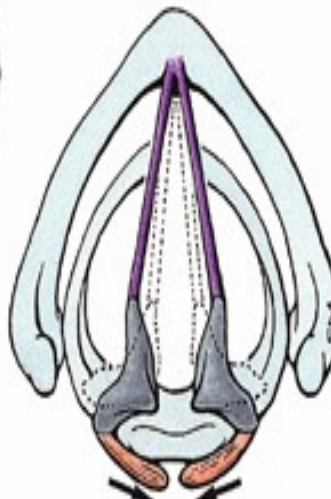
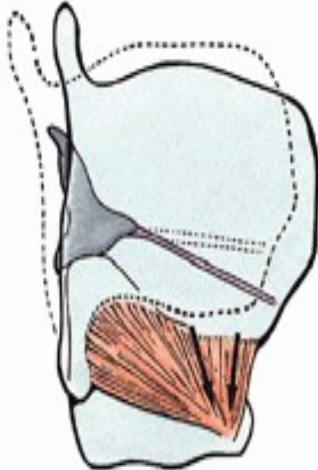
- consists of skeletal muscle (**the vocalis**), the **vocal ligament** (formed by a thick band of elastic fibers), and a covering **stratified squamous epithelium**.
- contraction of the laryngeal muscles change the length and tension of the vocal cords and the size and shape of the opening between the vocal cords (**rima glottidis**), which affects the pitch of the sounds caused by air passing through the larynx.
- inferior to the vocal cords, the lining epithelium changes to respiratory epithelium, which lines air passages down through the trachea and primary bronchi.
- Contain the the vocal ligaments under stratified squamous epithelium, vocalis muscle and the thyro-arytenoid muscle from medial to lateral.
 - The vocalis muscles produce minute adjustments of the vocal ligaments, selectively tensing and relaxing the anterior and posterior parts, respectively, of the vocal folds during animated speech and singing.

In vocalization

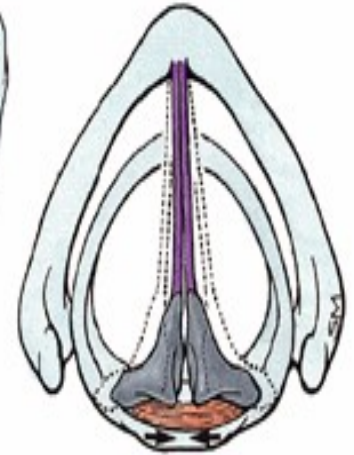
- The vocal cords are adducted to produce vocal sounds.



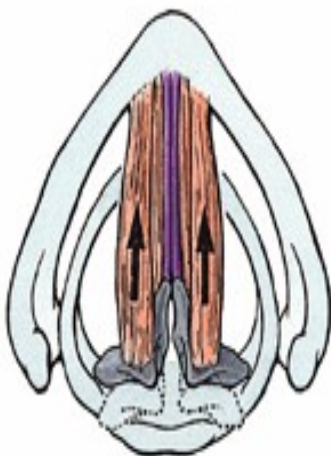
Lateral view
Cricothyroid



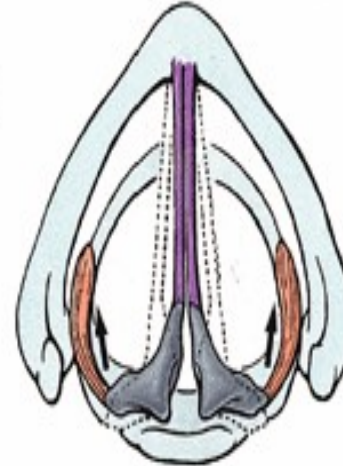
Superior view
Posterior cricoarytenoid



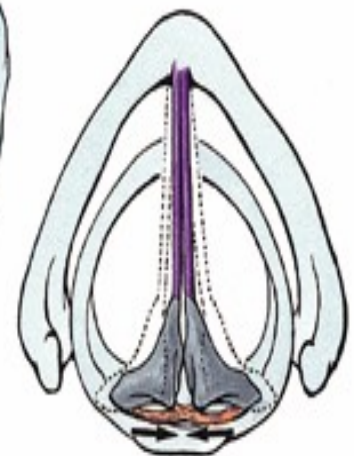
Superior view
Transverse arytenoid



Superior view
Thyroarytenoid



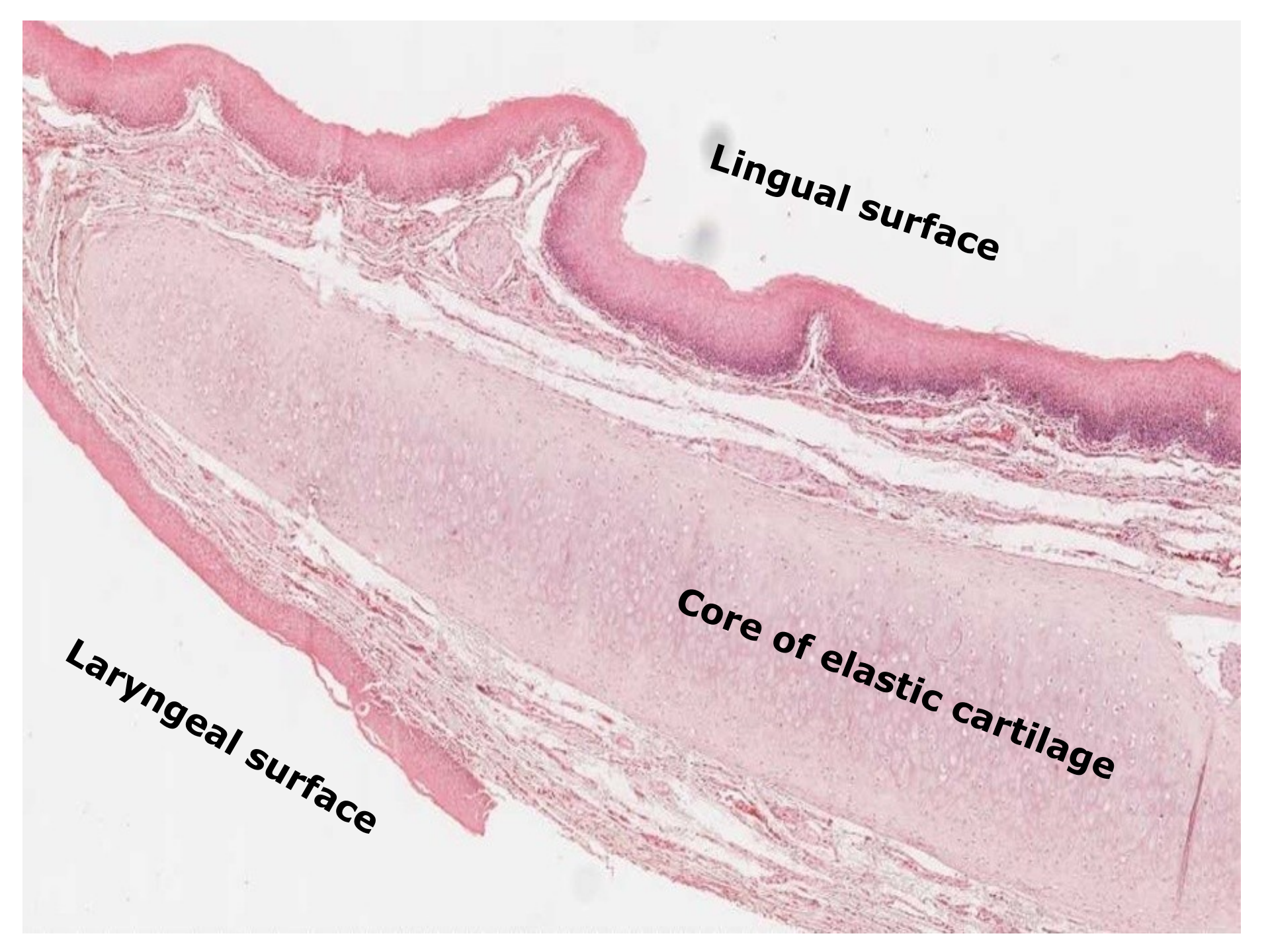
Superior view
Lateral cricoarytenoid



Superior view
Oblique arytenoid

EPIGLOTTIS

- Consists of a central core of elastic cartilage covered by mucosa on both its lingual (anterior) and laryngeal (posterior) surfaces.
- It has a tip which projects from the upper rim of the larynx extending upward into oropharynx behind tongue.
- The entire lingual surface and the apical portion of its laryngeal surface are covered with protective mucosa lined by non-keratinized stratified squamous epithelium.
 - At a variable point on its laryngeal surface, the epithelium undergoes a transition to the typical respiratory epithelium.
- Mixed mucous and serous glands are seen in the lamina propria beneath the epithelium on the laryngeal side.
- Its laryngeal/posterior surface may contain occasional **taste buds**.



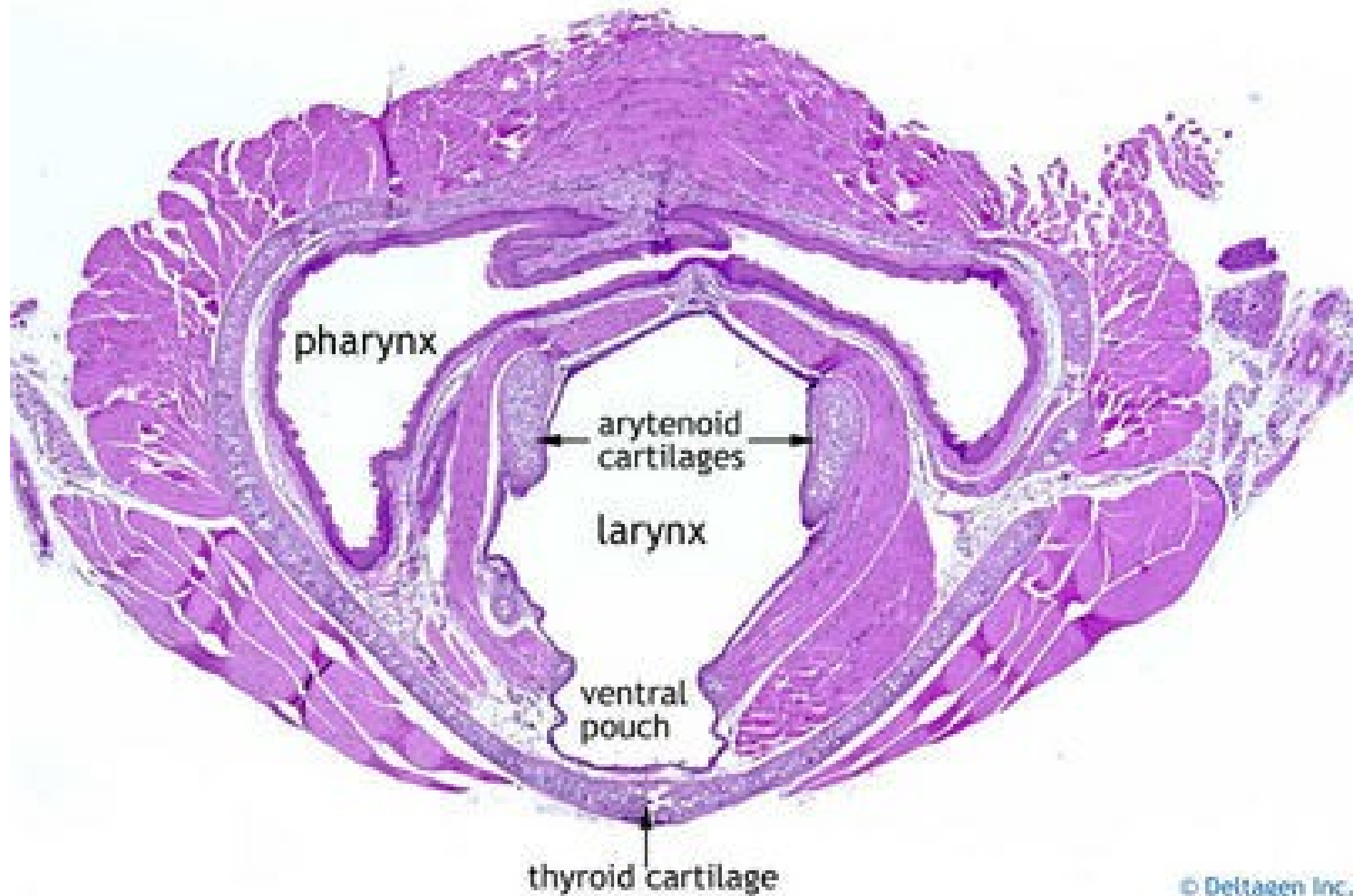
Lingual surface

This histological section shows the epiglottis, a structure that prevents food from entering the trachea. The lingual surface is the upper, convex side, while the laryngeal surface is the lower, concave side. The core of elastic cartilage provides the structural support and flexibility necessary for its function.

Laryngeal surface

Core of elastic cartilage

NORMAL LARYNX, LABELED

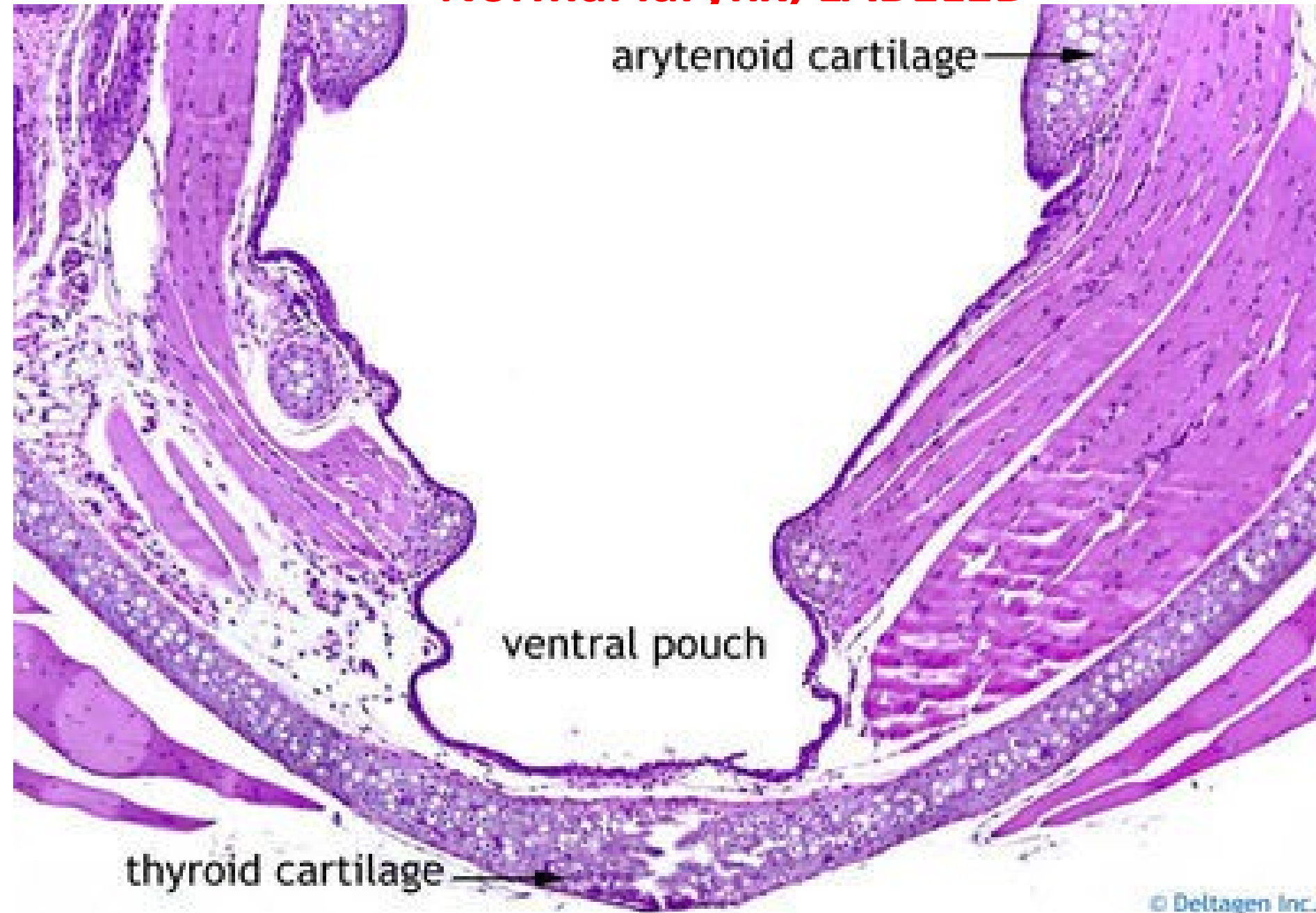


Normal larynx, LABELED

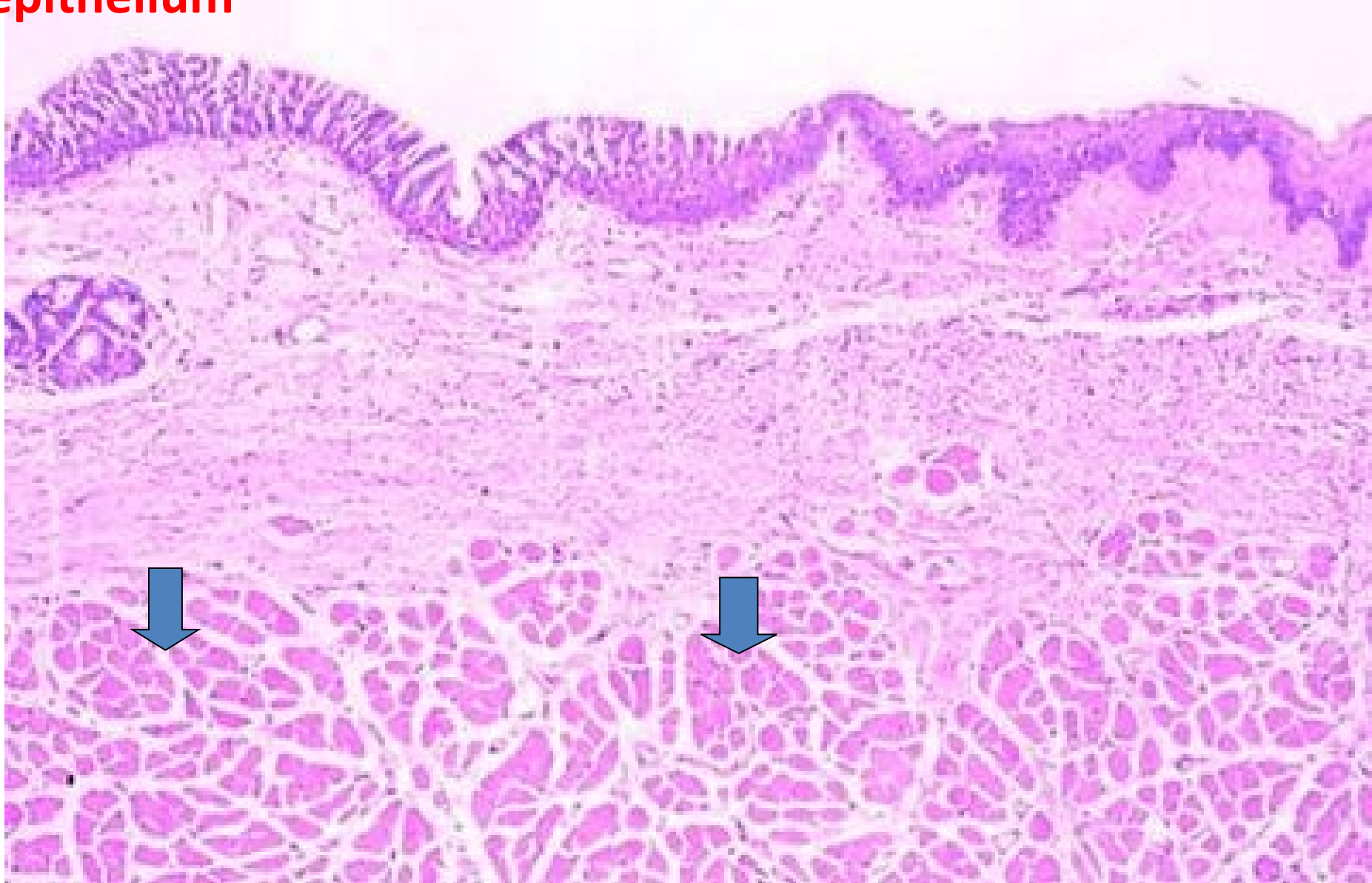
arytenoid cartilage →

ventral pouch

thyroid cartilage →

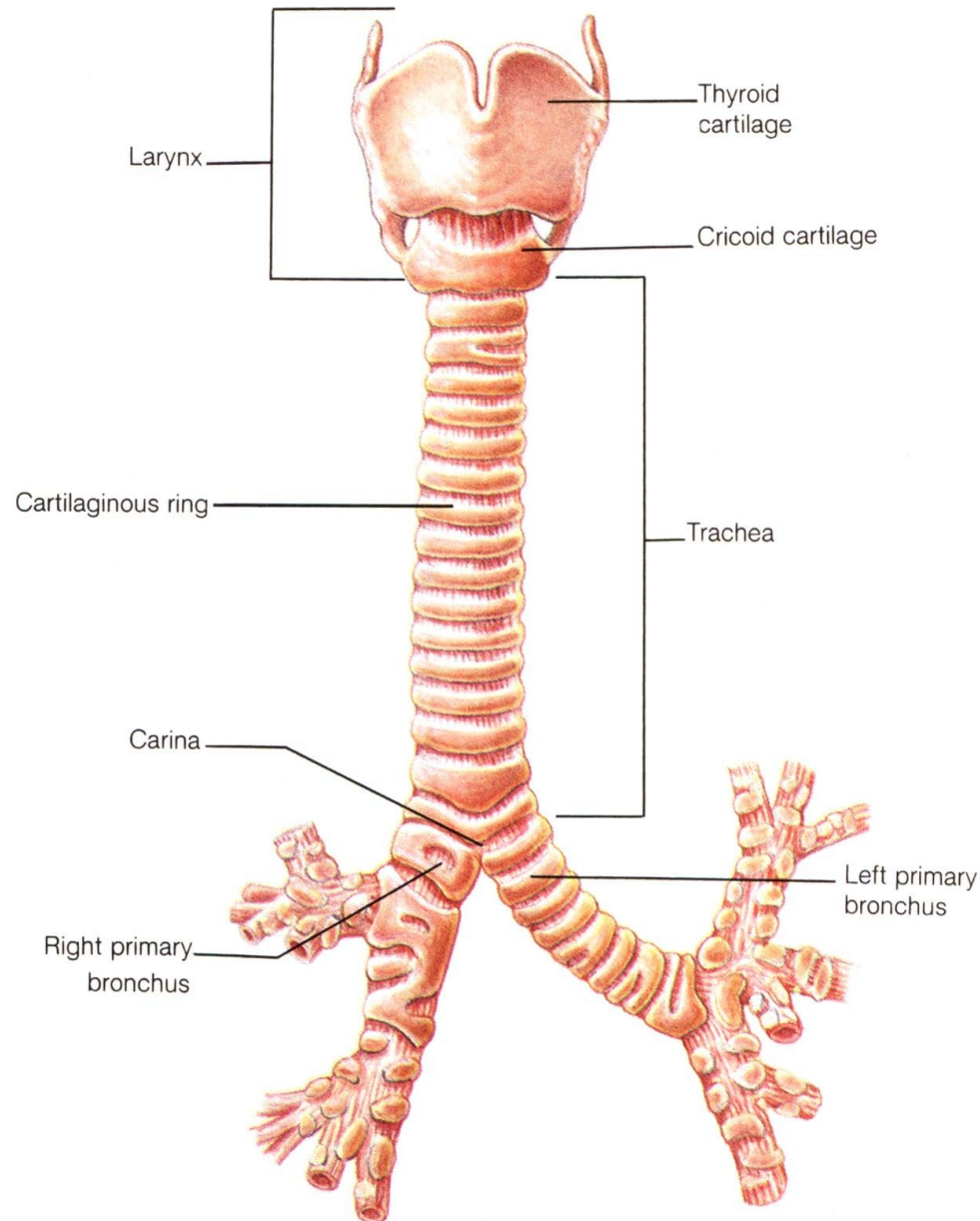


Larynx, a vocal cord at the junction of nonkeratinized stratified squamous & pseudostratified columnar respiratory epithelium

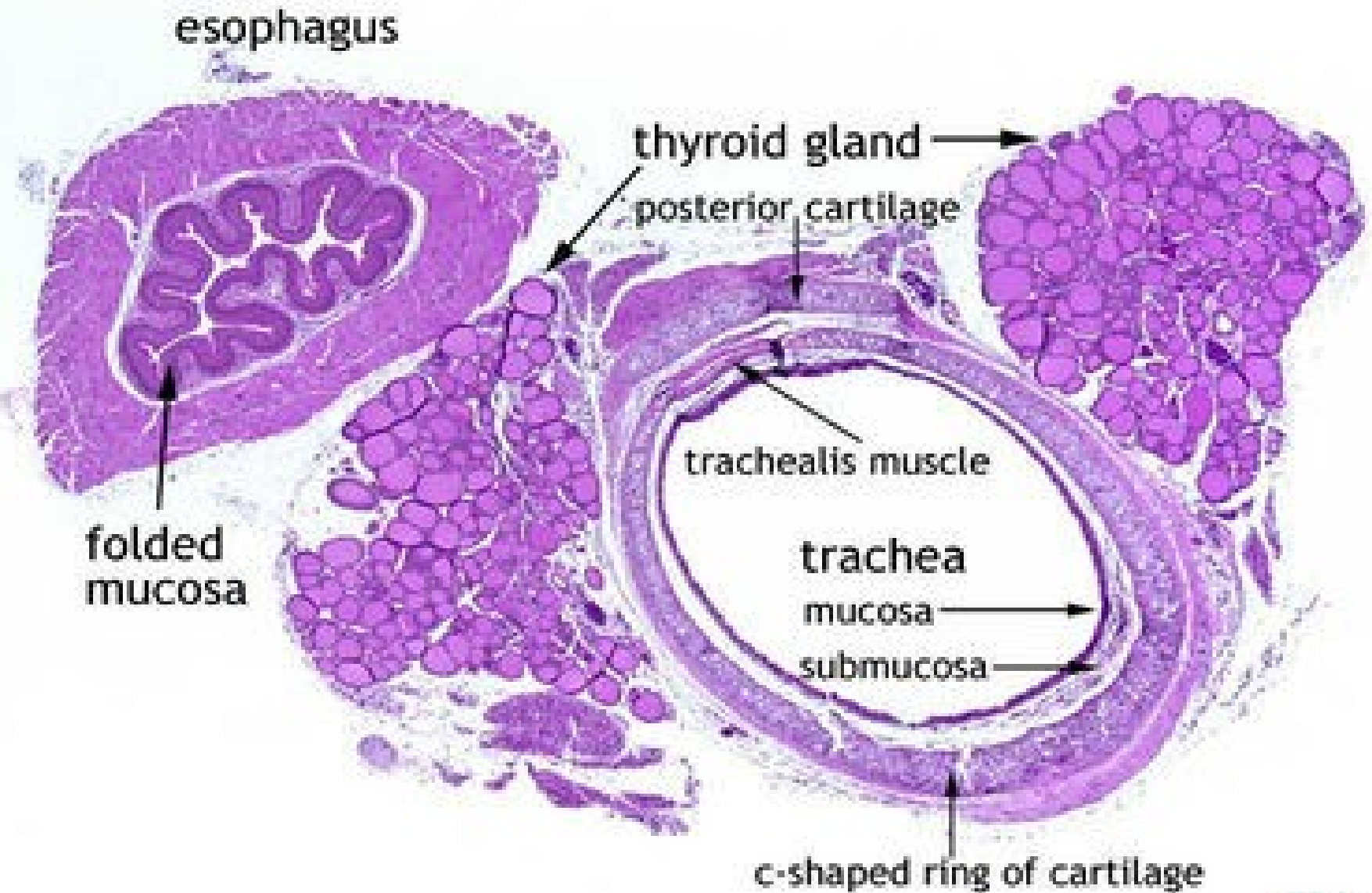


THE TRACHEA & MAIN BRONCHI

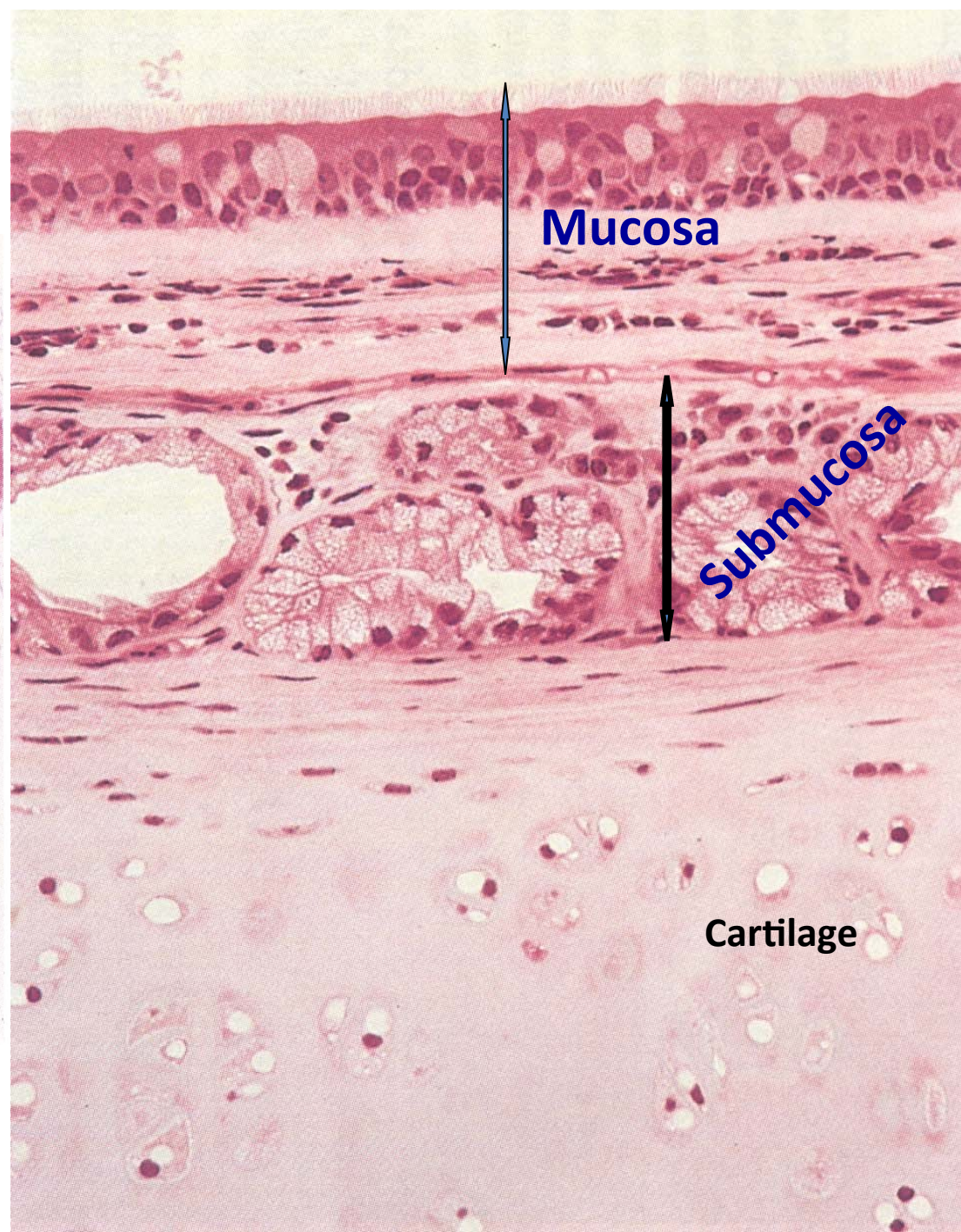
- The trachea is formed of 16-20 segments; each containing a U- or a C-shaped piece of hyaline cartilage joined at the free ends to one another by bands of smooth muscle called **trachealis** posteriorly.
- **The trachea and the two main (primary) bronchi are histologically similar.**



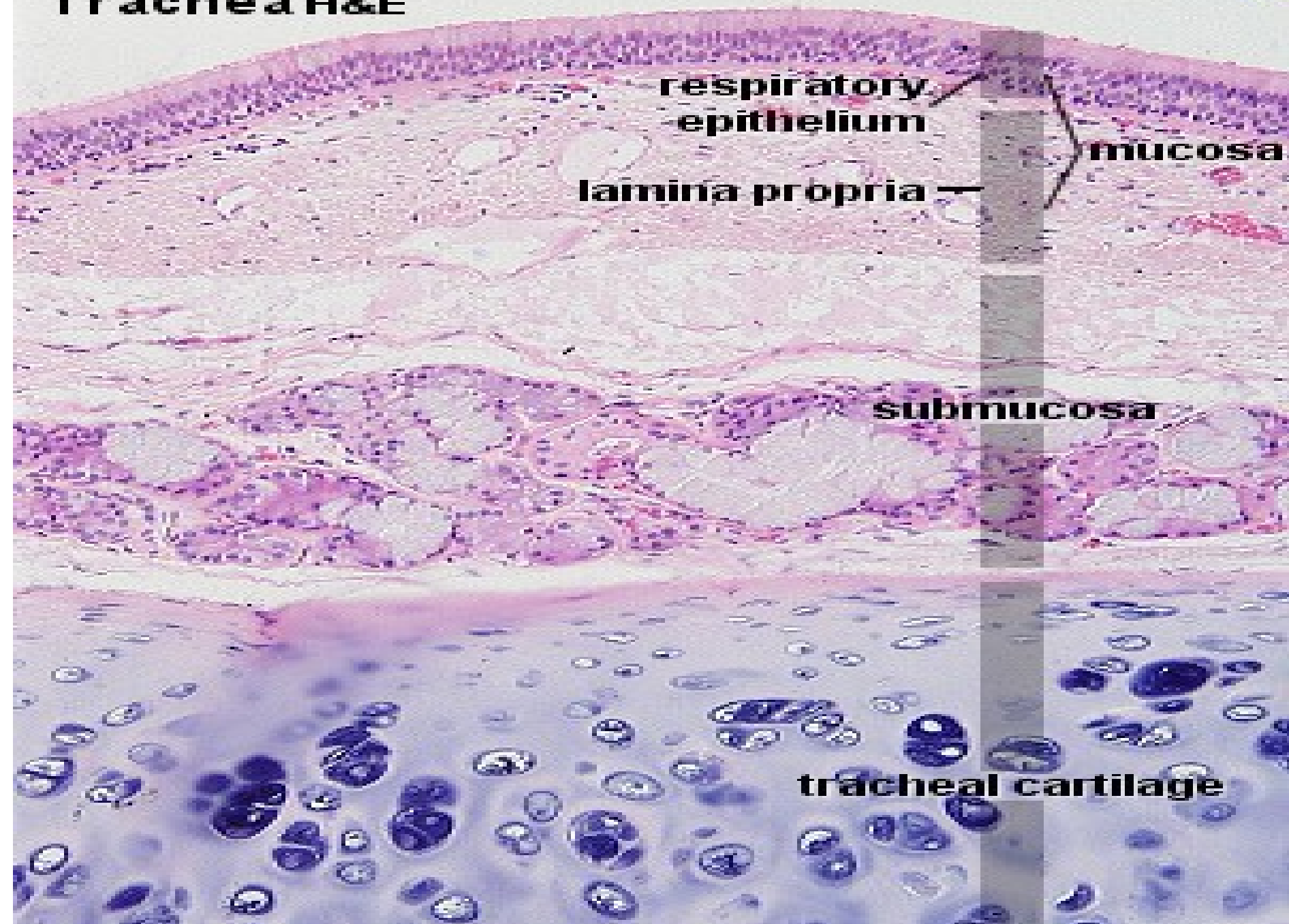
NORMAL TRACHEA, THYROID AND OESOPHAGUS



Trachea in T.S.



Trachea H&E



Types of cells in lining epithelium of trachea and bronchi:

- Three types of cells are seen by light microscope:

- Ciliated columnar cells.
- Goblet cells which produce mucous secretion.
- Basal cells: these are undifferentiated cells which can divide by mitosis and can give other types of cells.

- Two types of cells seen by electron microscope:

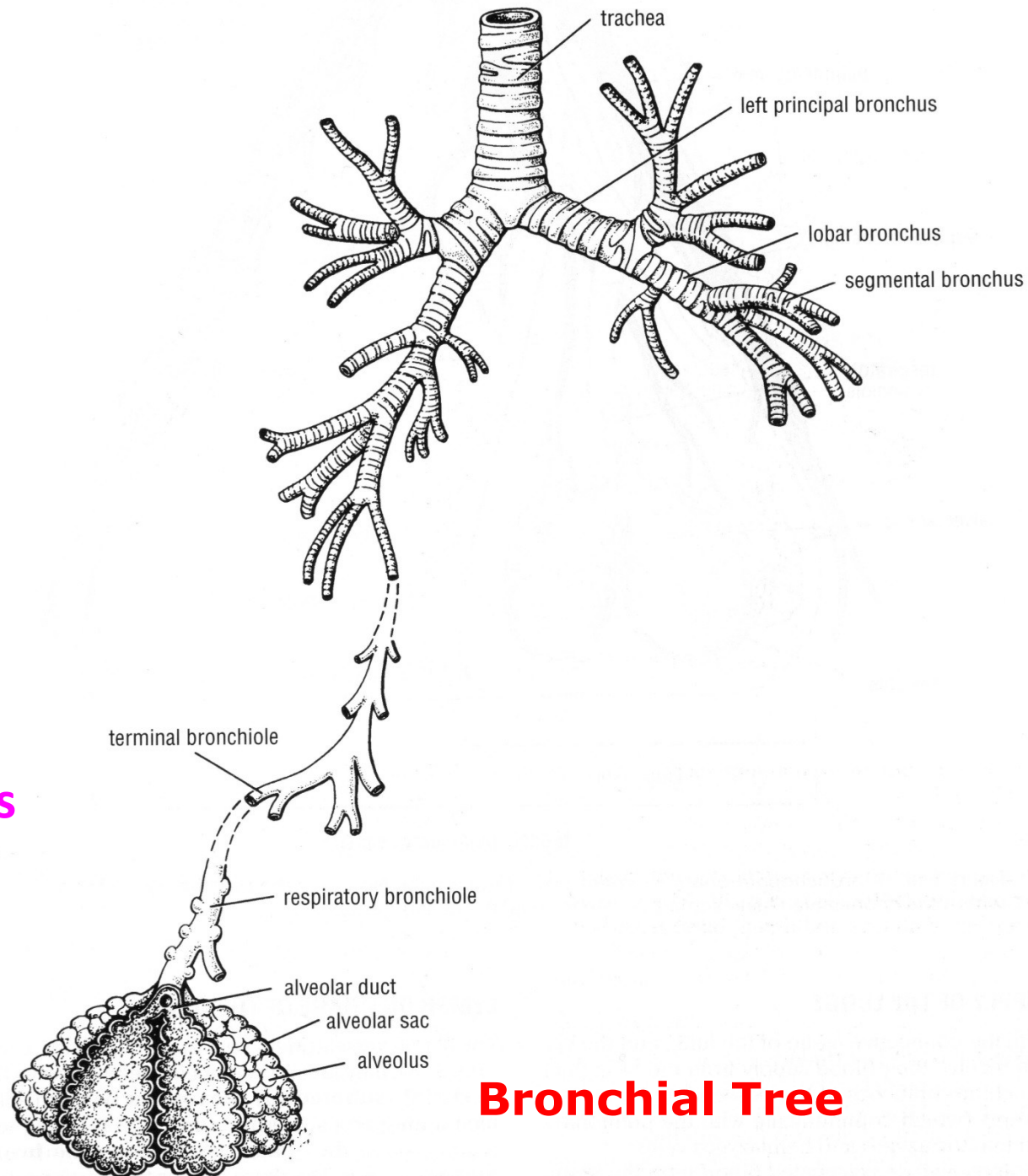
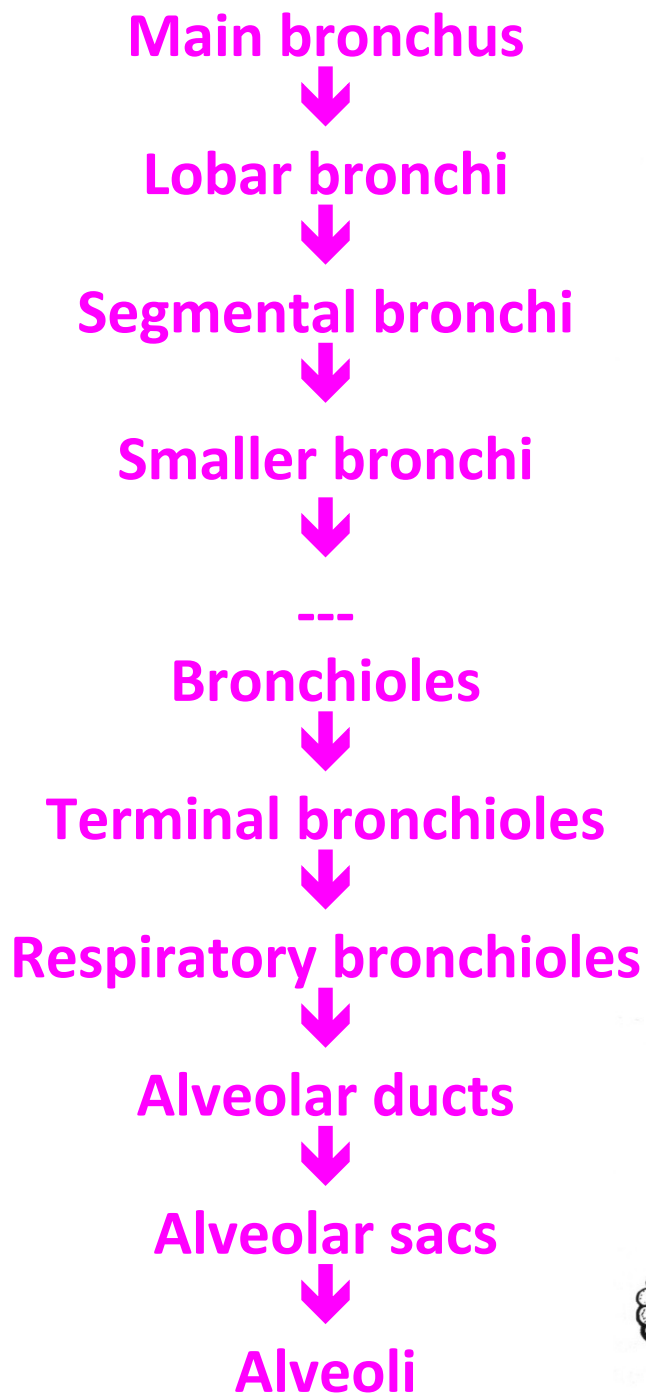
- Brush cells which contain glycogen granules and carry microvilli. They may be either immature columnar cells or degranuled goblet cells.
- Granule cells which are of 2 types according to function of granules:
 - Catecholamine secreting cells (neurosecretory cells).
 - Protein hormone secreting cells (enteroendocrine like cells). They secrete amines and amine precursors.

TRACHEAL EPITHELIUM, CILIATED & NON CILIATED CELLS, NOTE THE DIFFERENCE IN SIZE BETWEEN CILIA & MICROVILLI ON THE SURFACE OF NON CILIATED CELLS,SEM



Bronchial Tree & Lung

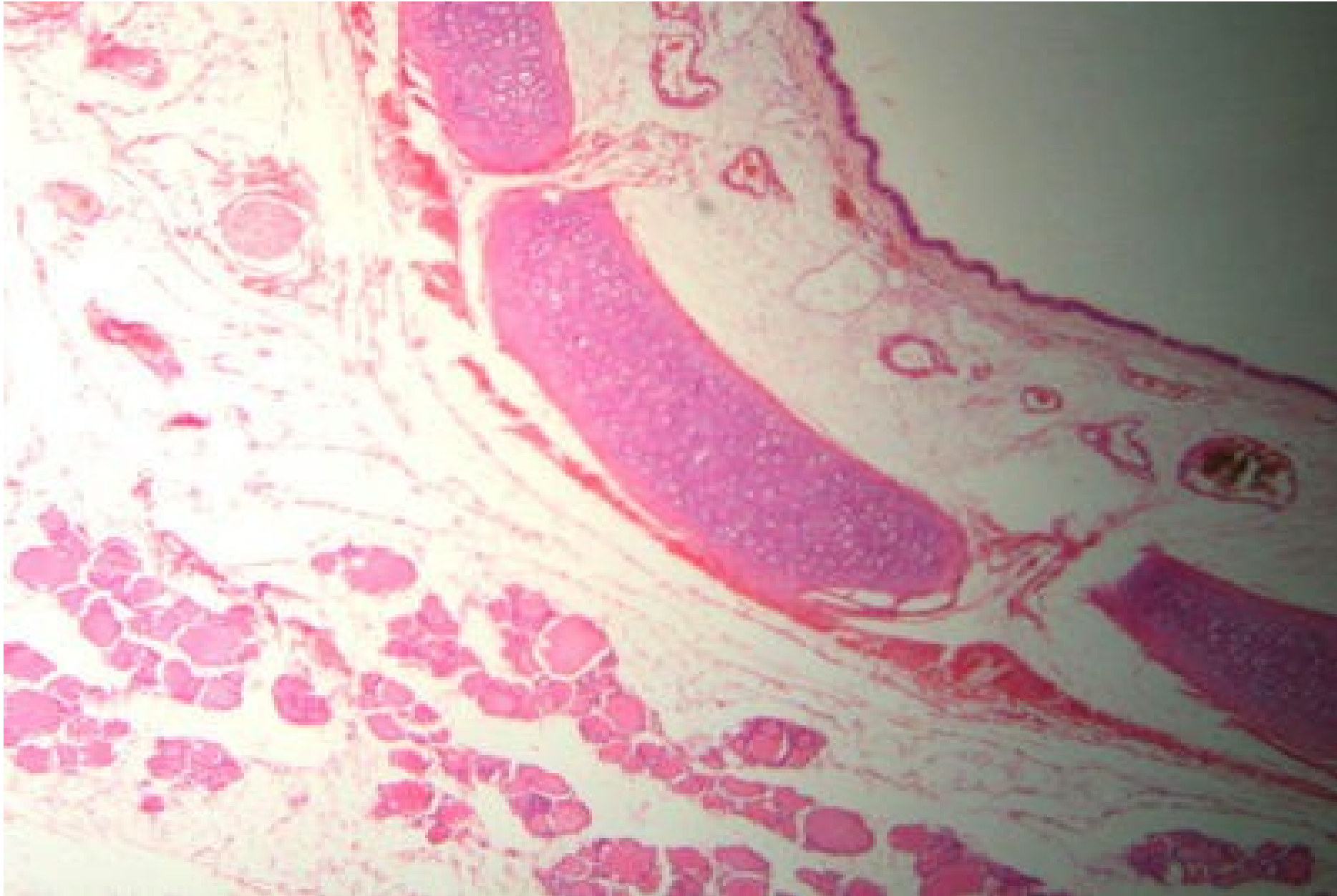
- The trachea divides into two primary bronchi that enter the lungs at the hilum, along with arteries, veins, and lymphatic vessels.
- At the hilum, the primary bronchi course downward and outward, giving rise to three secondary (lobar) bronchi in the right lung and two in the left lung, each of which supplies a pulmonary lobe.
- These lobar bronchi again divide, forming tertiary (segmental) bronchi.
- Each of the tertiary bronchi, together with the smaller branches it supplies, constitutes a bronchopulmonary segment—approximately 10–12% of each lung with its own connective tissue capsule and blood supply.
- The existence of such lung segments facilitates the specific surgical resection of diseased lung tissue without affecting nearby healthy tissue.



Bronchial Tree

- The tertiary bronchi give rise to smaller and smaller bronchi, whose terminal branches are called bronchioles.
- Each bronchiole enters a pulmonary lobule, where it branches to form five to seven terminal bronchioles.
- The pulmonary/lung lobules are pyramid-shaped, with the apex directed toward the pulmonary hilum.
- Each lobule is delineated by a thin connective tissue septum, best seen in the fetus.
- In adults, the pulmonary septa are frequently incomplete, resulting in a poor delineation of the lobules.
- Moving through the smaller bronchi and bronchioles toward the respiratory portion, the histologic organization of both the epithelium and the underlying lamina propria gradually becomes more simplified.

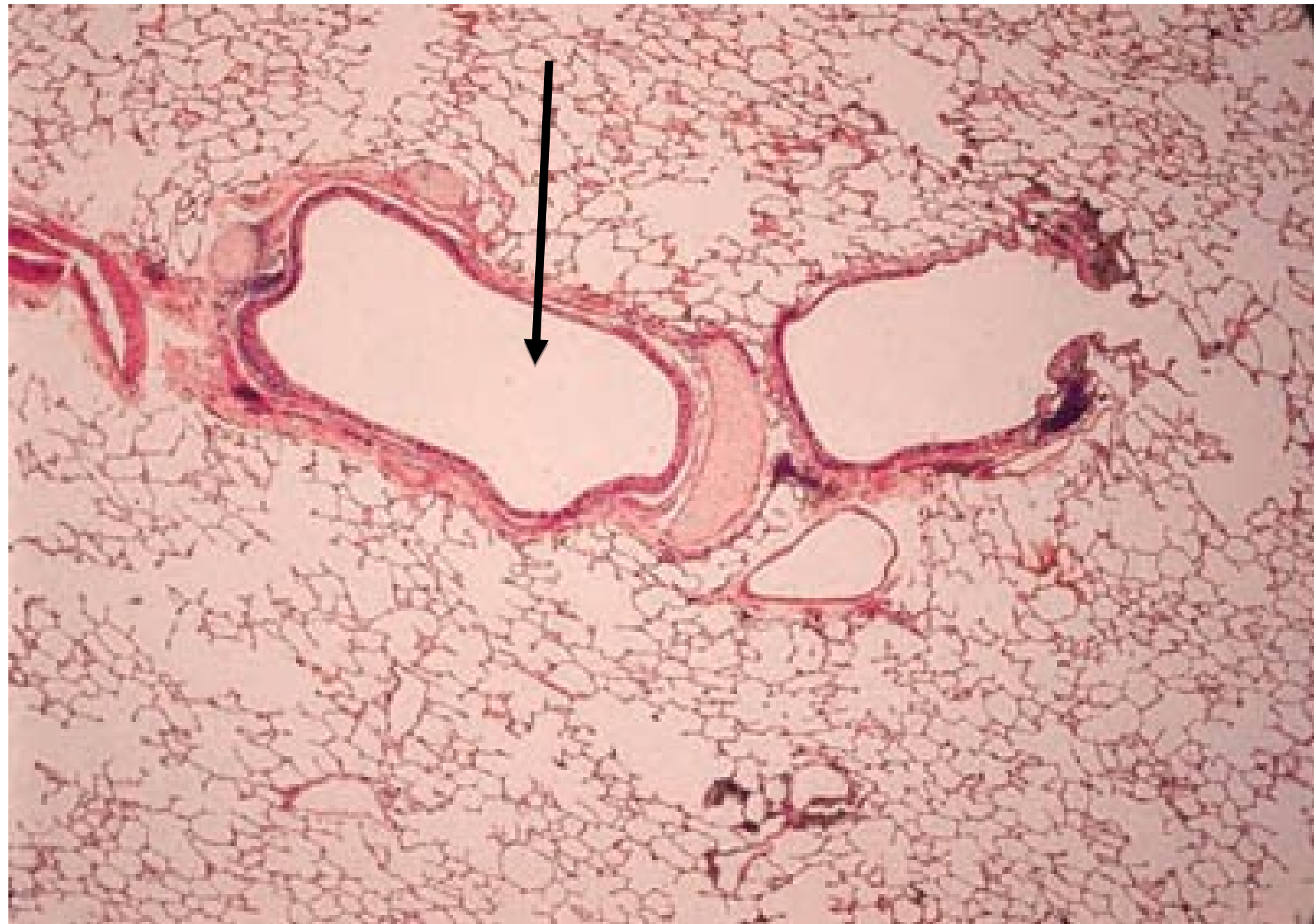
EXTRAPULMONARY BRONCHUS, INTERCONNECTED CARTILAGE PLATES AND SPARSE SUBMUCOSA IN ITS WALL



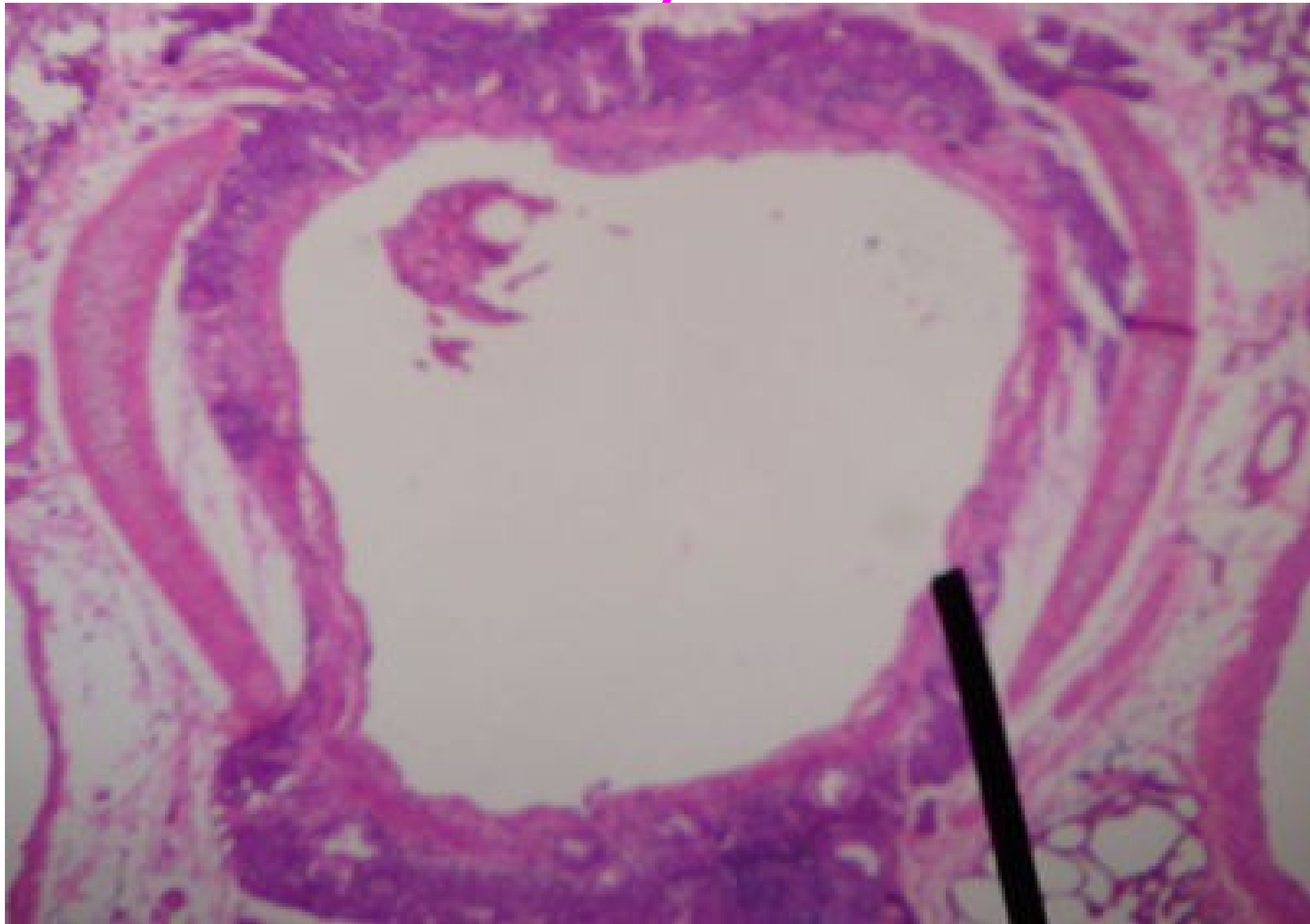
Secondary Bronchi

- The intrapulmonary secondary bronchi differs from extrapulmonary bronchi by the following, otherwise the structure is the same:
- **The mucosa is slightly folded.**
 - **The presence of muscularis mucosa:** a layer of crisscrossing bundles of spirally arranged smooth muscle beneath the lamina propria, which become more prominent in the smaller bronchial branches.
 - Contraction of this muscle layer is responsible for the folded appearance of the bronchial mucosa observed in histologic section.
- **Adventitia** (or the fibrocartilaginous layer) contains **interconnected cartilage plates** and dense fibroelastic CT.

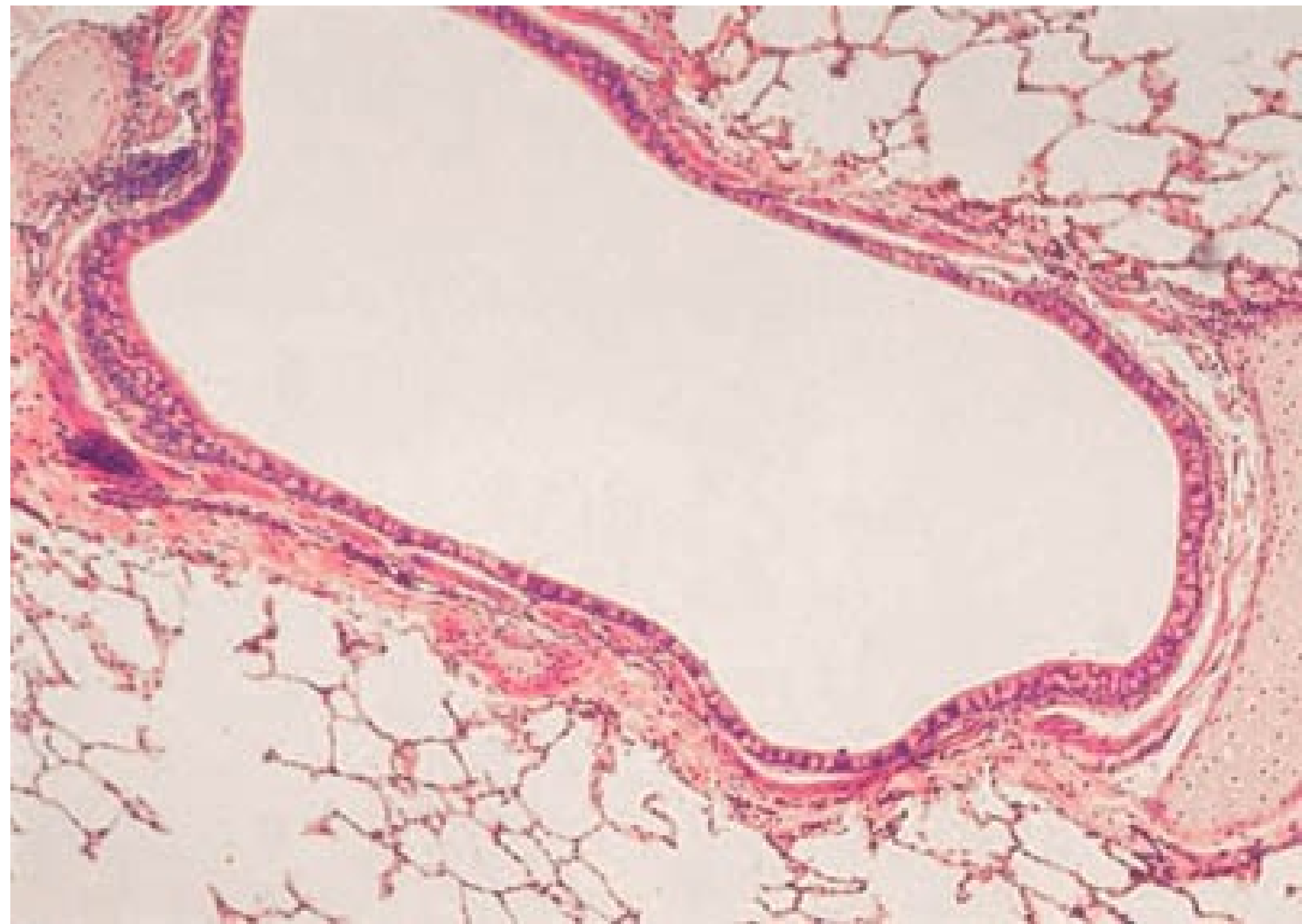
SECONDARY BRONCHUS T. S.



Secondary bronchus



INTRAPULMONARY BRONCHUS T.S.



Microscopic structure of the lung and the bronchial tree

- The main parenchymal lung tissue is formed of lobes, lobules, and lung acini housing the different parts of the bronchial tree which are:

1- The intrapulmonary (secondary & tertiary) bronchi

2- Bronchioles

3- Terminal bronchioles

4- Respiratory bronchioles

5. Alveolar duct, sacs and alveoli

THE INTRAPULMONARY BRONCHI

- The wall of an intrapulmonary bronchus is formed of the following:

1- Mucosa which is folded and formed of:

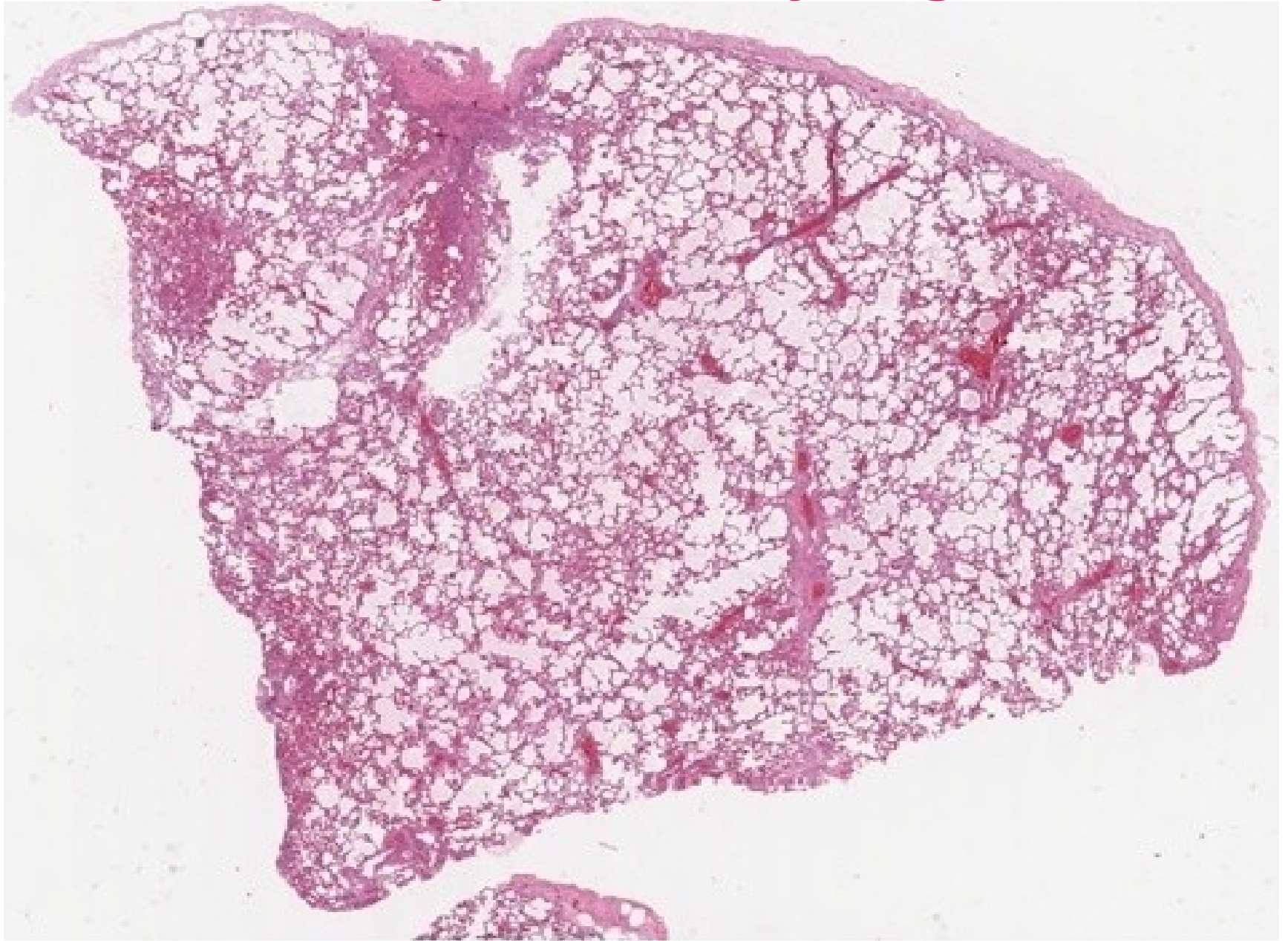
- Epithelium, which is of the same typical respiratory type.
- Lamina propria formed of loose connective tissue rich in blood vessels, lymphocytes, mast cells and elastic fibers.

2- Complete layer of smooth muscle fibers spirally arranged (criss-crossing fibers) which encircle the whole lumen.

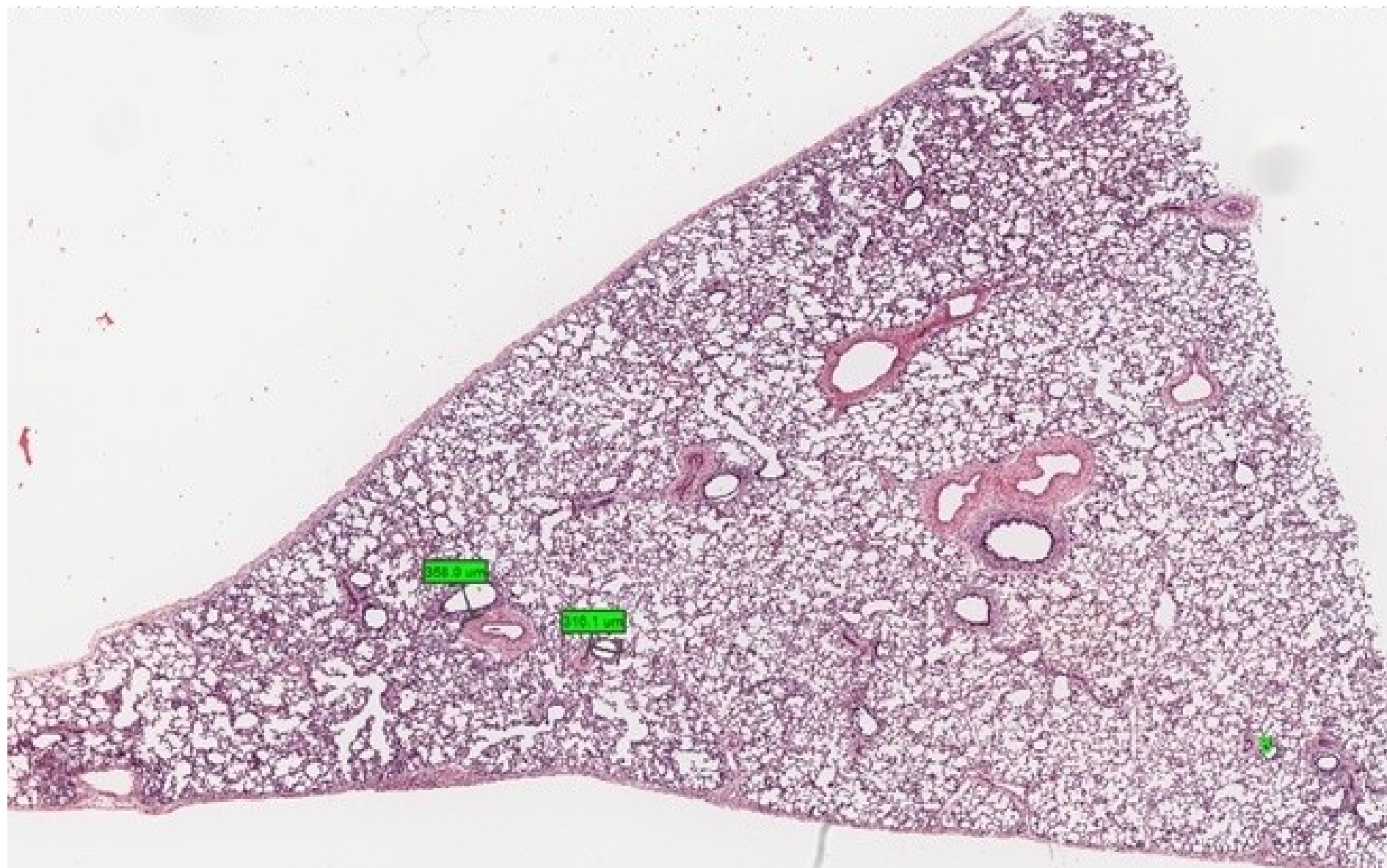
3- Layer of alternated hyaline cartilage plates with seromucous glands.

- **The tertiary bronchi** give rise to smaller and **smaller bronchi**, whose terminal branches are called **bronchioles**.
- Each of these tertiary bronchi, together with all its smaller branches it supplies, constitutes a **bronchopulmonary segment**—approximately 10–12% of each lung with its own connective tissue capsule and blood supply.
- **Each bronchiole** enters a **pulmonary lobule**, where it branches to form five to seven **terminal bronchioles**.
- The pulmonary lobules are pyramid-shaped, with the apex directed toward the pulmonary hilum.
- Each lobule is delineated by a thin connective tissue septum, best seen in the fetus.
- In adults these septa are frequently incomplete, resulting in a poor delineation of the lobules.
- Moving through smaller bronchi & bronchioles toward respiratory portion, the histologic organization of both the epithelium & underlying lamina propria gradually becomes more simplified.

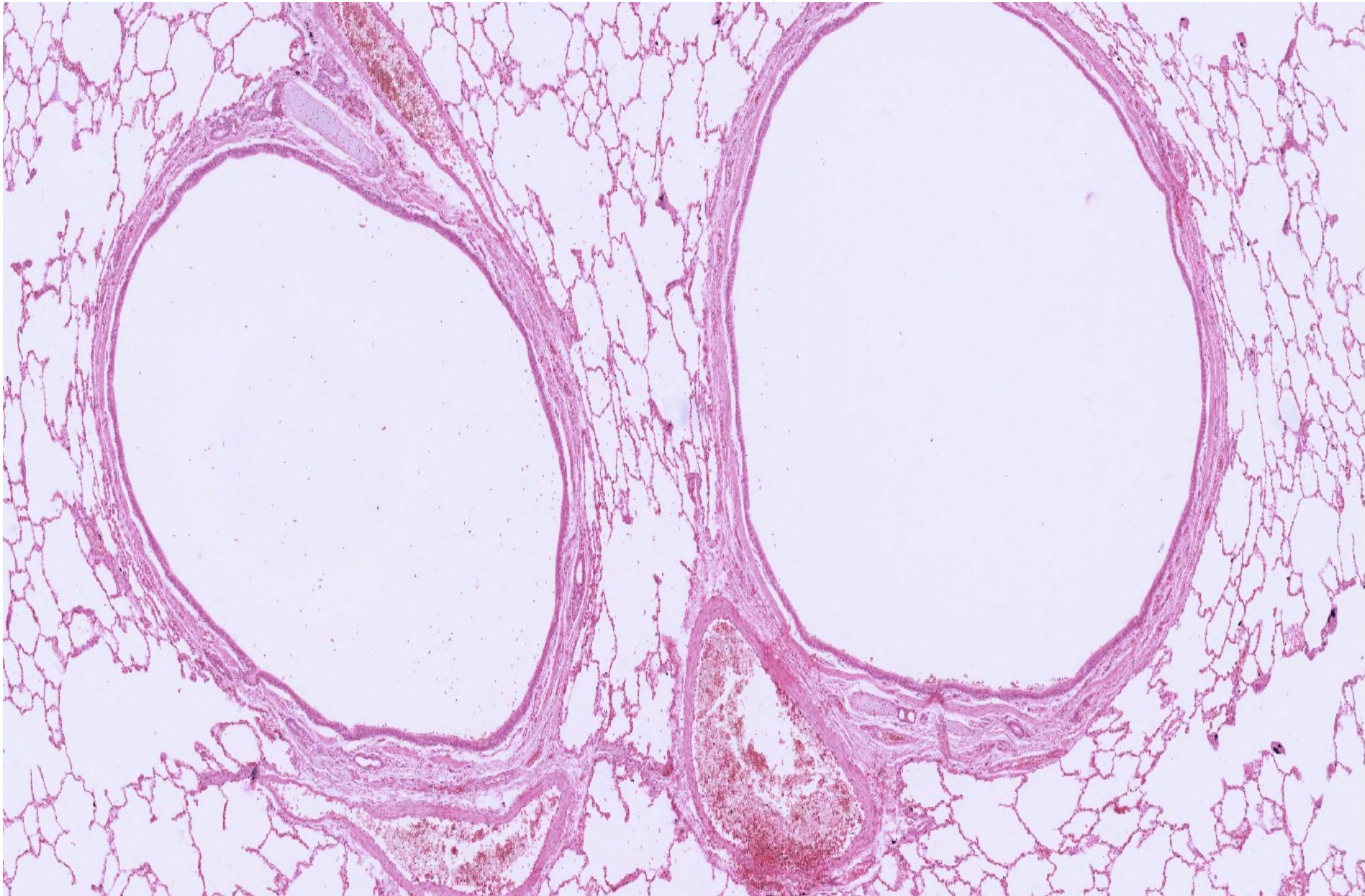
Bronchopulmonary segment



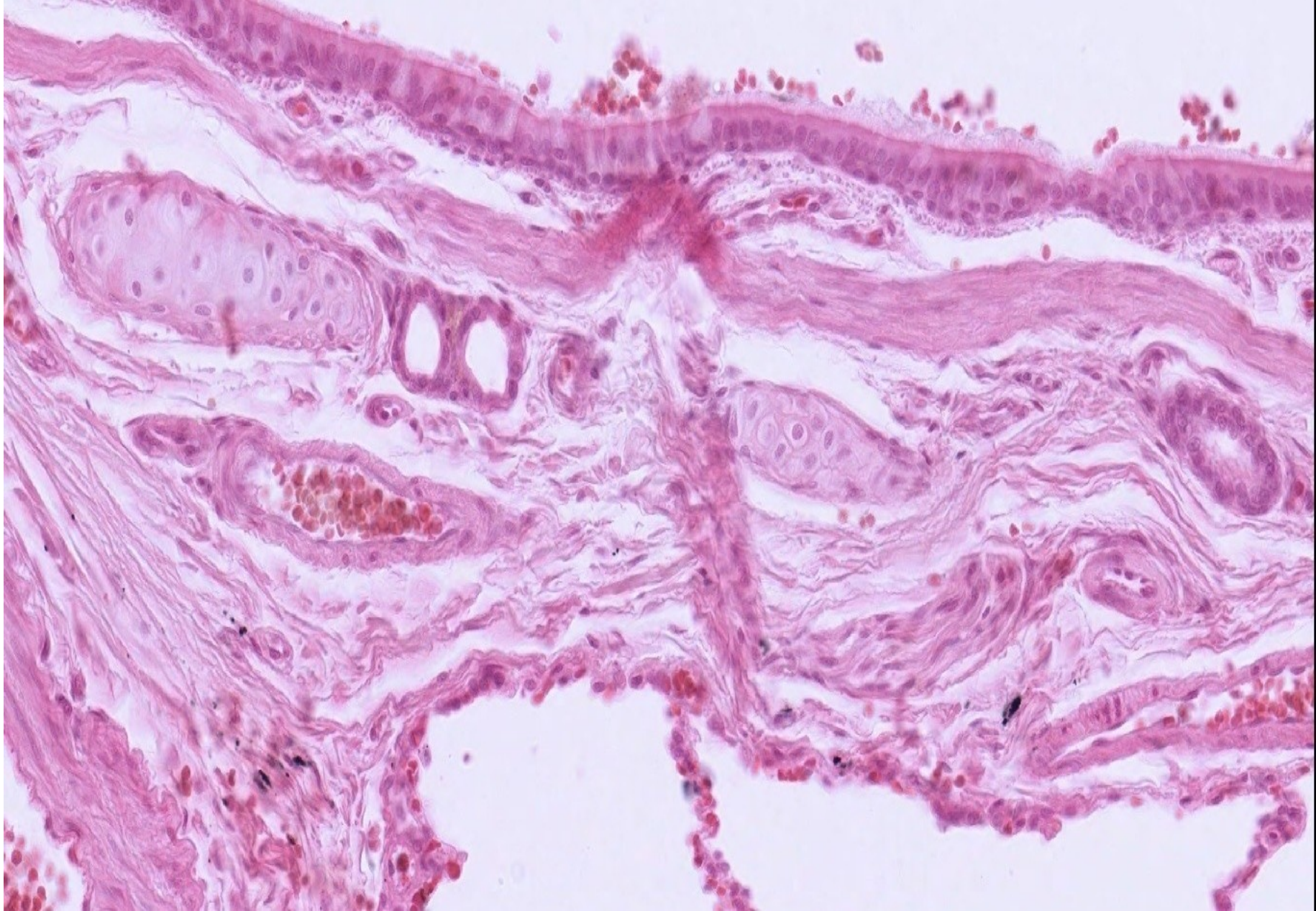
Bronchopulmonary segment



SEGMENTAL BRONCHI



Histology of Segmental Bronchus



Extra-pulmonary bronchus

1. It has a wide lumen.
2. Its mucosa is less folded.
3. **Epithelium** is pseudo-stratified columnar ciliated with many goblet cells
4. It has a definite submucosa
5. Presence of **C-shaped** cartilaginous rings in its wall
6. Smooth muscles are present posteriorly only at the ends of C-shaped cartilage
7. Presence of an elastic membrane in the submucosa
8. **Seromucous** glands are present in the submucosa

Intra-pulmonary bronchus

1. It has a narrow lumen.
2. Its mucosa is highly folded.
3. **Epithelium** is pseudo-stratified columnar ciliated with few goblet cells.
4. It has no submucosa
5. Presence of multiple plates of cartilage connected by C.T. in its wall
6. Smooth muscles are arranged spirally and encircling the whole lumen.
7. Elastic fibres are distributed between cartilage plates and in the adventitia
8. **Seromucous** glands and lymphatic nodules are present between cartilage plates and in the adventitia.

BRONCHIOLES

- are intralobular air passage ways lacking cartilage support and sero-mucous glands in their wall.
- are highly variable in size having outer diameter of 1mm or less.
- the largest bronchioles, traditionally called primary bronchioles are last divisions of bronchial tree which lacks cartilage support and seromucous glands.
- **bronchiolar wall is free of cartilages but consists of:**
 - 1. Mucosa:** folded and contains:
 - a) Epithelium:** shorter pseudostratified columnar ciliated epithelium with fewer goblet cells (typical **respiratory epithelium but with fewer goblet cells**) in the larger **(primary) bronchioles**, but the epithelium decreases in height and complexity to become **ciliated simple columnar and cuboidal epithelium** in the terminal bronchioles and respiratory bronchioles, respectively.

Bronchiolar Wall

- b) **The lamina propria:** is composed largely of smooth muscle, mast cells and elastic fibers.

2. Smooth muscle layer:

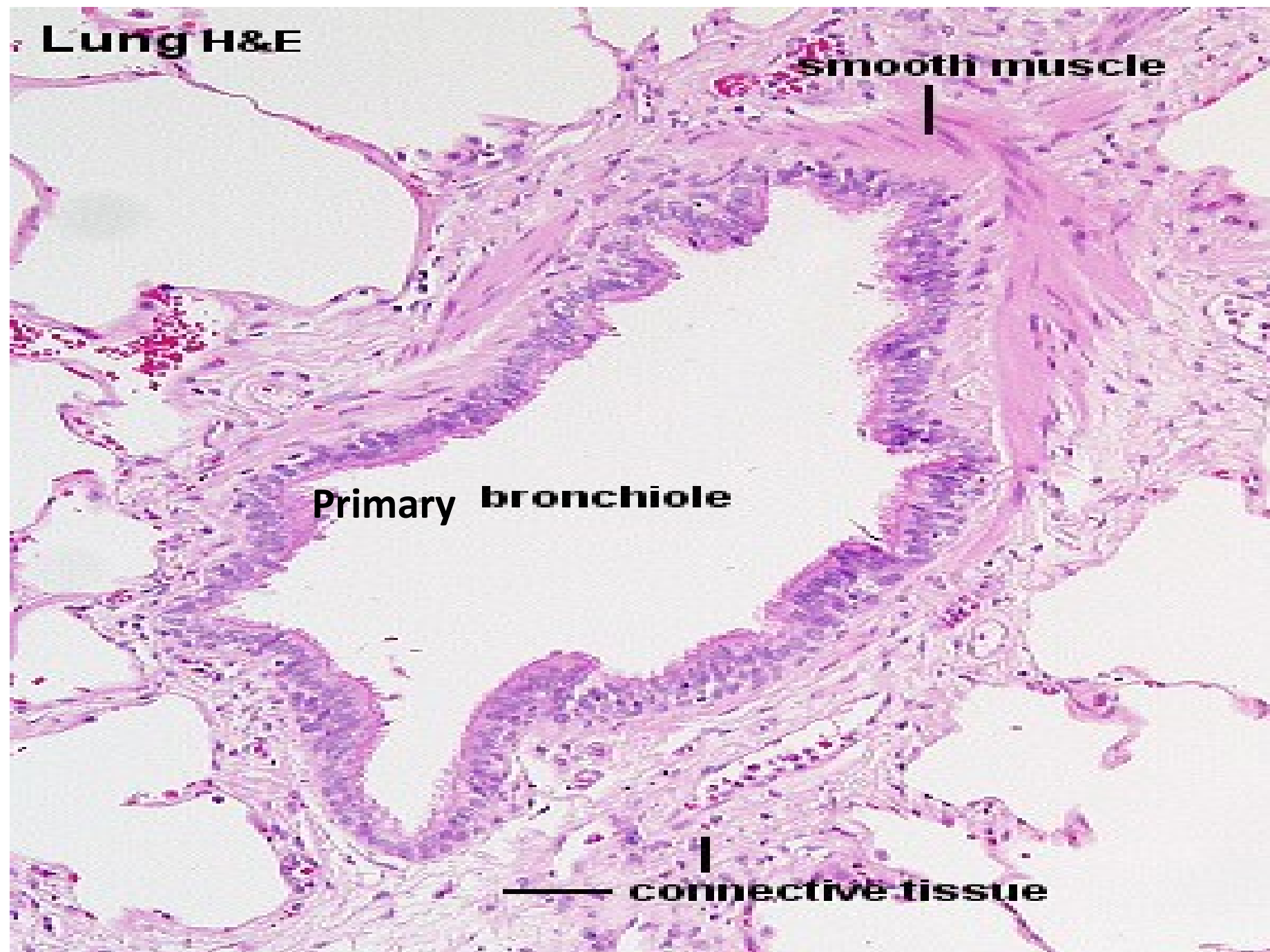
- relatively thick concentrically arranged smooth muscle layer.
- Sympathetic stimulation relaxes these muscles, hence epinephrine and sympathomimetic drugs are used as anti-asthmatic agents.

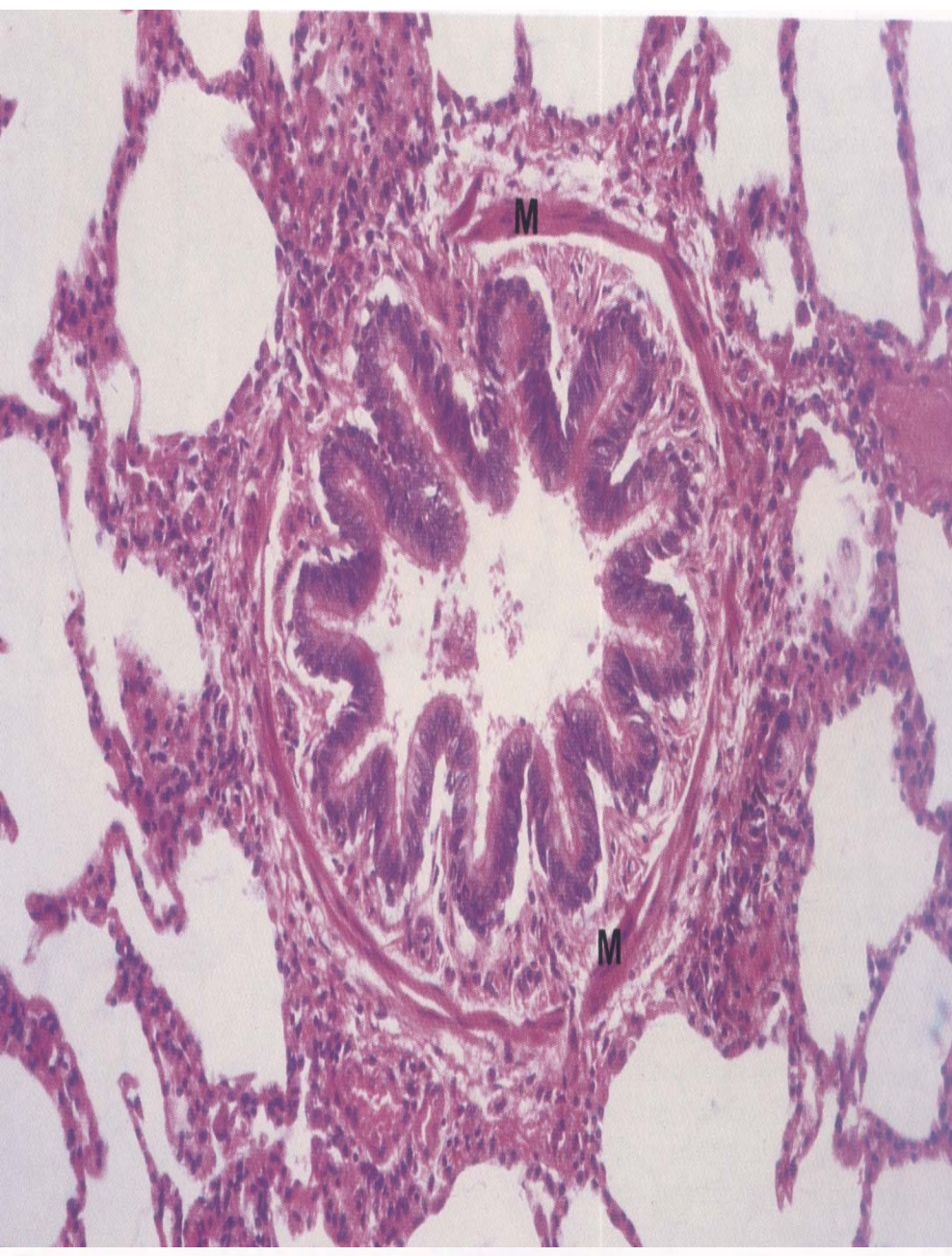
3. Adventitia:

- loose connective tissue without cartilage support
- blend with the surrounding lung parenchyma

No Submucosa and no seromucous glands.

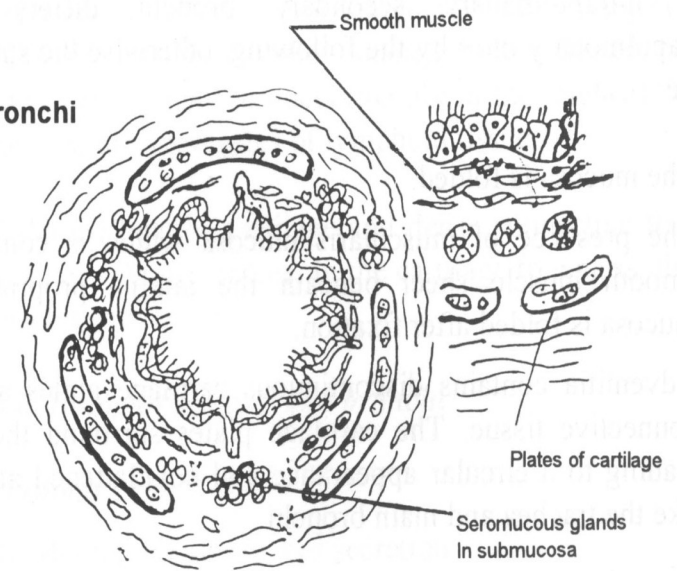
- The primary bronchioles may give rise to secondary and tertiary bronchioles and finally terminal bronchioles, which give respiratory bronchioles, that receive the alveoli.



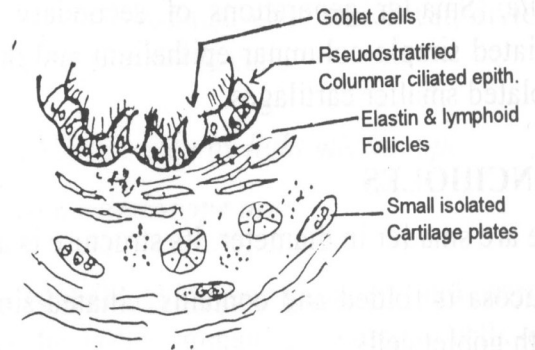


Bronchiole (*H & E* $\times 160$)

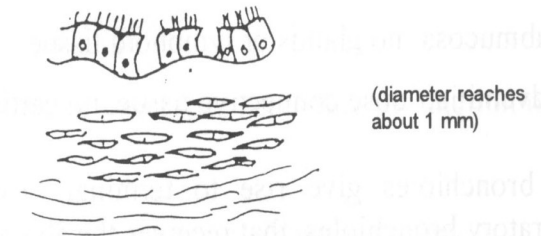
Secondary Bronchi



Smaller bronchi



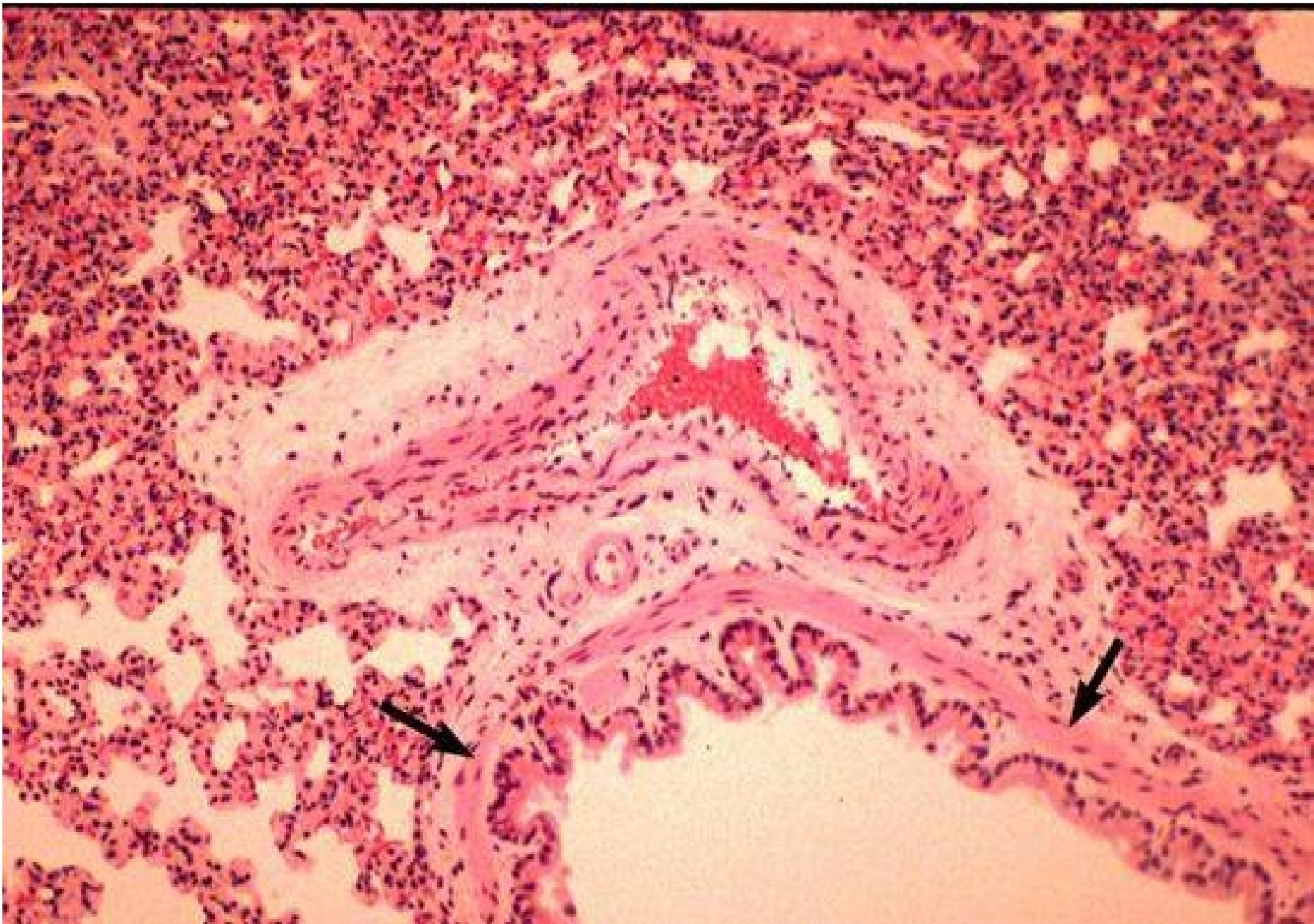
The Bronchioles



Terminal Bronchioles

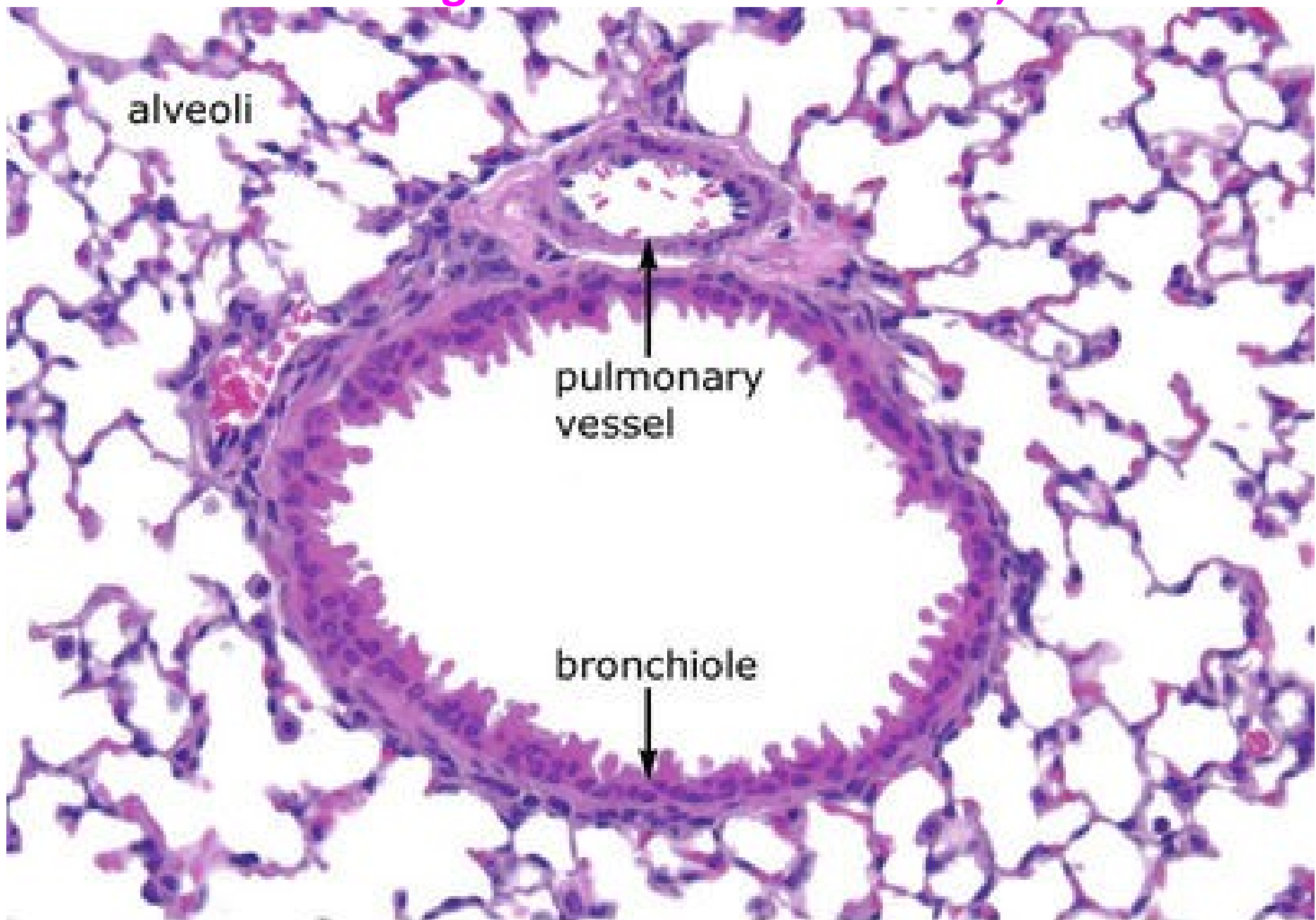


BRONCHIOLE, SMOOTH MUSCLE LAYER (ARROWS)



- Absence of cartilage from the wall of bronchioles is a potential hazard, since these airways can constrict to a point of closing the lumen if the tone of their muscles is increased.
- This is the problem of **asthma** which is an allergic condition to non-specific lung irritant. *Wheezing noises* and *difficulty in breathing* occurs during *expiration* rather than inspiration.
- The musculature of both the bronchi and the bronchioles is under the control of the vagus nerve and the sympathetic nervous system, in addition to the influence of neuroendocrine peptides and local factors.
- Stimulation of the vagus nerve decreases the diameter of these structures; sympathetic stimulation produces the opposite effect.

Normal lung labeled with bronchiole, 40x



Terminal Bronchioles

- are the **smallest** air tubes of purely conducting portion of the respiratory system.
- give rise to further branches known as respiratory bronchioles, which increasingly involve in gaseous exchange.

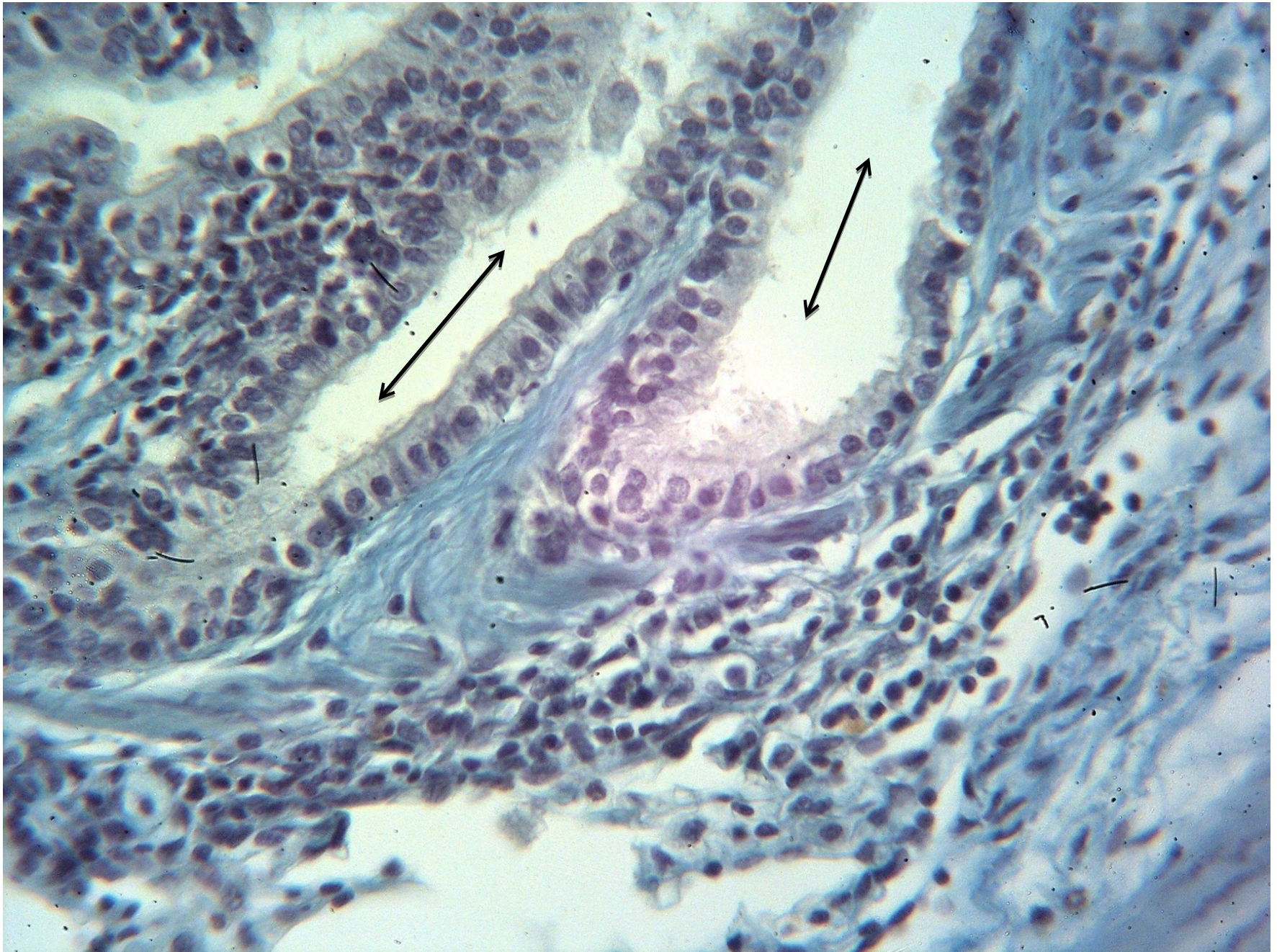
1. **Their mucosa** is characterized by:

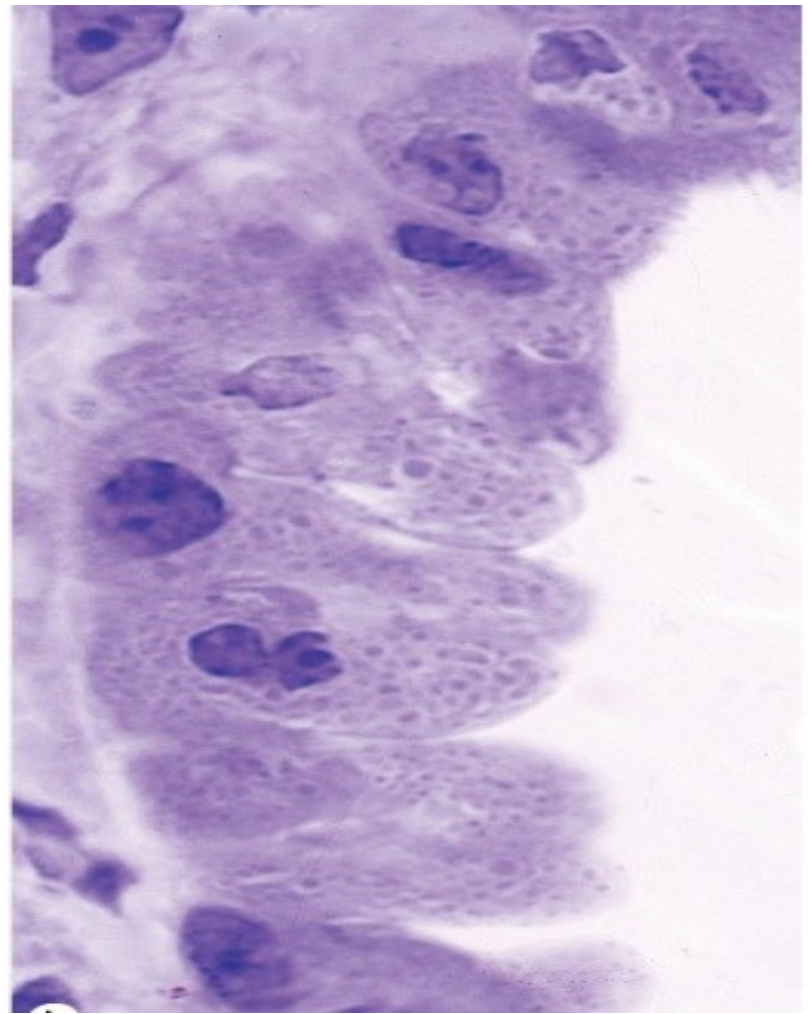
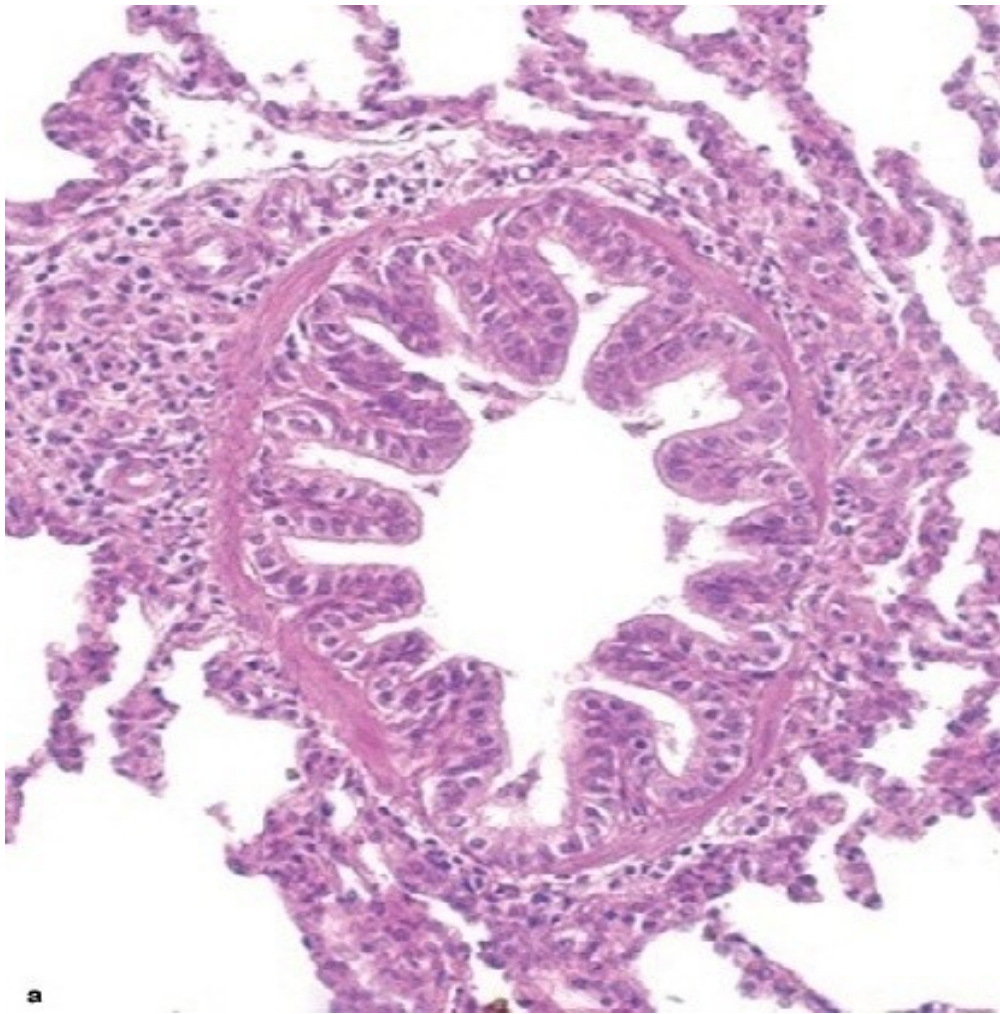
- ciliated simple columnar epithelium without goblet cells.
- only one or two layers of smooth muscle cells in LP.
- contain non ciliated, bronchiolar secretory cells known as club cells (formerly called Clara cells), which secrete surfactant-like product rich in phospholipids and lipoproteins and known as **club/Clara cell secretory protein (CC16), a 16-kilodalton** that alters the surface tension of the fluid layer covering the bronchiolar surface.
- CC16 is an abundant component of the airway secretion.

Terminal Bronchiole



Terminal Bronchioles, lined by simple columnar epithelium



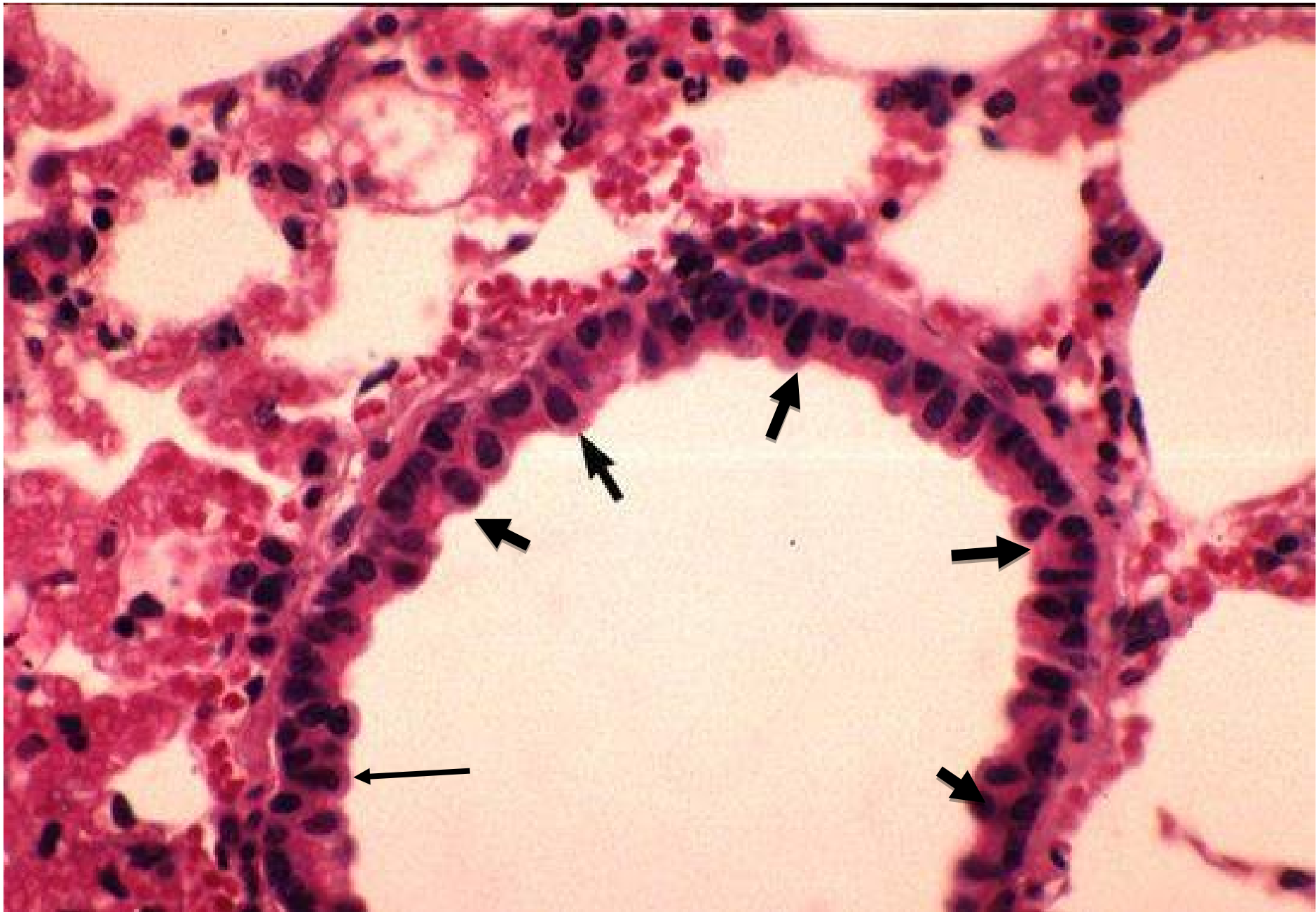


Cross-section of a terminal bronchiole showing: a lining columnar epithelium containing both ciliated cells and non ciliated bronchiolar secretory cells, and only one or two layers of smooth muscle cells. The X300. PT. (b): The nonciliated Clara cells with bulging domes of apical cytoplasm contain granules, as seen better in a plastic section.

Club (Clara) cells:

- are dome-shaped, non-ciliated, exocrine secretory cells lining the bronchiolar mucosa.
- found interspersed in the ciliated epithelium of bronchioles
- *originally described by their namesake of Dr. Max Clara, the histologist who first described them in 1937 by using lung tissue from executed victims of the Nazz's Third Reich (1933-1945)*
- *due to the ethical controversy surrounding the discovery of these cells, new name known as **Club cell** was adopted on January 1, 2013.*
- *Similarly, **Clara cell secretory protein, 16-kilodalton**, was also replaced by **club cell secretory protein (CC16)**.*
- Ultrastructurally they reveal: Long cisternae of RER & SER, scattered elongated mitochondria and small accumulation of glycogen.

CLUB (CLARA) CELLS



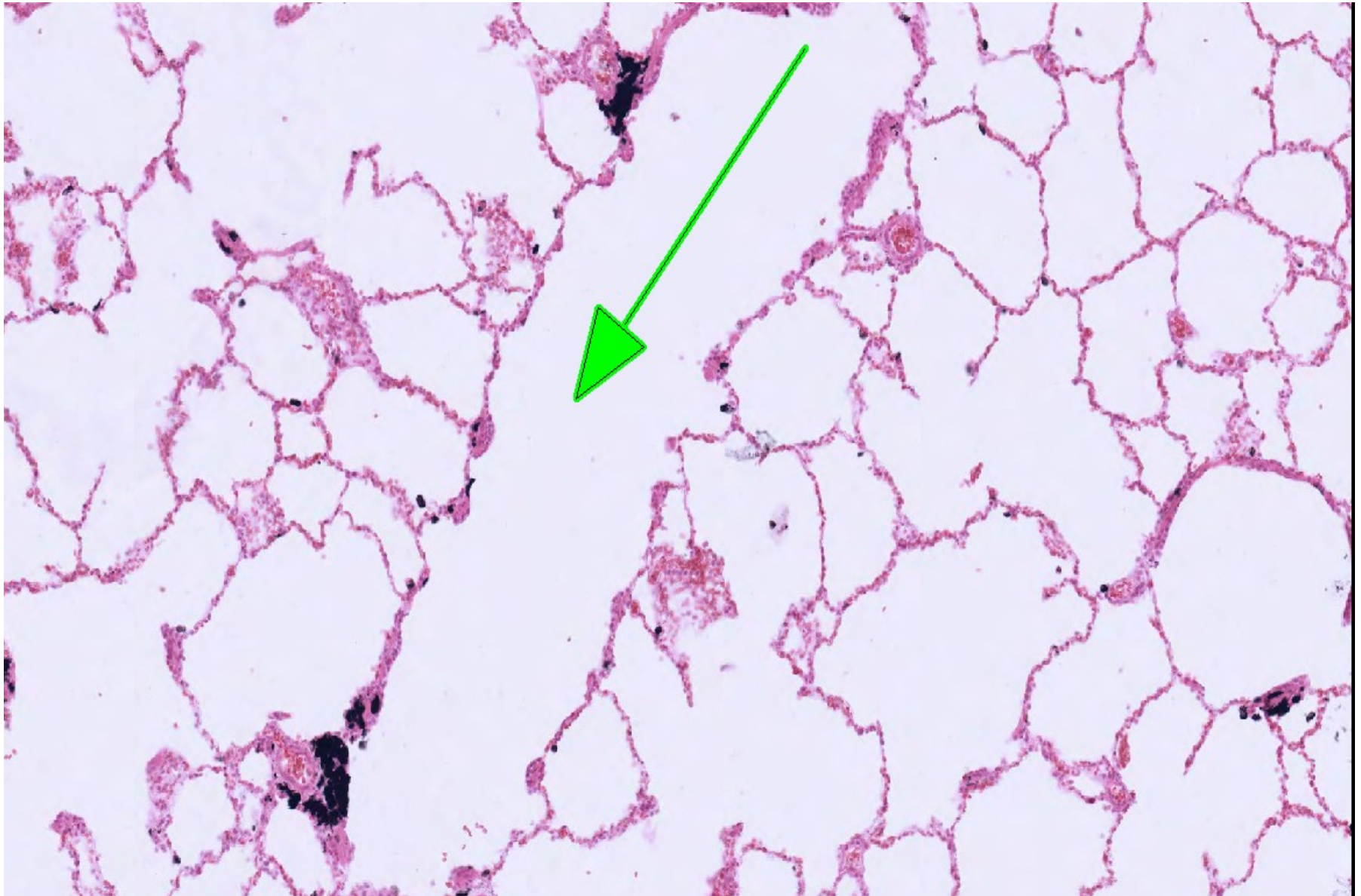
Bronchiolar epithelium consisted of ciliated & non ciliated cells, note the difference in size between cilia & microvilli on non ciliated cell surface, SEM



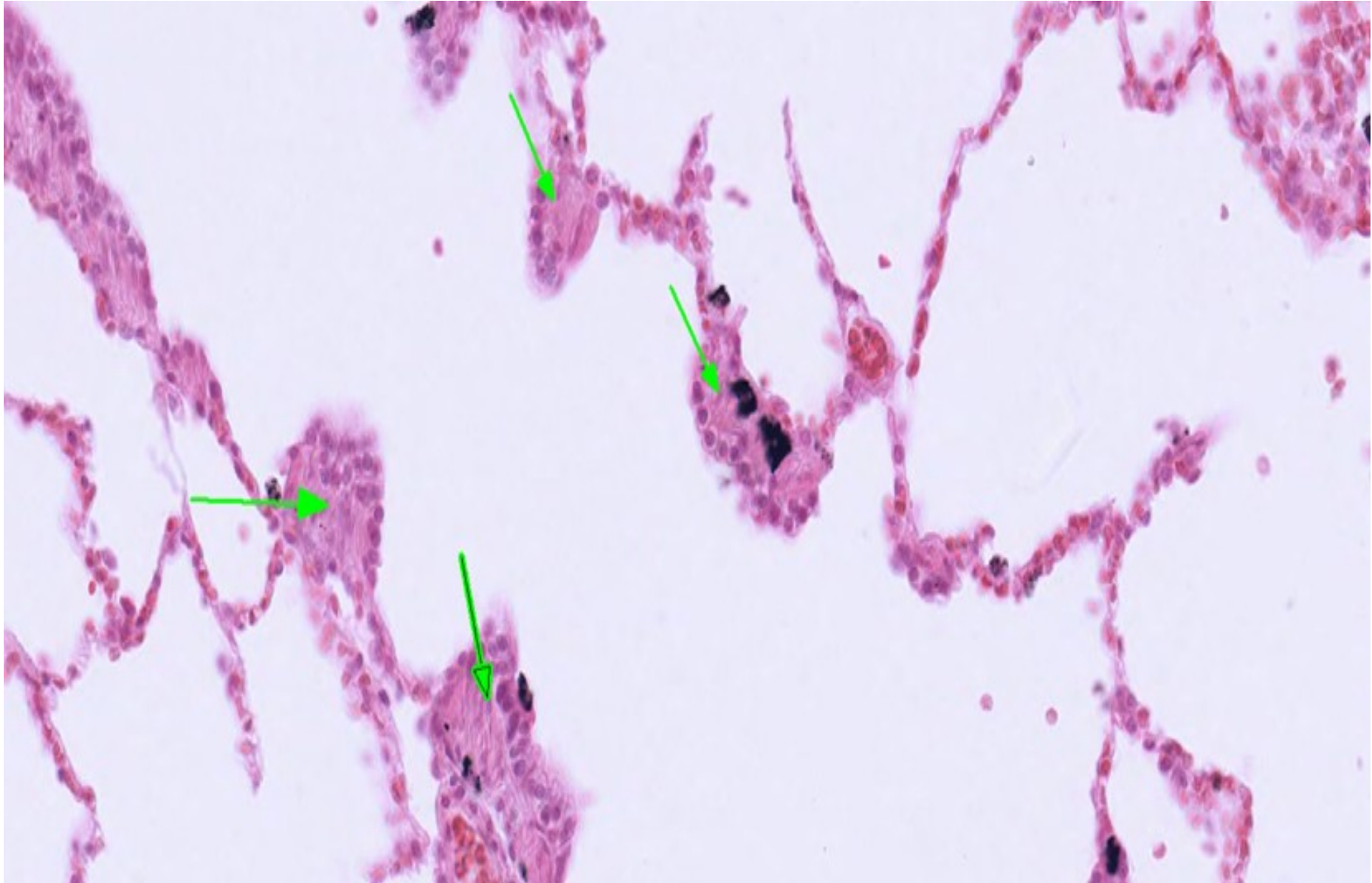
RESPIRATORY BRONCHIOLES

- Named for the presence of individual alveoli attached to them at irregular interval.
- Lined by ciliated simple cuboidal epithelium which is interrupted by the openings of saclike individual alveoli where gas exchange occur.
- Their wall is interrupted by the openings of many alveoli.
- are lined with simple cuboidal ciliated epithelium, some of the cells are ciliated, and others are non-ciliated *Clara cells*.
- At the junction between the bronchiole and the alveoli the cuboidal cells continue as simple squamous epithelium of pneumocytes type I.
- Some smooth muscle fibers and elastic fibers are present in the lamina propria under the bronchiolar epithelium.

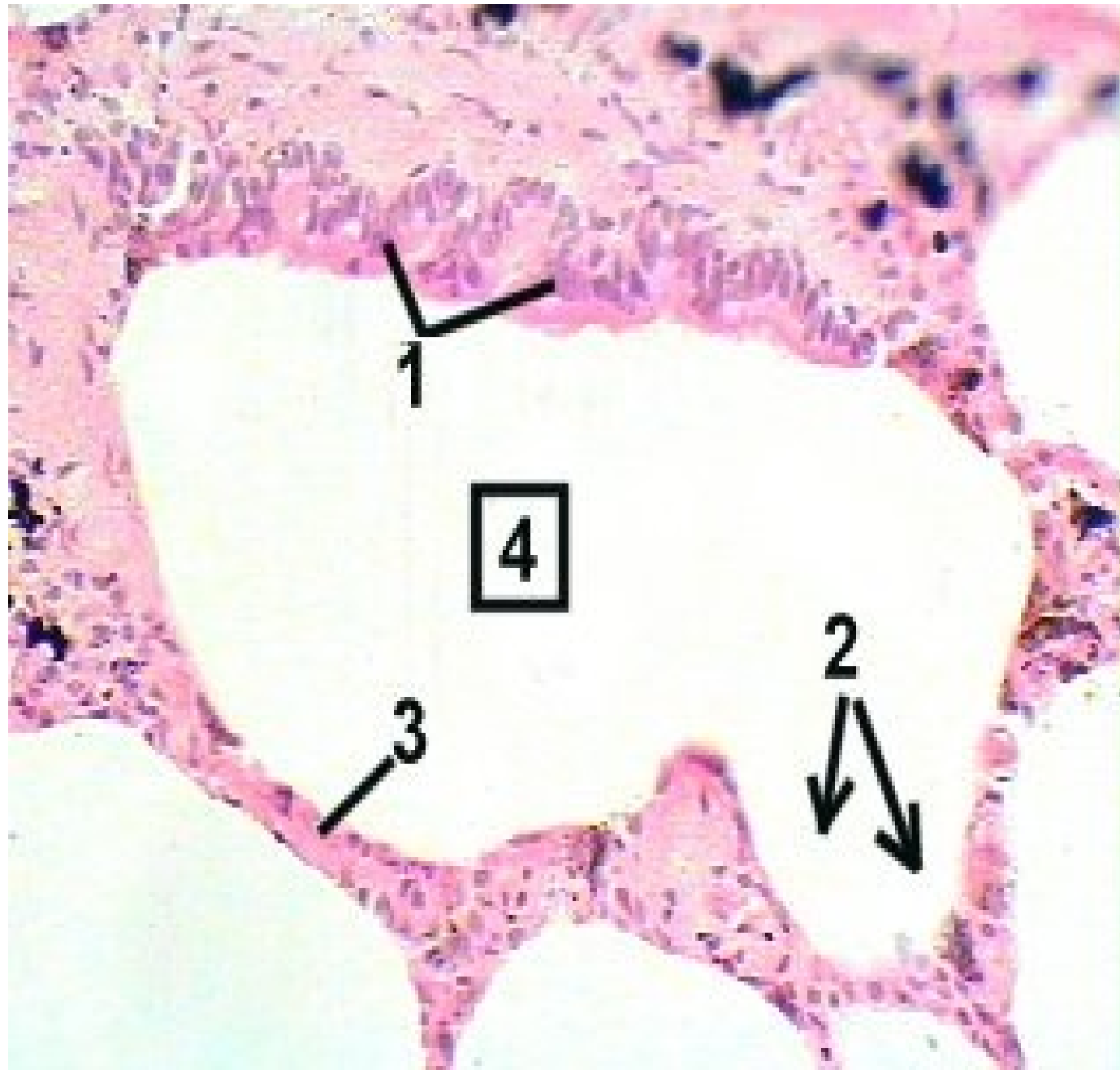
RESPIRATORY BRONCHIOLES



Respiratory Bronchiole

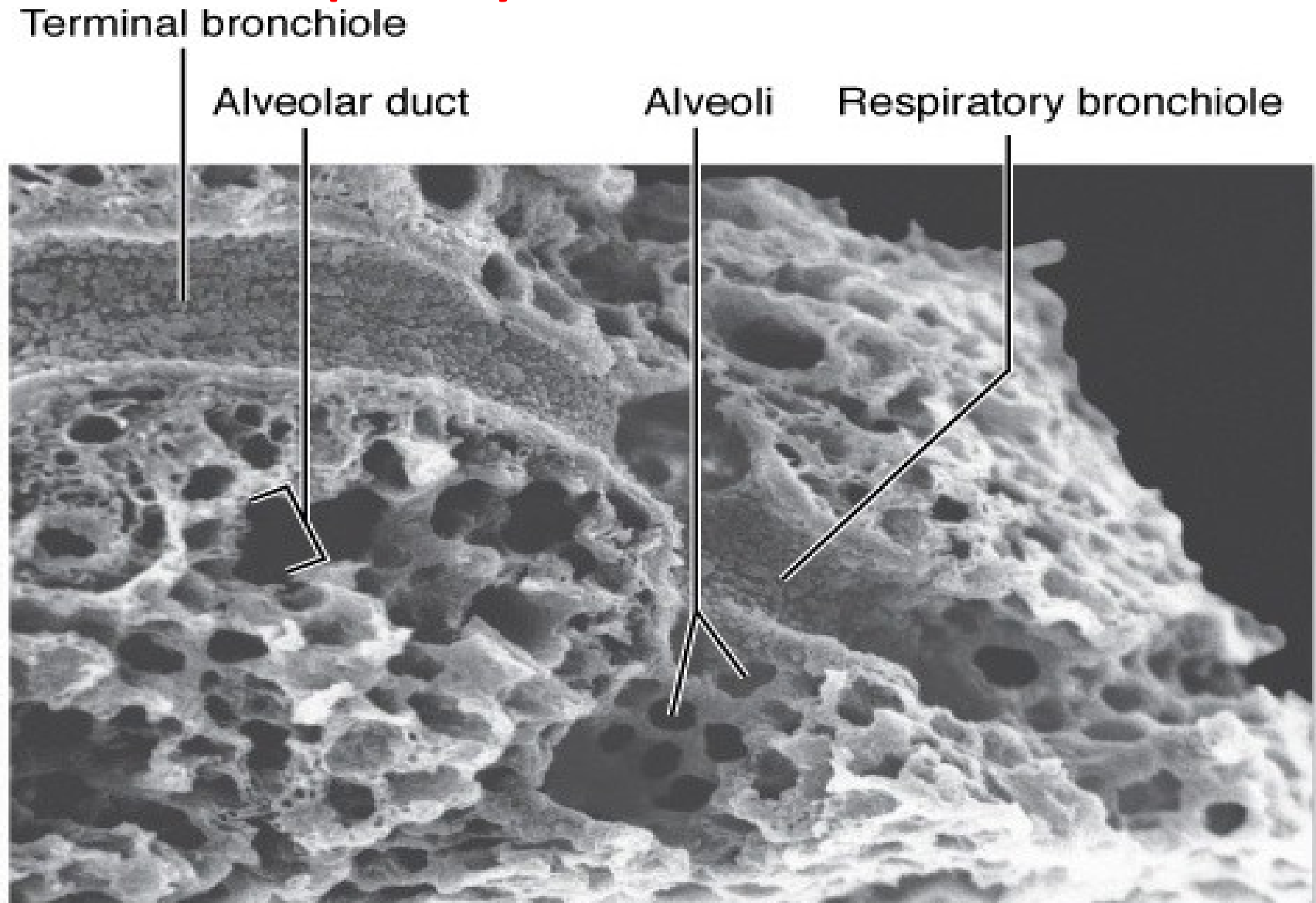


High Magnification of the Lung Demonstrating a Respiratory Bronchiole



1. High cuboidal epithelium
2. Alveolus
3. Interalveolar septum
4. Lumen

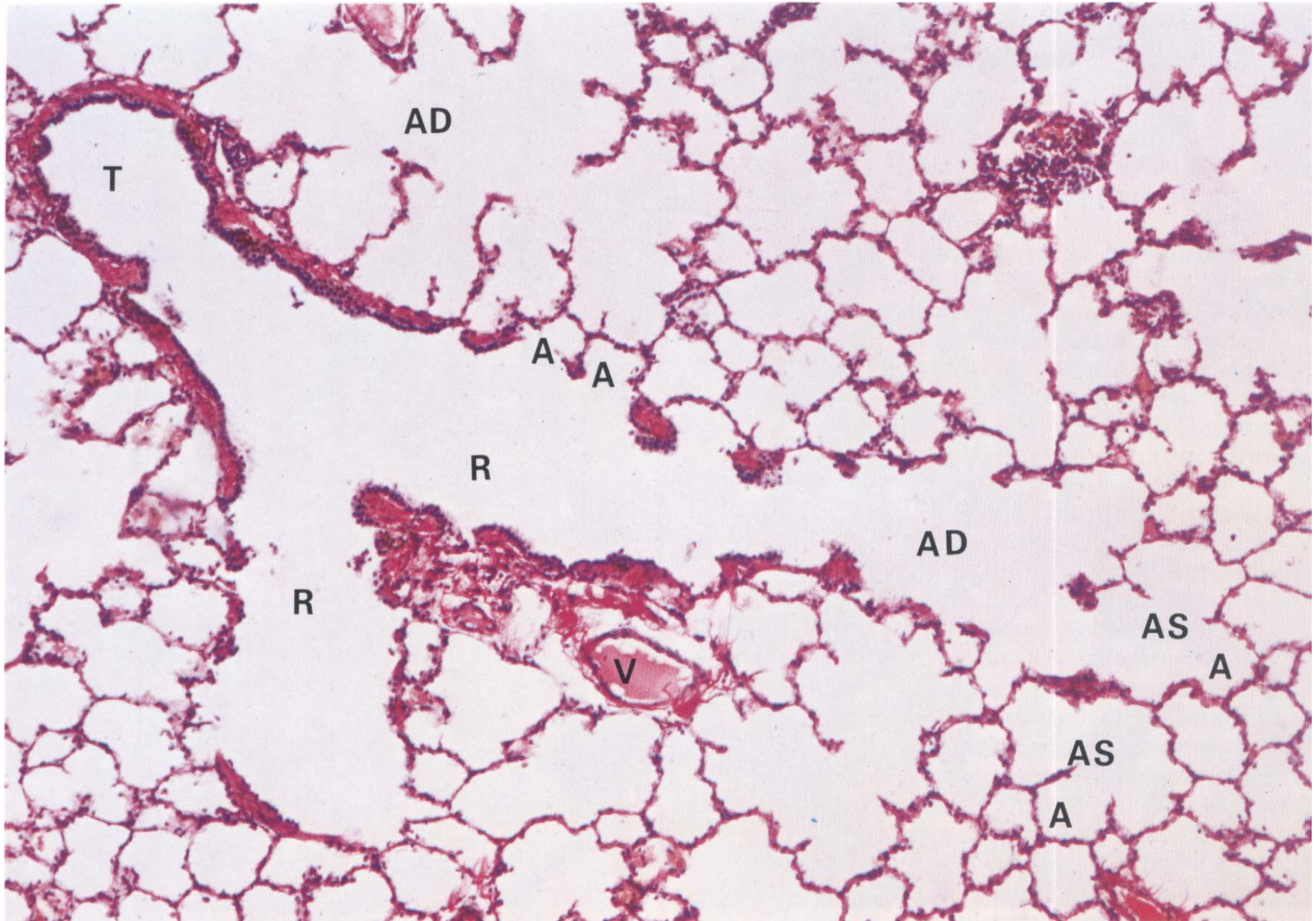
SEM shows in three dimensions the relationship of alveoli to terminal and respiratory bronchioles

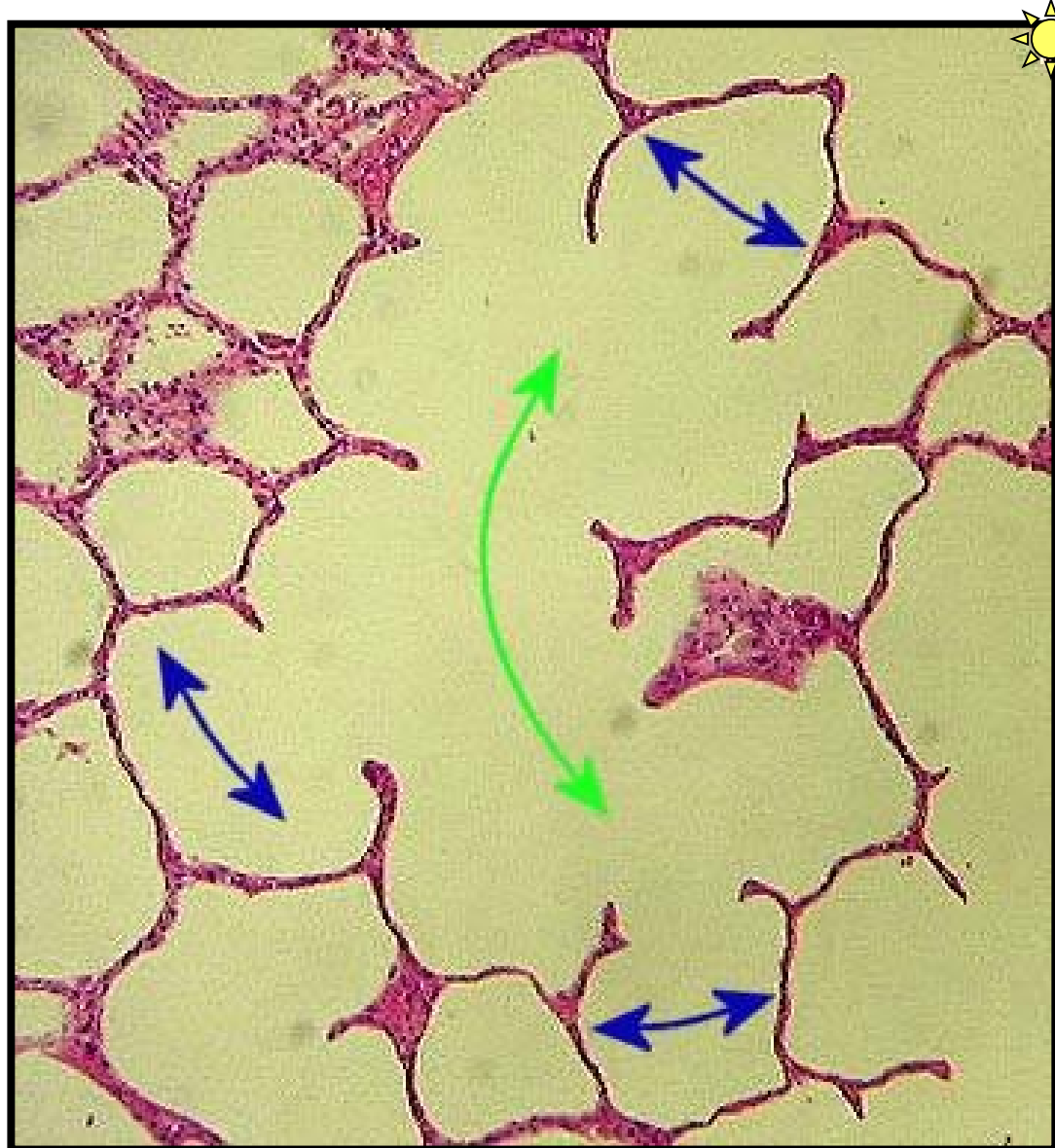
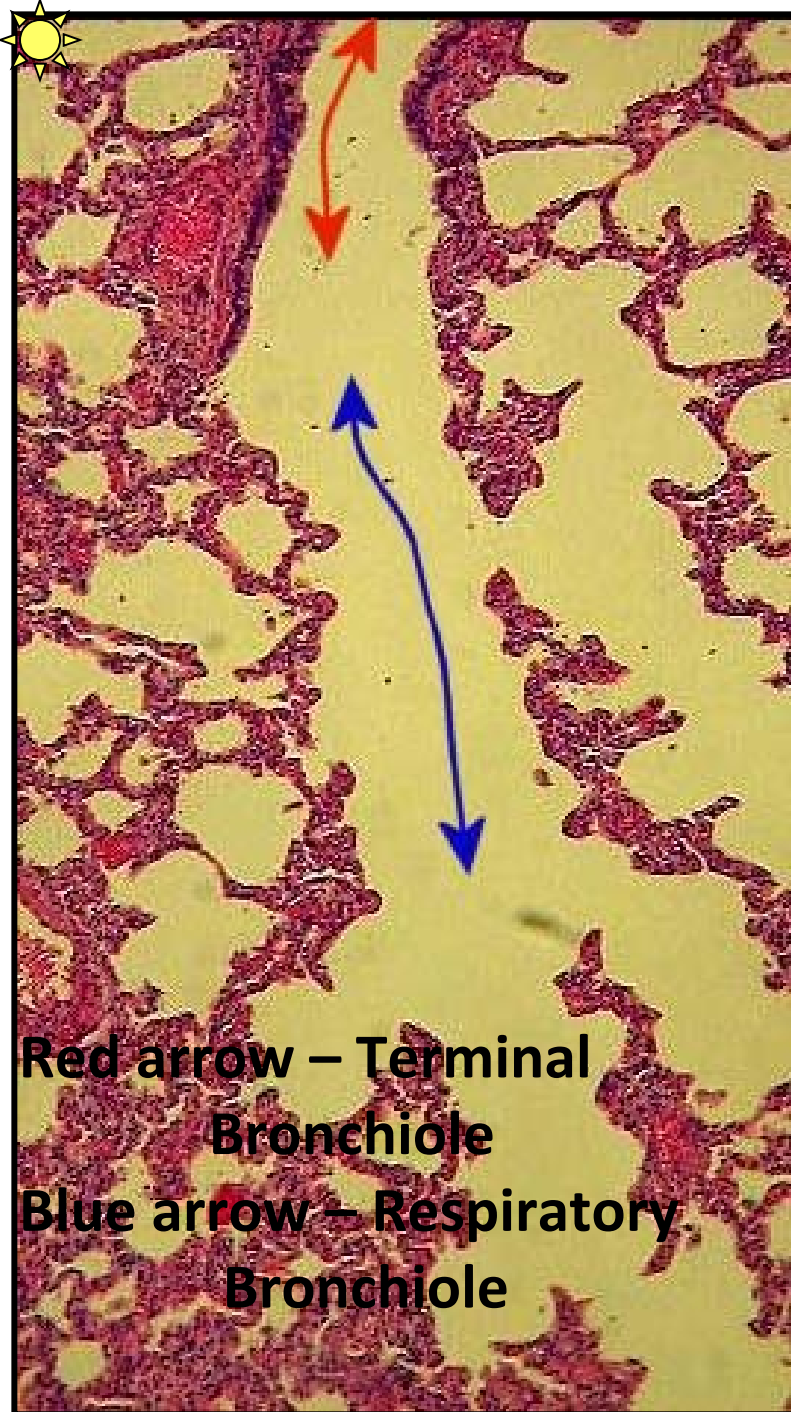


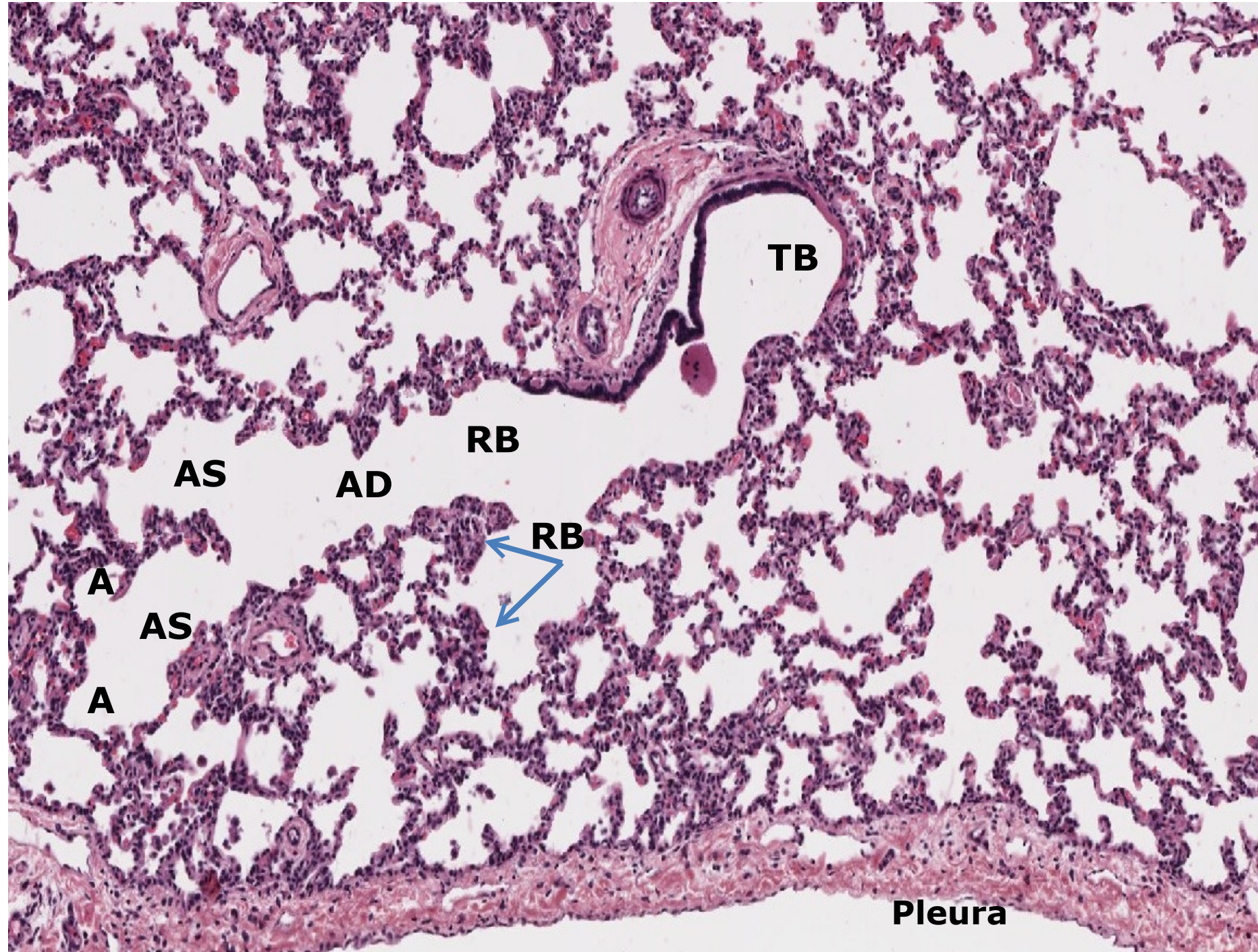
Respiratory Bronchiole thick segment and



Terminal portion of the respiratory tree (Lung Acinus)







TB

RB

AD

AS

RB

A

AS

A

Pleura

Alveoli and alveolar cells



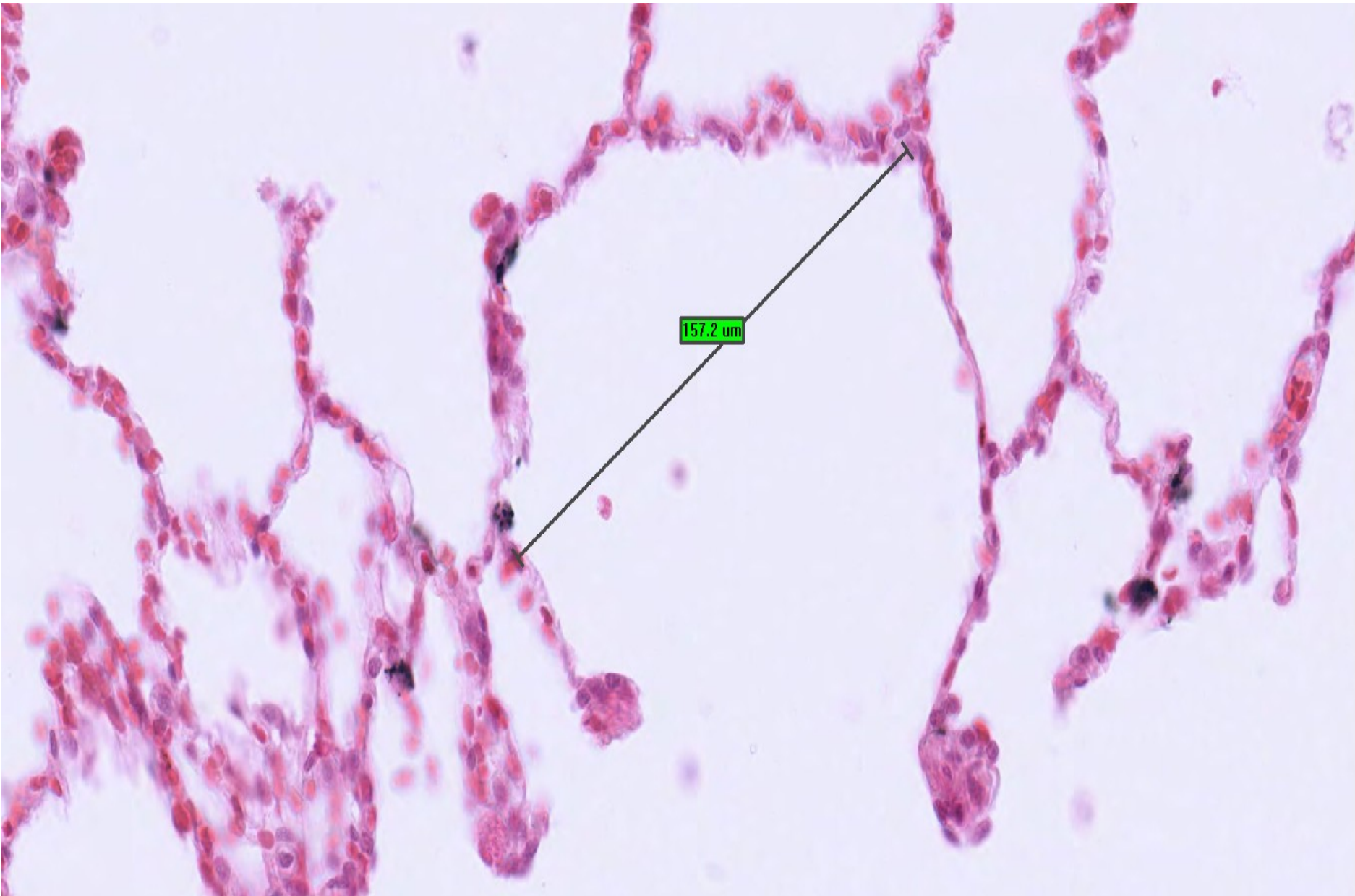
ALVEOLI

- cup-shaped/saclike blind-ended out pouching (up to 200µm in diameter) of the respiratory bronchioles, alveolar ducts, and alveolar sacs.
- about 150 to 250 million alveoli are found in each adult lung; their combined internal surface area is approximately 75 m², roughly the size of a tennis court.
- constitute the bulk of lung parenchyma
- lined by very thin simple squamous epithelium supported by complete basal lamina and a network of reticular and elastic fibers.
 - This epithelium is known as alveolar epithelium and it is formed mainly by Type-I alveolar cells (pneumocytes I cells).

ALVEOLI

- an alveolar sac consists of two or more alveoli that share a common opening, and alveolar sac and alveoli are responsible for the spongy structure of the lungs.
- alveoli resemble small pockets that are open on one side, similar to the honeycombs of a beehive.

An Alveolus

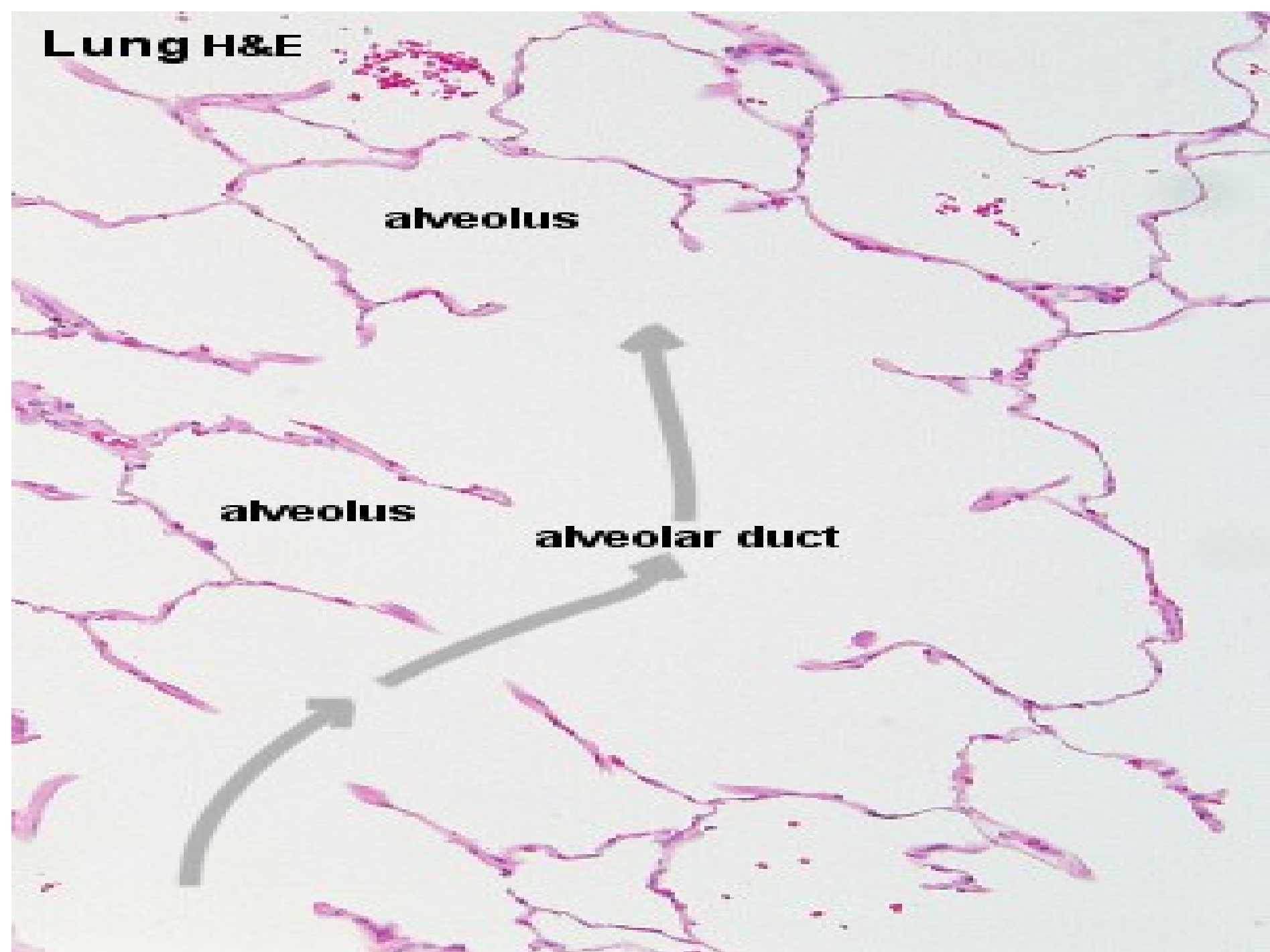


Lung H&E

alveolus

alveolus

alveolar duct



ALVEOLI

- enveloped by a rich network of pulmonary capillaries forming capillary basket around the alveolus.
- The structure of alveolar wall is also extremely thin due to highly attenuated Type I pneumocytes.
 - This arrangement provides a vast interface (which has been calculated to be approximately 140 m^2) of minimal thickness for gaseous (O_2 and CO_2) exchange between the air in the alveoli and the blood in the surrounding pulmonary capillary basket.
- Between two neighboring alveoli is thin CT called an **interalveolar septum (lung interstitium)**.
- These fibrous septa contain the cells and ECM of CT, notably the elastic and reticular fibers, which is highly vascularized with the richest capillary network in the body.

Lung respiratory zone



Erythrocyte

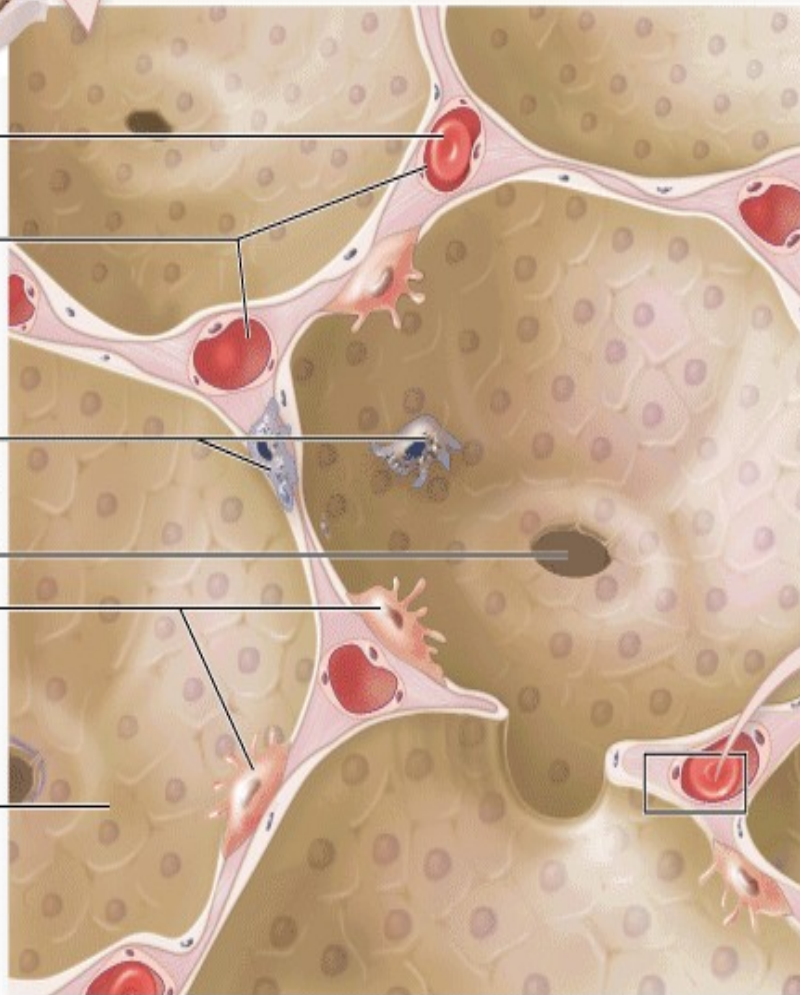
Pulmonary capillaries

Alveolar macrophages

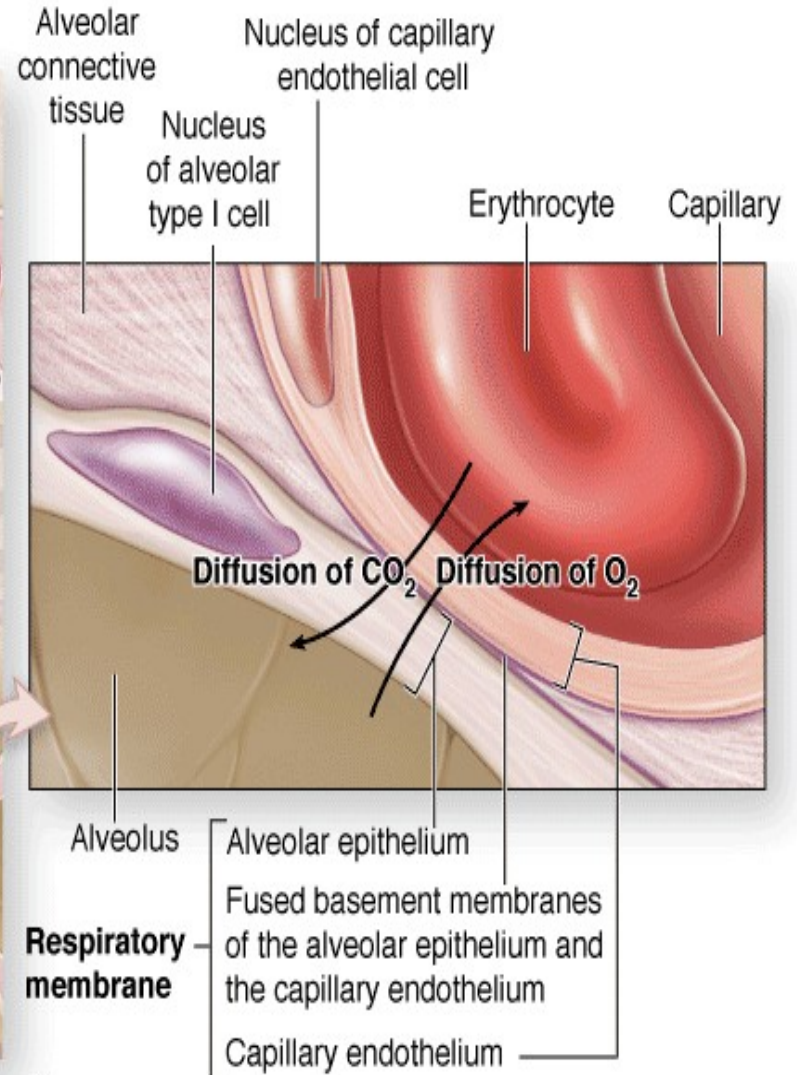
Alveolar pore

Alveolar type II cells

Alveolar type I cell

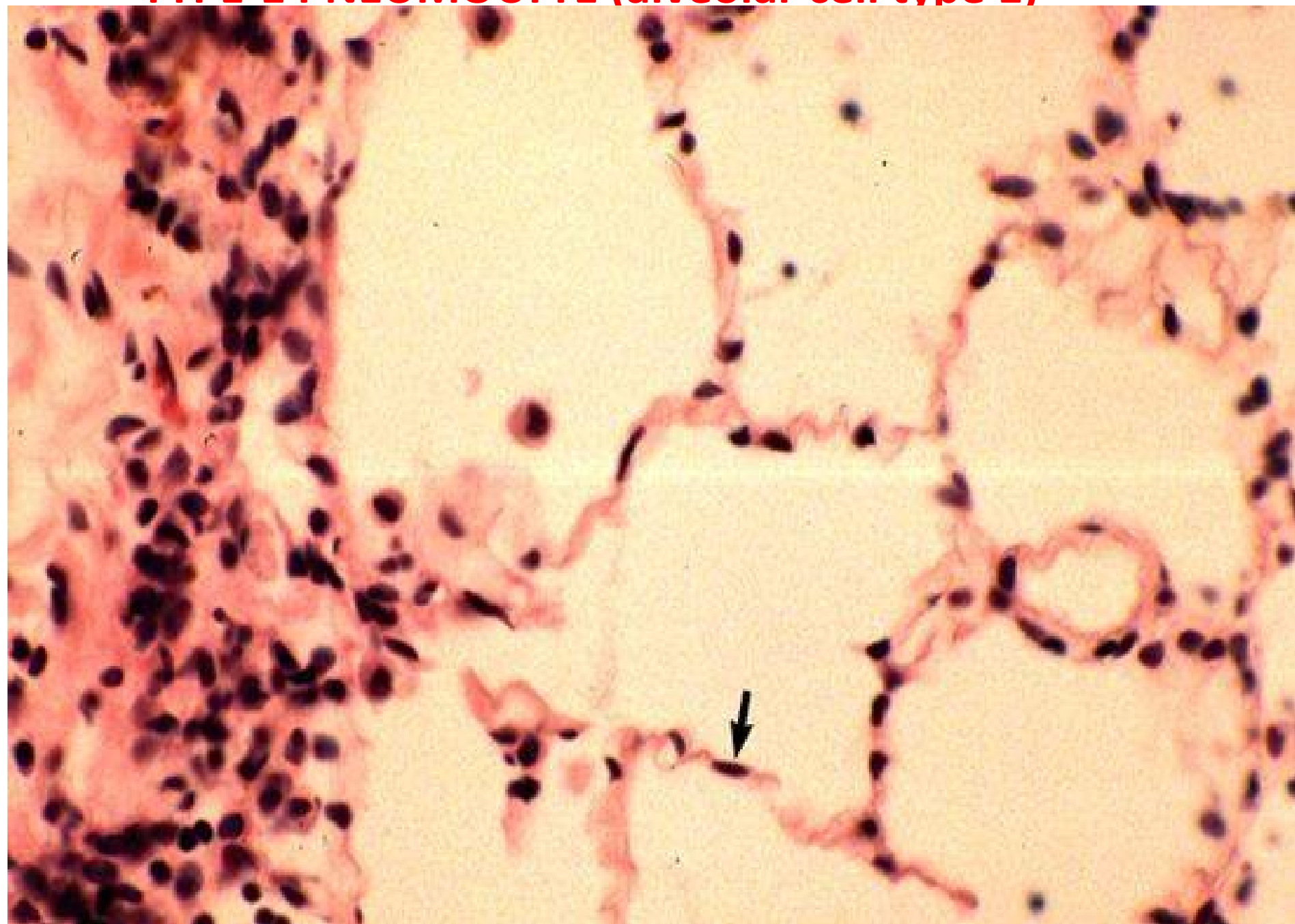


a

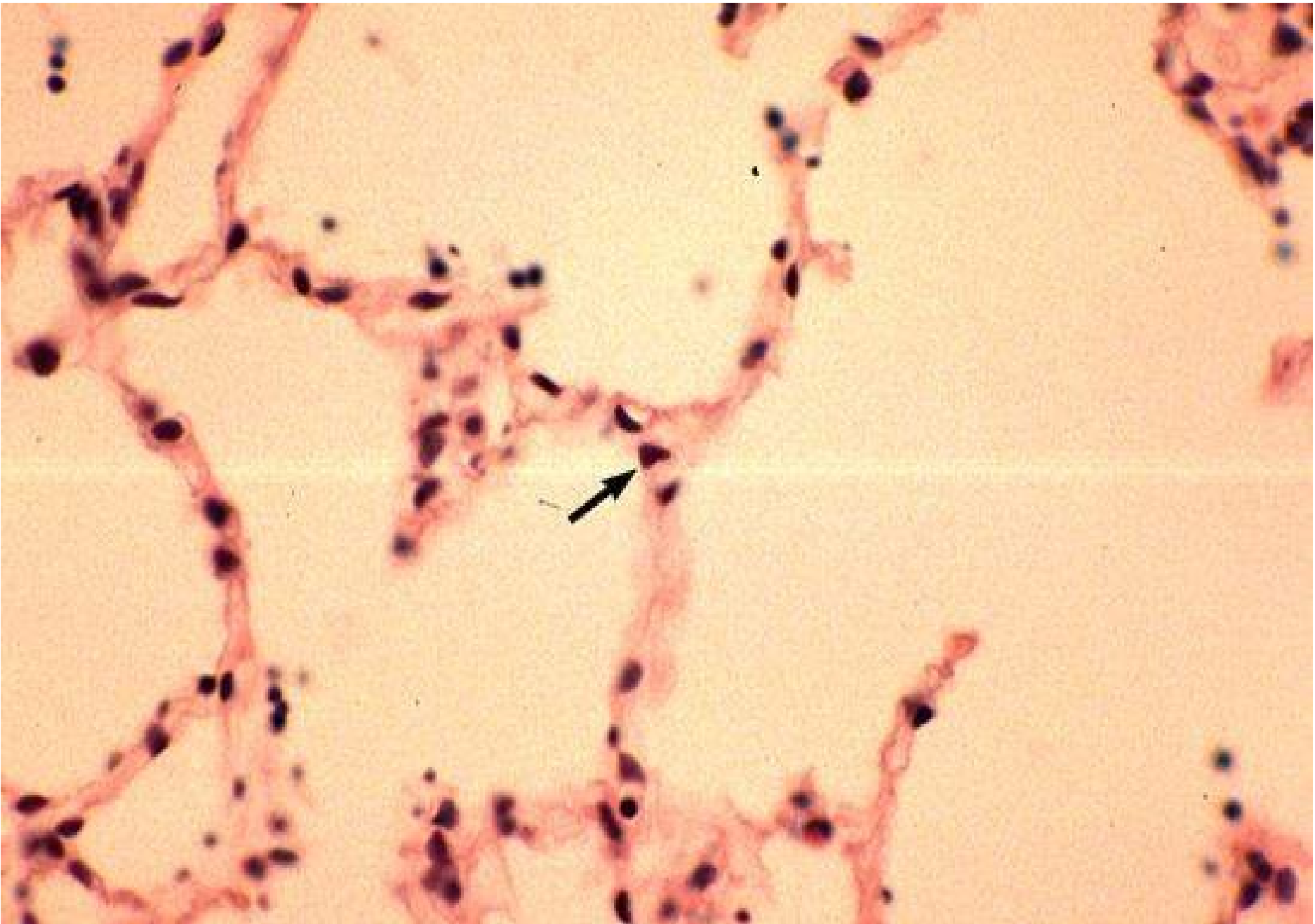


b

TYPE-1 PNEUMOCYTE (alveolar cell type 1)

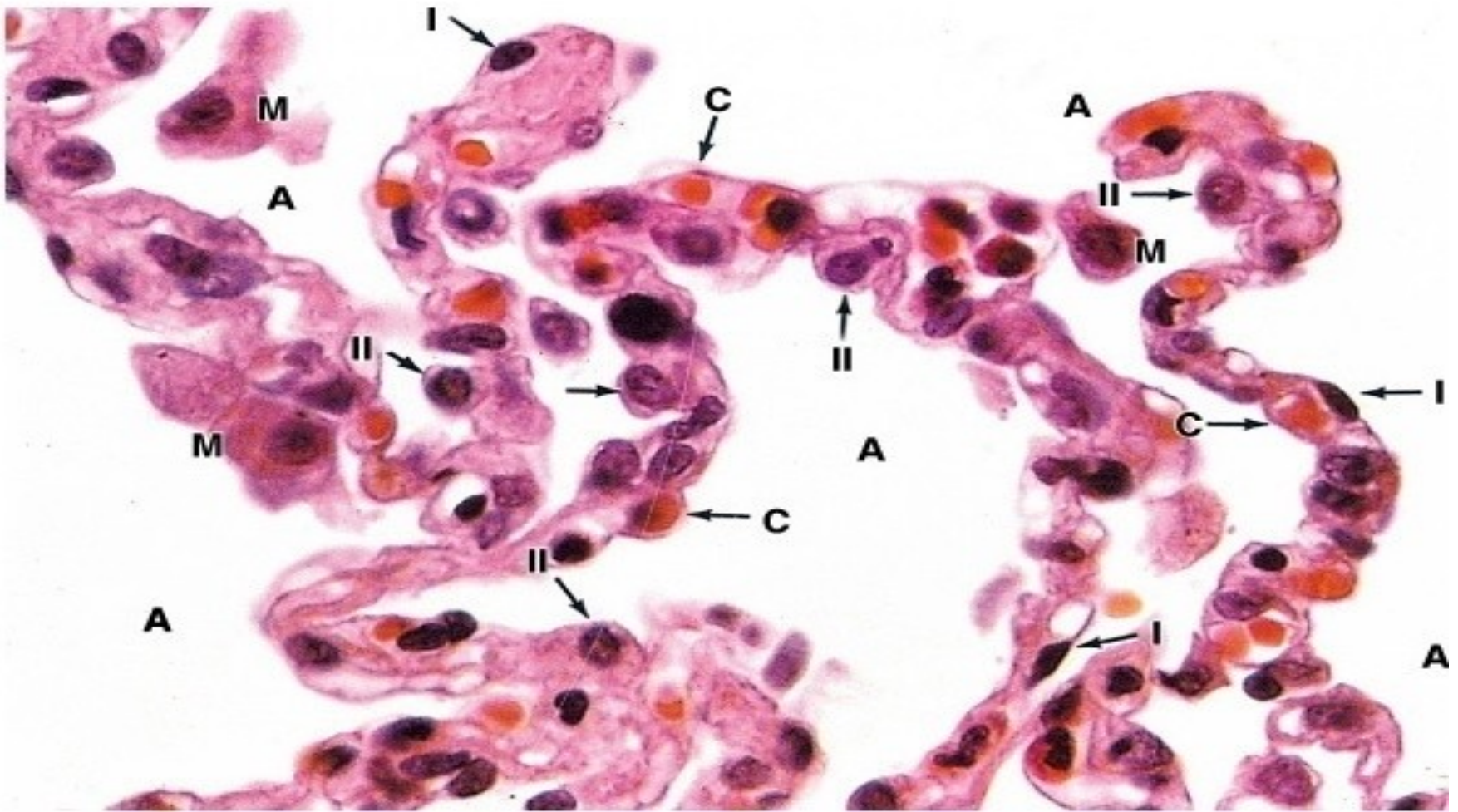


TYPE-2 PNEUMOCYTE (alveolar cell type II)



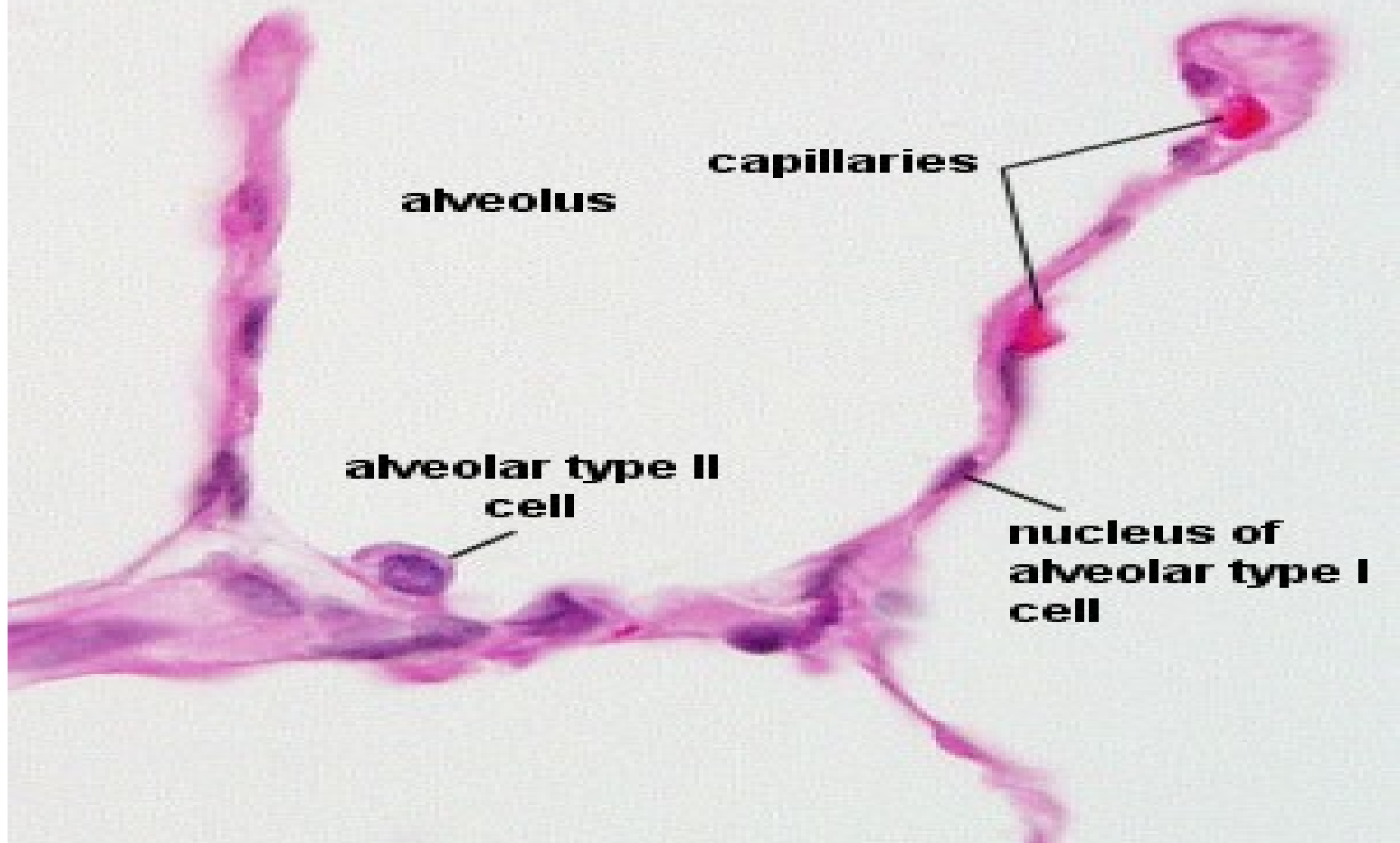
Alveolar Cells Type II (Type II Pneumocytes)

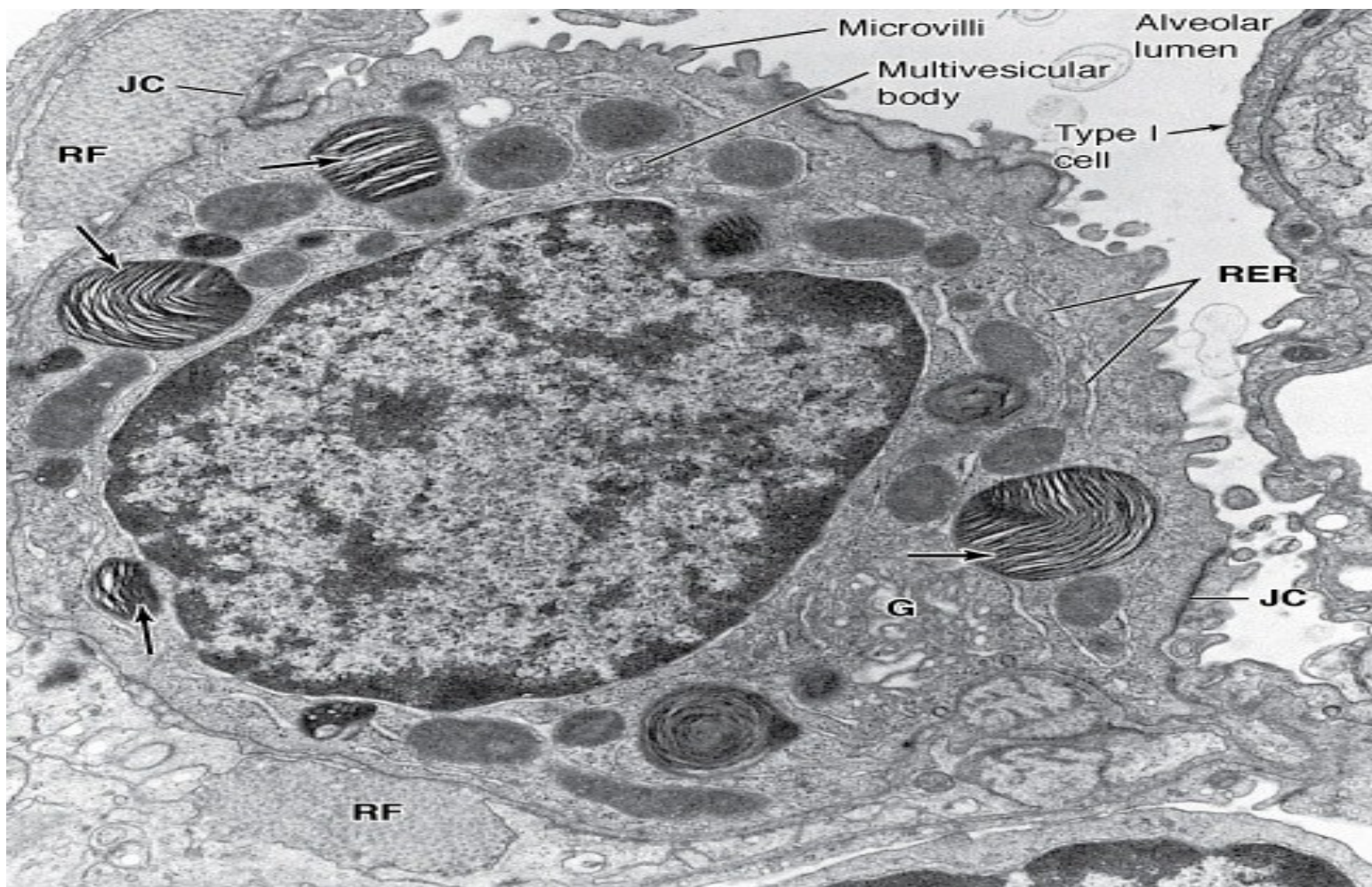
- In histologic sections, they exhibit a characteristic **vesicular** or **foamy** peripheral **cytoplasm**, where **membrane bound secretory** vesicles called **lamellar bodies** are located.
- Lamellar bodies, which average 1–2 μ m in diameter, contain concentric or parallel lamellae limited by a unit membrane (phospholipid monolayer).



Alveolar walls: The space between alveoli (A) contains several cell types. As seen here the capillaries (C) contain erythrocytes and leukocytes. The alveoli are lined mainly by squamous type I alveolar cells (I), which line almost the entire alveolus surface and across which gas exchange occurs. Type II alveolar cells line a bit of each alveolus and are large rounded cells, often bulging into the alveolus (II). Also present are alveolar macrophages (M), sometimes called dust cells, which may be in the alveoli or in the interalveolar septa.

Lung H&E





TEM of a type II alveolar cell protruding into the alveolar lumen shows unusual cytoplasmic features. Arrows indicate lamellar bodies which store newly synthesized pulmonary surfactant after processing of its components in rough ER (RER) and the Golgi apparatus (G). Smaller multivesicular bodies with intraluminal vesicles are also often present. Short microvilli are present and the type II cell is attached via junctional complexes (JC) with the adjacent, very thin type I epithelial cell. The ECM contains prominent reticular fibers and elastic fibers (RF). X17,000.

Respiratory distress syndrome (RDS)

- In premature delivery, type II cells are immature and do not produce the necessary pulmonary surfactant. This leads to fatal respiratory difficulty in new born (***respiratory distress syndrome***).
- principally associated with prematurity and is the leading cause of mortality among premature infants. The incidence of respiratory distress syndrome varies inversely with gestation age.
- The immature lung is deficient in both the amount and composition of surfactant. In the normal newborn, the onset of breathing is associated with a massive release of stored surfactant, which reduces the surface tension of the alveolar cells.
- This means that less inspiratory force is needed to inflate the alveoli, and thus the work of breathing is reduced.

Respiratory distress syndrome (RDS)

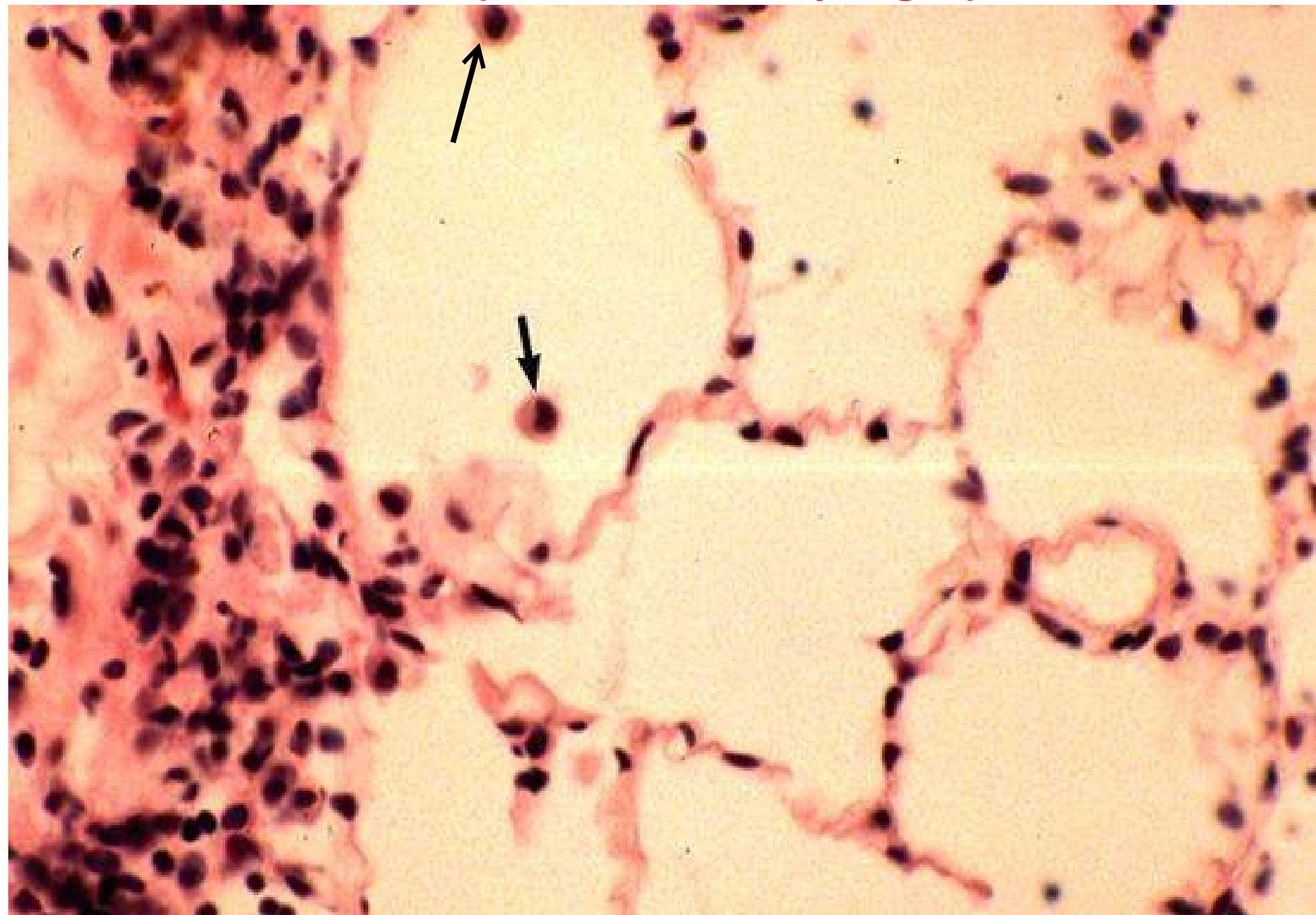
- Microscopically, the alveoli are collapsed and the respiratory bronchioles and alveolar ducts are dilated and contain edema fluid.
- RDS is treated by introducing synthetic or animal-derived surfactant into the lungs through a tube.
- Surfactant also has bactericidal effects, aiding in the removal of potentially dangerous bacteria that reach the alveoli.
- **Pulmonary Surfactant:** Type II pneumocytes secrete phospholipid surfactant that decreases the alveolar surface tension forces to a minimal level thus preventing the alveoli from collapse.
- The presence of this secretion is important for the *newborn* to obtain their first breath of air.

- **The distance between the two adjacent alveoli is called the interalveolar septum, and is formed of the following:**
 - 1- Thin alveolar epithelium (Type I and II pneumocytes)
 - 2- Basal lamina of the alveolar epithelium
 - 3- supporting connective tissue (lung interstitium) containing:
 - **Fibres:** reticular and elastic.
 - **Lung stromal cells:** fibroblasts, fibrocytes, mast cells, leukocytes, alveolar macrophages (dust cells).
 - Other septal cells contain bundles of actin and myosin filaments which contract in response to hypoxia (their role is unknown).
 - **Blood capillaries (continuous, non-fenestrated);**
 - 4- Basal lamina of the alveolar epithelium.
 - 5- Thin alveolar epithelium
- The air in the alveoli is separated from the blood inside the capillaries by the **blood air (pulmonary) barrier.**

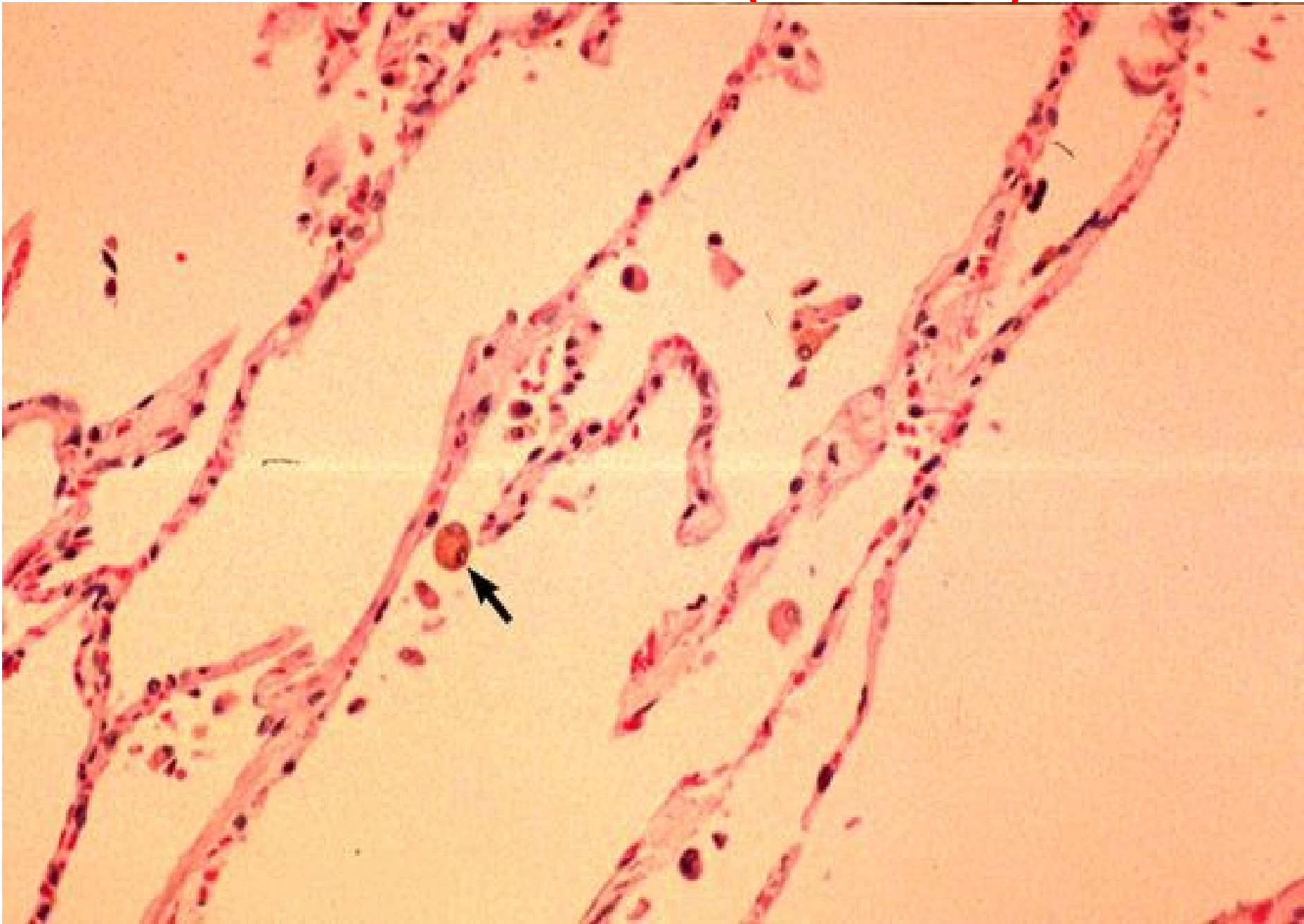
Alveolar macrophages (also called dust cells)

- are phagocytic cells which phagocytose erythrocytes lost from damaged capillaries and air-borne particulate matter that has entered alveoli.
- are found in lumina of the alveoli and in the interalveolar septum
- Some debris within these cells was most likely passed from the alveolar lumen into the interstitium following the pinocytotic activity of type I alveolar cells.
- Active macrophages in lung are often slightly darker due to their content of dust and carbon from air and complexed iron (hemosiderin) from erythrocytes.
- Filled macrophages have various fates: most migrate into bronchioles where they move up the mucociliary escalator for removal in the pharynx; others exit the lungs in the lymphatic drainage, while some remain in the interalveolar connective tissue septa for years.
- Alveolar lining fluids are also removed via the conducting passages as a result of ciliary activity

DUST CELLS (Alveolar macrophages)



ALVEOLAR MACROPHAGES (DUST CELLS)



Cells of the interalveolar septum

- 1-Capillary endothelial cells (30%)
- 2-Pneumocytes type I (8%)
- 3-Pneumocytes type II (Septal cells) (16%)
- 4-Interstitial cells (Fibroblast, fibrocytes, leukocytes & mast cells)(36%)
- 5-Alveolar macrophages (DUST cells) (10%)

Alveolar Pores (of Kohn)

Functions:

- 1- They can equalize the pressure inside the alveoli
- 2- They allow collateral air circulation if the bronchiole is obstructed

Hans Kohn (1866–1935), pathologist, Berlin, Germany

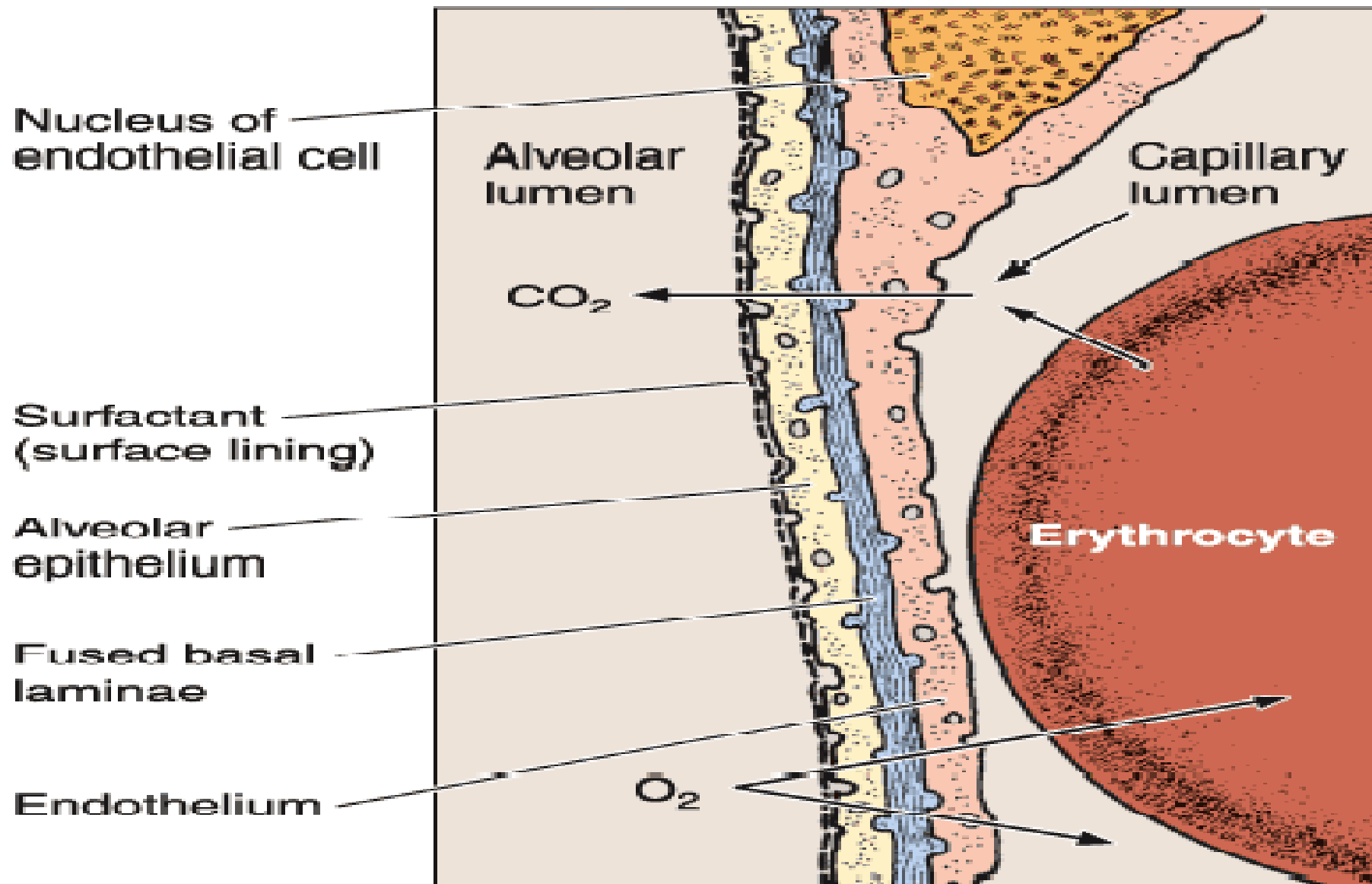
Pleura

- It is a serous membrane covering the lungs.
- It is formed of two layers parietal and visceral which are continuous at the hilum.

- Air within the alveolus is separated from capillary blood by three components referred to collectively as the **respiratory membrane** or **blood-air barrier/pulmonary barrier**.
- The **pulmonary barrier** consists of
 1. An alveolar epithelium and its basal lamina
 2. Inter-alveolar CT septum
 3. A capillary endothelium and its basal lamina

When the pulmonary barrier is extremely thin it consists of only:

1. Cytoplasm of the alveolar epithelial cells
2. Fused Basal lamina of alveolar epithelial cells and Basal lamina of the capillary endothelial cells and
3. Cytoplasm of the endothelial cells.

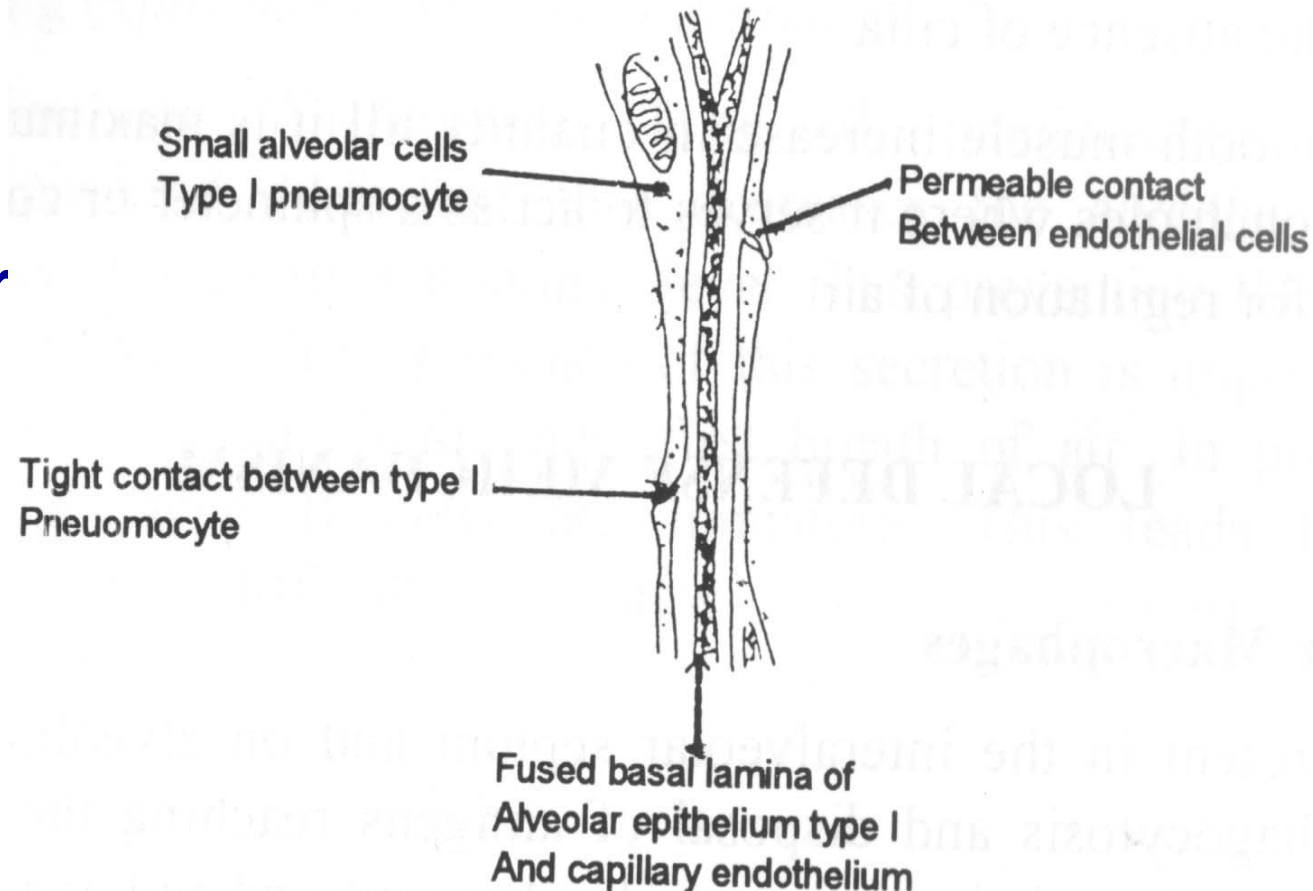


Portion of the interalveolar septum showing the blood air barrier. To reach the erythrocyte, O₂ traverses the surface lining, the alveolar epithelium cytoplasm, and the plasma. In some locations, there is loose interstitial tissue between the epithelium and the endothelium.

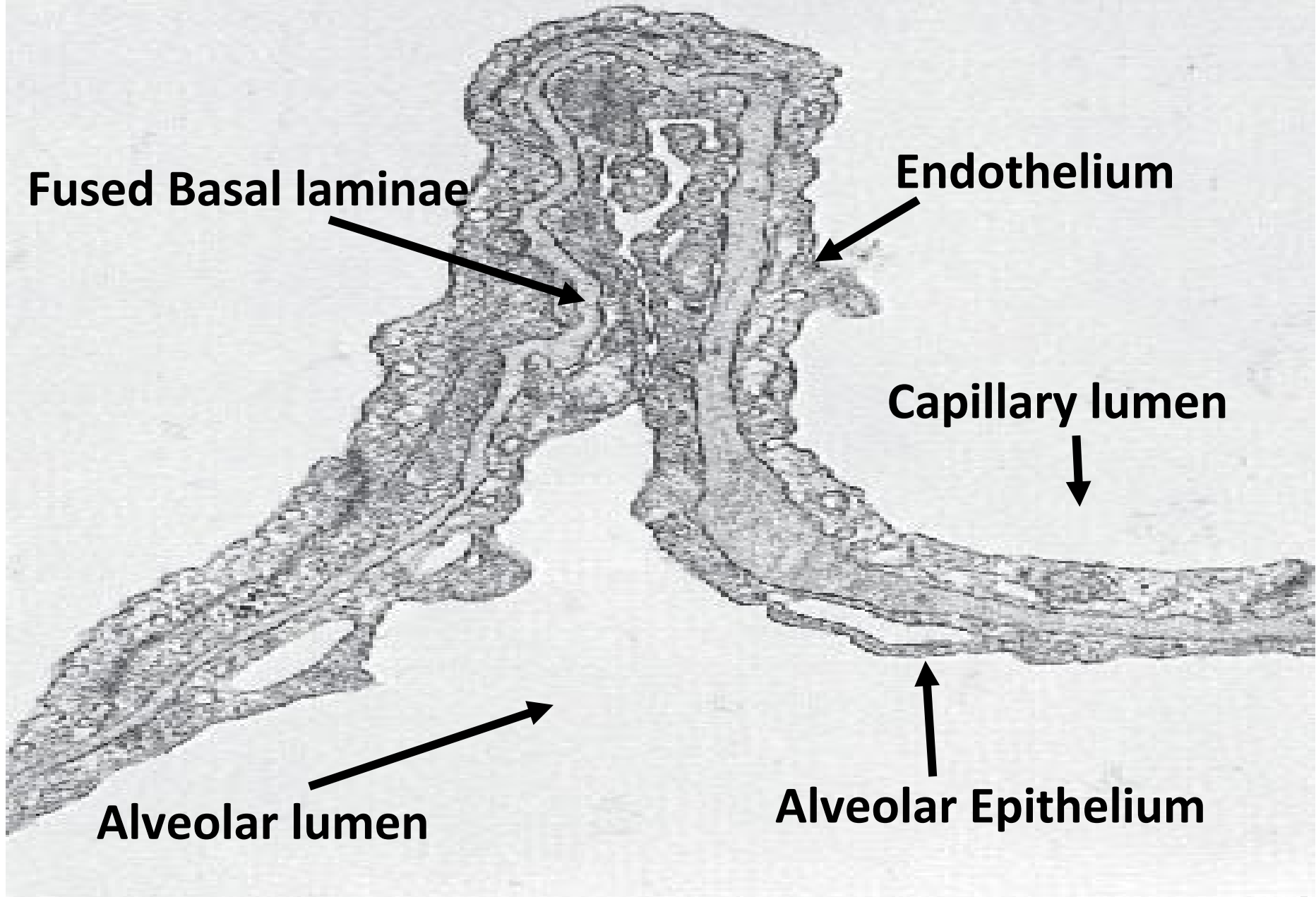
Blood-Air barrier: (Alveolar Membrane, 0.1 to 1.5 μm thick):
consists of:

- Simple squamous epithelium of Type I pneumocyte cells of the alveolus.
- Common or fused basement membrane.
- Endothelial lining of capillaries.

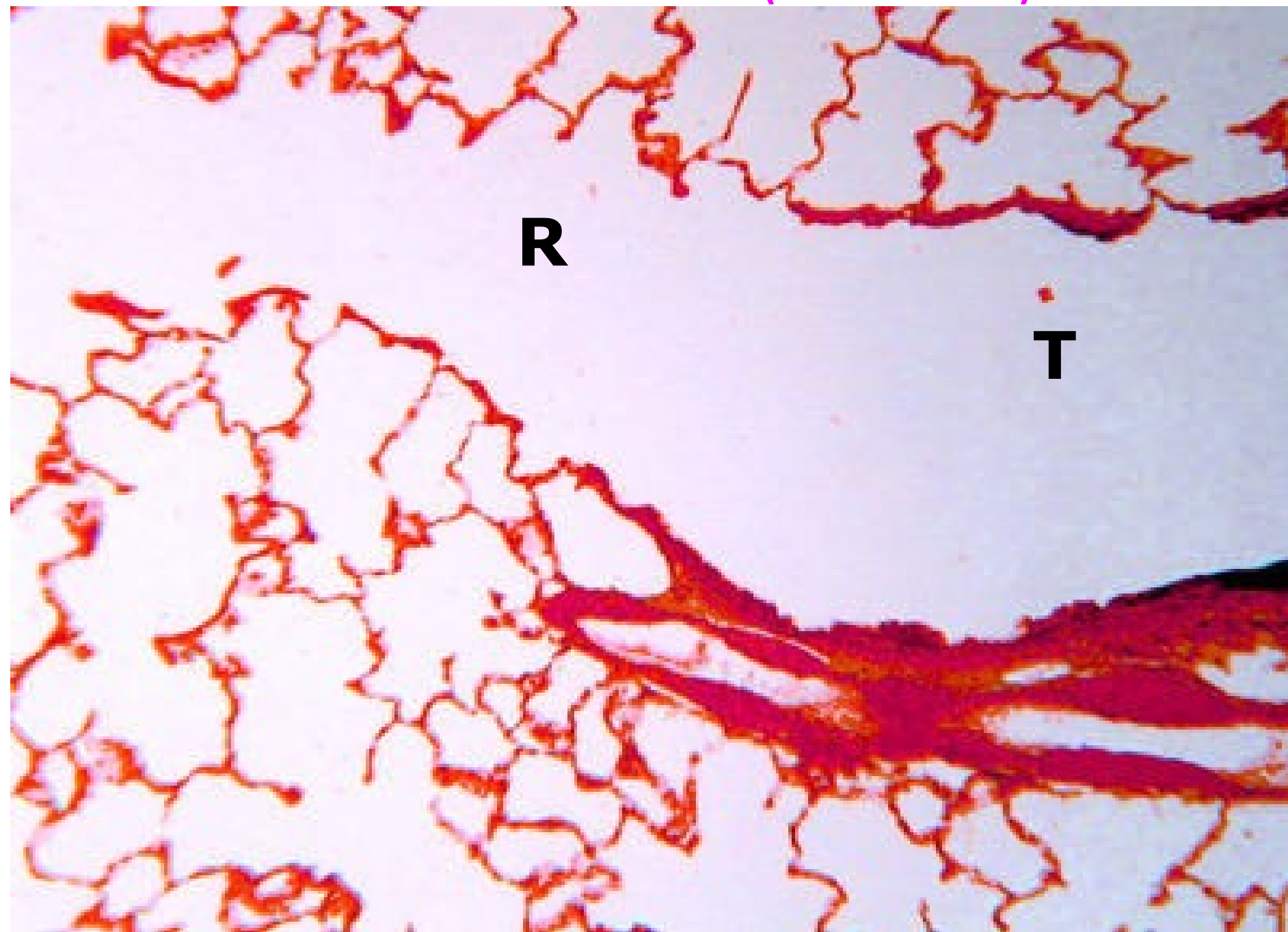
- The alveoli inter-communicate through alveolar pores, which equilibrate air pressure within a lobule, but may cause spread of infection.



Pulmonary membrane



LUNG BY HEMATOXYLIN & EOSIN (HIGH POWER)



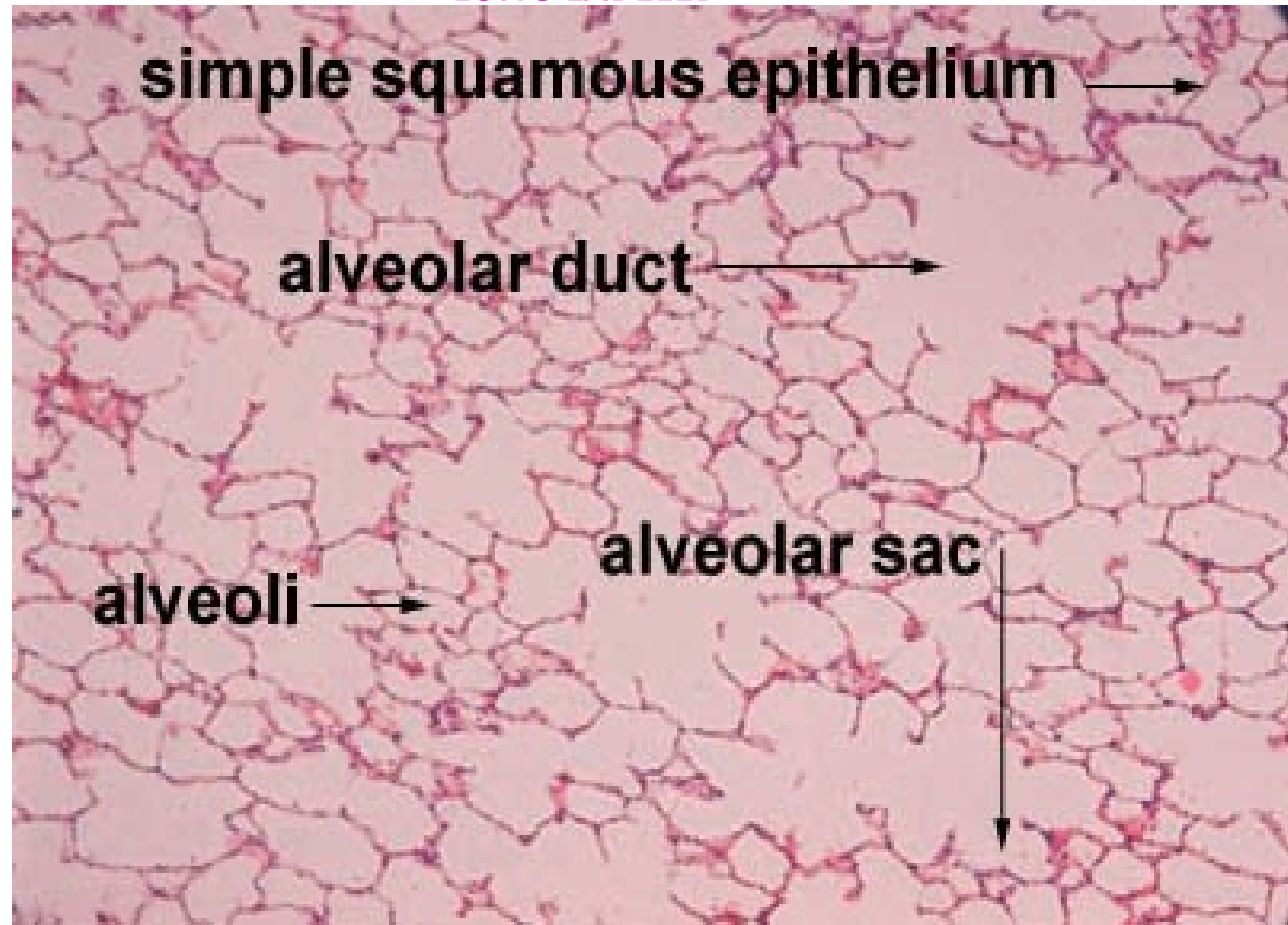
LUNG LABELED

simple squamous epithelium →

alveolar duct →

alveoli →

alveolar sac ↓



Secretory epithelial cell (septal cell) containing
Multilamellar bodies (type II pneumocyte)

Alveolus

Reticular fibres

Elastic fibres

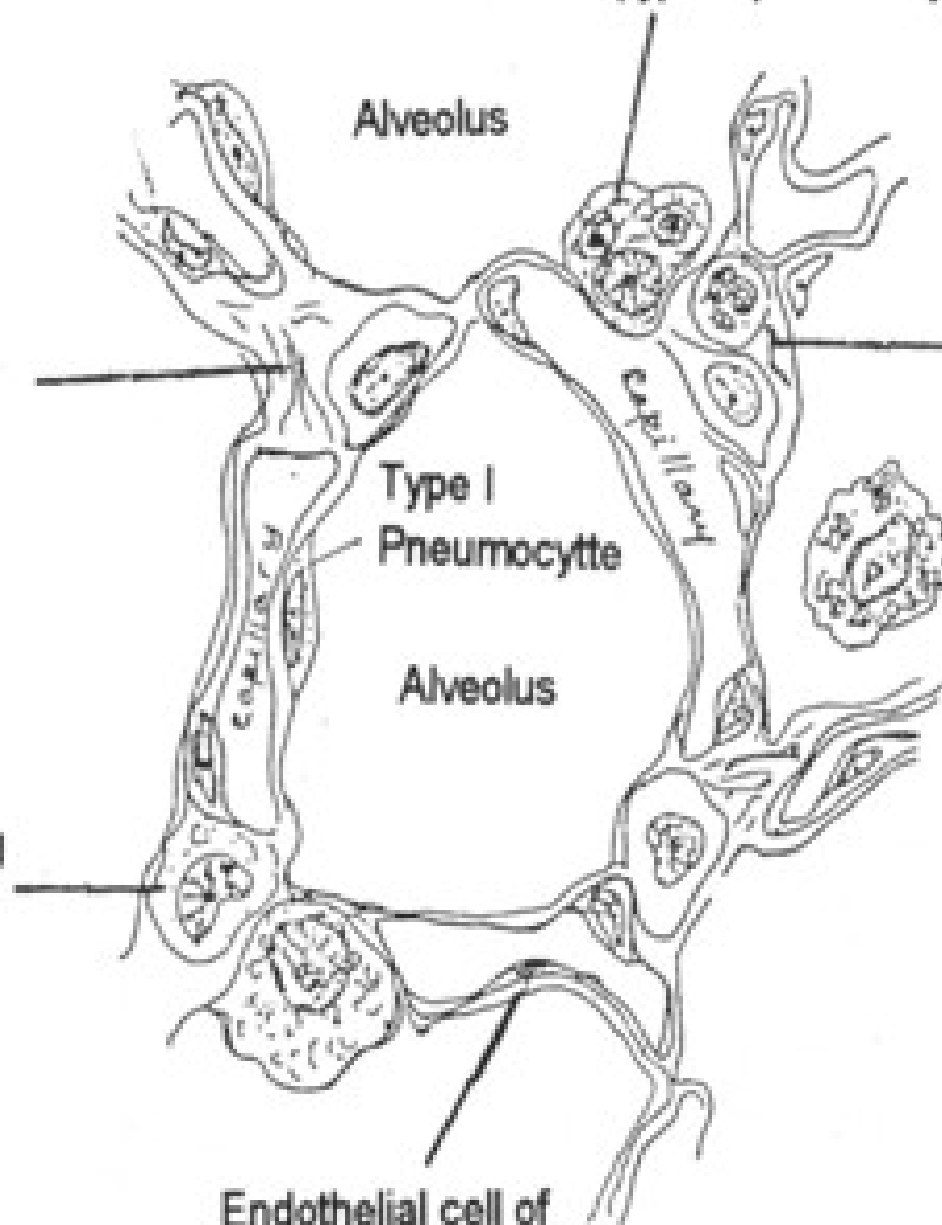
Type I
Pneumocyte

Alveolus

Alveolar Phagocyte
(macrophage)

Monocyte transforming
into macrophage

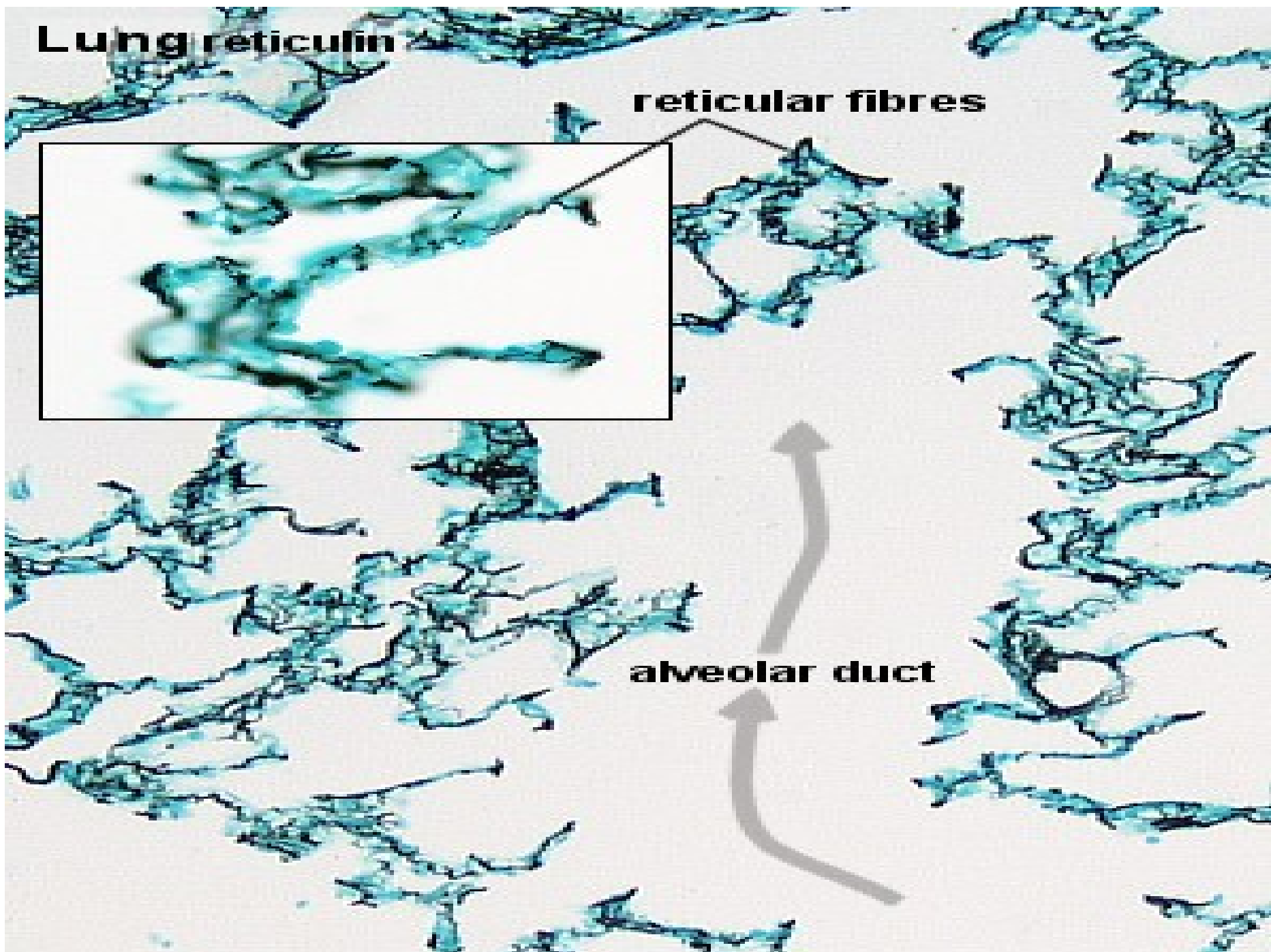
Endothelial cell of
Capillary



Lung reticulin

reticular fibres

alveolar duct



Summary of structures of the Respiratory Tree

- The structure of respiratory system changes according to the functional need:
- The air is inhaled through nose or mouth where it is humidified, warmed and the suspended particles trapped by the hair of nasal vestibule and the mucus present on the mucosa. These particles are ultimately expelled out by the cilia.
- As we go down the respiratory tree the cartilage decreases till it is completely absent in the bronchioles. In trachea it is present as a single semilunar plate while in secondary and tertiary bronchi, it is present in few discontinuous patches.
- The height of the epithelium decreases down the tree till it becomes interrupted simple cuboidal in the respiratory bronchioles.
- The cilia decrease in number as we go down till they are completely absent in the alveolar ducts.
- **The goblet cells disappear within the terminal bronchiole & beyond.**
- The smooth muscle increase in quantity till it is maximum in the bronchioles where it serves to act as a sphincter or control valve for regulation of air.

REVISION

Trachea

- Pseudostratified columnar, ciliated epithelium (Tallest)
- Goblet cells
- Sero-mucous glands
- Cartilage rings

Bronchi

- Pseudostratified columnar, ciliated epithelium (shorter)
- Goblet cells (fewer)
- Sero-mucous glands
- Cartilage
- Smooth muscle

Bronchioles

- Pseudostratified columnar, ciliated epithelium (shorter)
- Sparse goblet cells, no glands & no cartilages
- Smooth muscle (relatively abundant)

Terminal bronchioles

- Simple columnar ciliated epithelium with secretory club/Clara cells

Respiratory Bronchioles

- Simple cuboidal, ciliated epithelium (shorter) with club/Clara cells
- Receive alveolar openings

Alveoli

Type I Pneumocyte

- flattened for gas exchange
- covers 95% of alveolar surface (but only accounts for about 40% of pneumocyte population)

Type II Pneumocyte

- Cuboidal, secretory cells
- produces surfactant
- constitutes about 60% of the pneumocytes (but only 5% of the surface area)

Bronchus

1. It is long, wide and its lumen is usually patent (not collapsed).
2. Its mucosa is folded and is lined with ps. st. col. cil. with goblet cells.
3. The C.T. adventitia is rich in elastic fibres and lymphoid follicles.
4. The smooth muscles are arranged in a spiral manner.
5. Presence of muco-serous glands outside the muscles.
6. Presence of plates of cartilage in a layer of fibrous tissue.

Bronchiole

1. It is short, narrow and its lumen may be collapsed as in bronchial asthma.
2. Its mucosa is highly folded and is lined with simple col. ciliated but with No goblet cells.
3. The C.T. corium has No lymphoid follicles.
4. The smooth muscles are well developed and circularly arranged.
5. No glands.
6. No cartilage.

Regeneration in the Alveolar Lining

- Inhalation of toxic gases or similar materials can kill both Type I and Type II pneumocytes lining pulmonary alveoli.
- Death of the first cells results in increased mitotic activity in the remaining type II cells, the progeny of which become both cell types.
- The normal turnover rate of type II cells is estimated to be 1% per day and results in a continuous renewal of both alveolar cells.
- With increased toxic stress Clara cells can also divide and give rise to alveolar cells.

Lung Development and Maturation

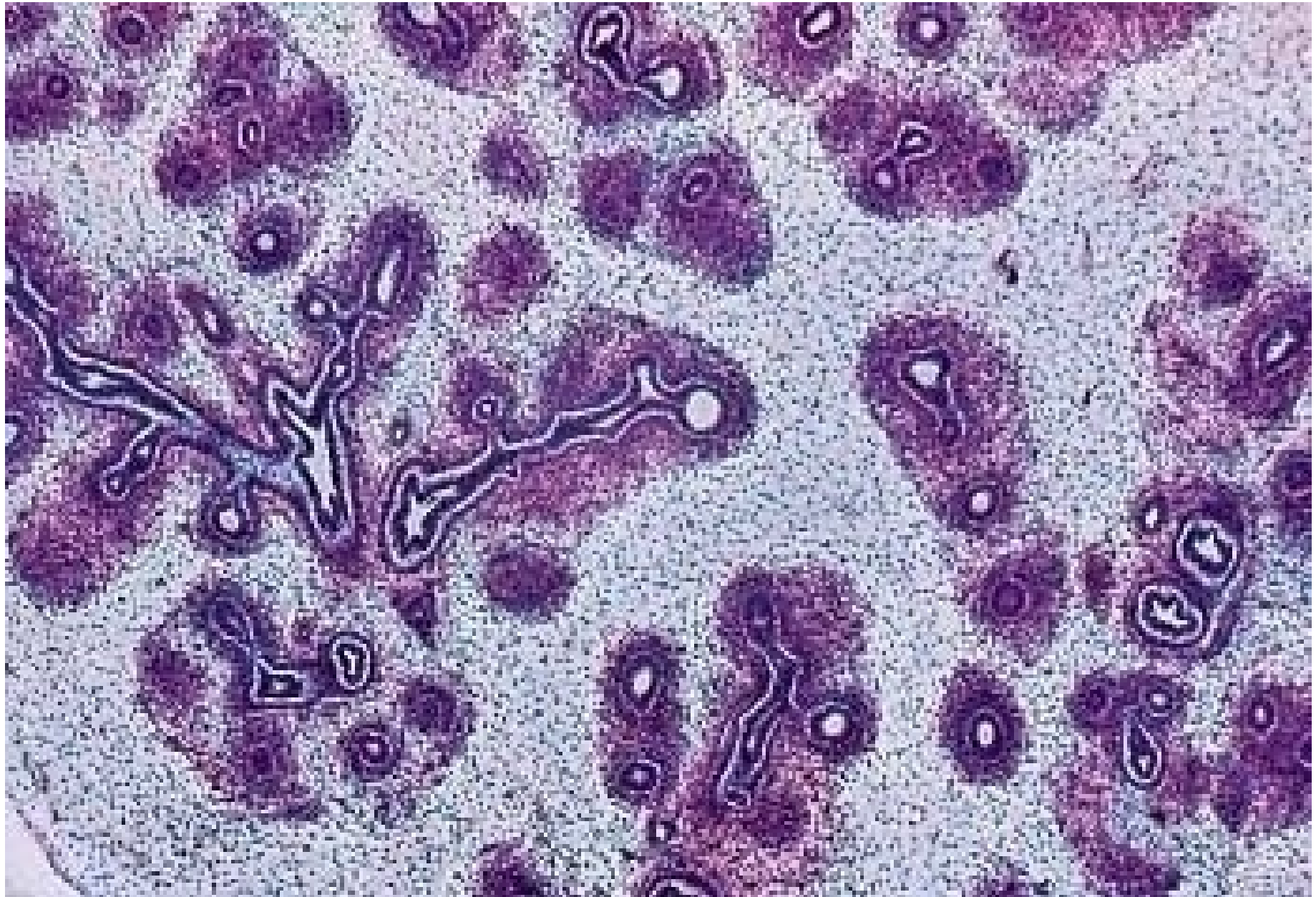
1. Pseudoglandular period:

- lasts from 5th week to 15th or 16th weeks.
- duct system develops, named so for its resemblance to compound exocrine glands.
- bronchi grow and branch to give rise to bronchial tree and large bronchioles
- alveoli are not present

2. Canalicular period:

- Lasts from 16th or 17th to 24th weeks.
- bronchi and bronchioles begin to form terminal bronchioles and then respiratory bronchioles with developing primitive alveoli lined by columnar epithelium.
- lung tissue is vascularized.

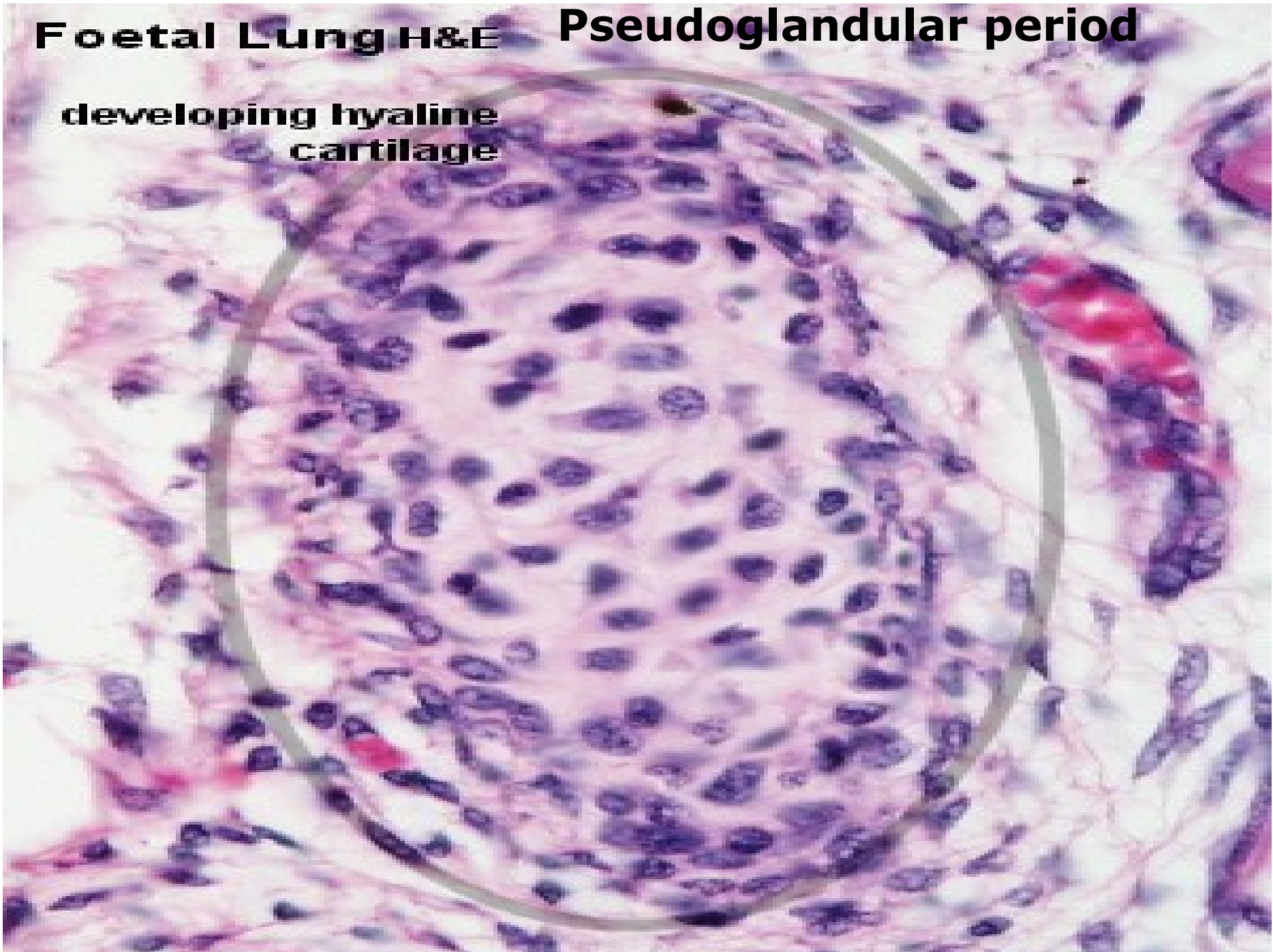
Developing lung, Pseudoglandular Period



Foetal Lung H&E

Pseudoglandular period

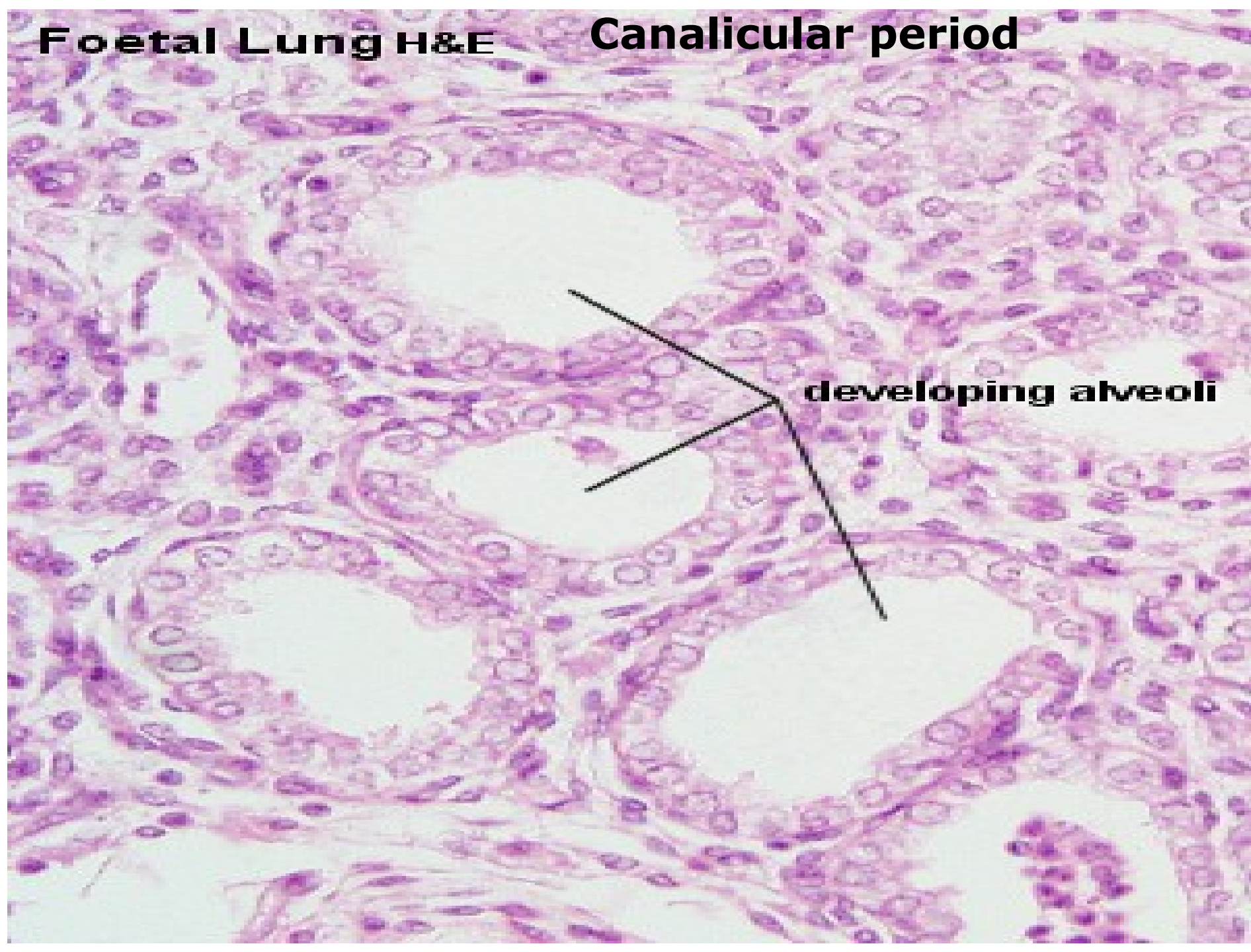
**developing hyaline
cartilage**



Foetal Lung H&E

Canalicular period

developing alveoli



3. Terminal Saccular period:

- 25th weeks to 32nd weeks.
- development of terminal sacs which further differentiate into alveolar sac and attached alveoli.
- differentiation of alveolar epithelium
- surfactant appears in this period and coincides with the appearance of lamellar bodies in Type II pneumocytes.

4. Alveolar period (both prenatal and postnatal):

- from 33rd week of gestation to the first 8 or 10 years
- characterized by maturation and tremendous increase in number of alveoli.
- at birth, only about 1/8th of alveoli found in the adult are present. i.e. more than 88% of alveoli develop postnatally.

CLINICAL NOTES

A. Hyaline membrane disease (*respiratory distress syndrome*)

- is frequently observed in premature infants who lack adequate amounts of pulmonary surfactant.
- is characterized by labored breathing, which results from difficulty in expanding the alveoli due to high alveolar surface tension (caused by inadequate levels of surfactant).
- is often be circumvented, if detected before birth, by the administration of glucocorticoids, which induce synthesis of surfactant.

B. Asthma

- is marked by excessive contraction of smooth muscles in the wall of bronchioles, causing their constriction, a decrease in their diameter.
- is associated with extremely difficult expiration of air, accumulation of mucus in the passageways, and infiltration of inflammatory cells into the region.
- is often progressive and associated with allergic reactions during which vasoconstrictive substances are periodically released within the lungs.
- is treated by administering drugs such as epinephrine and isoproterenol, & other sympathomimetic drugs, which function to relax the bronchiolar smooth muscles, thus dilating the passageways.

C. Emphysema

- a chronic lung disease characterized by enlargement of the air spaces distal to the bronchioles into large cyst-like sacs, with destruction of the interalveolar wall.
- usually develops gradually and results in respiratory insufficiency by reducing the surface available for gas exchange.

- is marked by decreased lung elasticity, which is unable to recoil adequately during expiration.
- in time, the lungs become expanded and enlarge the thoracic cavity (“**barrel chest**”).
- its major cause is **cigarette smoking** and exposure to other substances that inhibit **α_1 -antitrypsin** secretion, a protein that normally protects the lungs from the action of **elastase** produced by alveolar macrophages.
- Irritation produced by cigarette smoking stimulates the destruction, or impairs the synthesis of elastic fibers and other components of the interalveolar septum.
- can be a hereditary condition resulting from a defective α_1 -antitrypsin. In such cases, gene therapy with recombinant α_1 -antitrypsin is being used.

D. Lung Tumor/Cancer

- Is squamous cell carcinoma, the principal lung tumor type,
- usually results from the effects of cigarette smoking on the bronchial and bronchiolar epithelium.
- Chronic smoking induces the transformation of the respiratory epithelium into a stratified squamous epithelium
 - This transformation also known as **metaplasia**, is an initial step in its eventual differentiation into a lung tumor or cancer.

E. Chronic Obstructive Pulmonary Disease (COPD)

- **Chronic lung diseases** such as and asthma are associated with changes in the abundance of CC16 in airway fluid and serum.
- CC16 is used as a measurable pulmonary marker in bronchoalveolar lavage fluid and serum.
- Secretion of CC16 into the bronchial tree decreases during lung injury (because of damage to the Clara cells), whereas serum levels of CC16 may increase because of leakage across the air–blood interface.